

Chapter II

The Theory of Valuations

§ 1. The p -adic Numbers

The p -adic numbers were invented at the beginning of the twentieth century by the mathematician *KURT HENSEL* (1861–1941) with a view to introduce into number theory the powerful method of power series expansion which plays such a predominant rôle in function theory. The idea originated from the observation made in the last chapter that the numbers $f \in \mathbb{Z}$ may be viewed in analogy with the polynomials $f(z) \in \mathbb{C}[z]$ as functions on the space X of prime numbers in $\mathbb{Z}$, associating to them their “value” at the point $p \in X$, i.e., the element

$$f(p) := f \bmod p$$

in the residue class field $\kappa(p) = \mathbb{Z}/p\mathbb{Z}$.

This point of view suggests the further question: whether not only the “value” of the integer $f \in \mathbb{Z}$ at p , but also the higher derivatives of f can be reasonably defined. In the case of the polynomials $f(z) \in \mathbb{C}[z]$, the higher derivatives at the point $z = a$ are given by the coefficients of the expansion

$$f(z) = a_0 + a_1(z - a) + \cdots + a_n(z - a)^n,$$

and more generally, for rational functions $f(z) = \frac{g(z)}{h(z)} \in \mathbb{C}(z)$, with $g, h \in \mathbb{C}[z]$, they are defined by the Taylor expansion

$$f(z) = \sum_{v=0}^{\infty} a_v(z - a)^v,$$

provided there is no pole at $z = a$, i.e., as long as $(z - a) \nmid h(z)$. The fact that such an expansion can also be written down, relative to a prime number p in $\mathbb{Z}$, for any rational number $f \in \mathbb{Q}$ as long as it lies in the local ring

$$\mathbb{Z}_{(p)} = \left\{ \frac{g}{h} \mid g, h \in \mathbb{Z}, p \nmid h \right\},$$

leads us to the notion of p -adic number. First, every positive integer $f \in \mathbb{N}$ admits a **p -adic expansion**

$$f = a_0 + a_1p + \cdots + a_np^n,$$

with coefficients a_i in $\{0, 1, \dots, p-1\}$, i.e., in a fixed system of representatives of the “field of values” $\kappa(p) = \mathbb{F}_p$. This representation is clearly unique. It is computed explicitly by successively dividing by p , forming the following system of equations:

$$\begin{aligned} f &= a_0 + pf_1, \\ f_1 &= a_1 + pf_2, \\ &\vdots \\ f_{n-1} &= a_{n-1} + pf_n, \\ f_n &= a_n. \end{aligned}$$

Here $a_i \in \{0, 1, \dots, p-1\}$ denotes the representative of $f_i \bmod p \in \mathbb{Z}/p\mathbb{Z}$. In concrete cases, one sometimes writes the number f simply as the sequence of digits $a_0, a_1a_2 \dots a_n$, for instance

$$\begin{aligned} 216 &= 0,0011011 & (2\text{-adic}), \\ 216 &= 0,0022 & (3\text{-adic}), \\ 216 &= 1,331 & (5\text{-adic}). \end{aligned}$$

As soon as one tries to write down such p -adic expansions also for negative integers, let alone for fractions, one is forced to allow infinite series

$$\sum_{v=0}^{\infty} a_v p^v = a_0 + a_1 p + a_2 p^2 + \dots$$

This notation should at first be understood in a purely formal sense, i.e., $\sum_{v=0}^{\infty} a_v p^v$ simply stands for the sequence of partial sums

$$s_n = \sum_{v=0}^{n-1} a_v p^v, \quad n = 1, 2, \dots$$

(1.1) Definition. Fix a prime number p . A **p -adic integer** is a formal infinite series

$$a_0 + a_1 p + a_2 p^2 + \dots,$$

where $0 \leq a_i < p$, for all $i = 0, 1, 2, \dots$. The set of all p -adic integers is denoted by $\mathbb{Z}_p$.

The p -adic expansion of an arbitrary number $f \in \mathbb{Z}_{(p)}$ results from the following proposition about the residue classes in $\mathbb{Z}/p^n\mathbb{Z}$.

(1.2) Proposition. The residue classes $a \bmod p^n \in \mathbb{Z}/p^n\mathbb{Z}$ can be uniquely represented in the form

$$a \equiv a_0 + a_1 p + a_2 p^2 + \dots + a_{n-1} p^{n-1} \bmod p^n$$

where $0 \leq a_i < p$ for $i = 0, \dots, n-1$.

Proof (induction on n): This is clear for $n = 1$. Assume the statement is proved for $n - 1$. Then we have a unique representation

$$a = a_0 + a_1 p + a_2 p^2 + \cdots + a_{n-2} p^{n-2} + g p^{n-1},$$

for some integer g . If $g \equiv a_{n-1} \pmod{p}$ such that $0 \leq a_{n-1} < p$, then a_{n-1} is uniquely determined by a , and the congruence of the proposition holds. $\square$

Every integer f and, more generally, every rational number $f \in \mathbb{Z}_{(p)}$ the denominator of which is not divisible by p , defines a sequence of residue classes

$$\bar{s}_n = f \pmod{p^n} \in \mathbb{Z}/p^n \mathbb{Z}, \quad n = 1, 2, \dots,$$

for which we find, by the preceding proposition,

$$\begin{aligned} \bar{s}_1 &= a_0 \pmod{p}, \\ \bar{s}_2 &= a_0 + a_1 p \pmod{p^2}, \\ \bar{s}_3 &= a_0 + a_1 p + a_2 p^2 \pmod{p^3}, \quad \text{etc.}, \end{aligned}$$

with uniquely determined coefficients $a_0, a_1, a_2, \dots \in \{0, 1, \dots, p-1\}$ which keep their meaning from one line to the next. The sequence of numbers

$$s_n = a_0 + a_1 p + a_2 p^2 + \cdots + a_{n-1} p^{n-1}, \quad n = 1, 2, \dots,$$

defines a p -adic integer

$$\sum_{v=0}^{\infty} a_v p^v \in \mathbb{Z}_p.$$

We call it the **p -adic expansion** of f .

In analogy with the Laurent series $f(z) = \sum_{v=-m}^{\infty} a_v (z-a)^v$, we now extend the domain of p -adic integers into that of the formal series

$$\sum_{v=-m}^{\infty} a_v p^v = a_{-m} p^{-m} + \cdots + a_{-1} p^{-1} + a_0 + a_1 p + \cdots,$$

where $m \in \mathbb{Z}$ and $0 \leq a_v < p$. Such series we call simply **p -adic numbers** and we write $\mathbb{Q}_p$ for the set of all these p -adic numbers. If $f \in \mathbb{Q}$ is any rational number, then we write

$$f = \frac{g}{h} p^{-m} \quad \text{where } g, h \in \mathbb{Z}, \quad (gh, p) = 1,$$

and if

$$a_0 + a_1 p + a_2 p^2 + \cdots$$

is the p -adic expansion of $\frac{g}{h}$, then we attach to f the p -adic number

$$a_0 p^{-m} + a_1 p^{-m+1} + \cdots + a_m + a_{m+1} p + \cdots \in \mathbb{Q}_p$$

as its p -adic expansion.

In this way we obtain a canonical mapping

$$\mathbb{Q} \longrightarrow \mathbb{Q}_p,$$

which takes $\mathbb{Z}$ into $\mathbb{Z}_p$ and is injective. For if $a, b \in \mathbb{Z}$ have the same p -adic expansion, then $a - b$ is divisible by p^n for every n , and hence $a = b$. We now identify $\mathbb{Q}$ with its image in $\mathbb{Q}_p$, so that we may write $\mathbb{Q} \subseteq \mathbb{Q}_p$ and $\mathbb{Z} \subseteq \mathbb{Z}_p$. Thus, for every rational number $f \in \mathbb{Q}$, we obtain an identity

$$f = \sum_{v=-m}^{\infty} a_v p^v.$$

This establishes the arithmetic analogue of the function-theoretic power series expansion for which we were looking.

Examples: a) $-1 = (p-1) + (p-1)p + (p-1)p^2 + \cdots$.

In fact, we have

$$-1 = (p-1) + (p-1)p + \cdots + (p-1)p^{n-1} - p^n,$$

$$\text{hence } -1 \equiv (p-1) + (p-1)p + \cdots + (p-1)p^{n-1} \pmod{p^n}.$$

$$\text{b) } \frac{1}{1-p} = 1 + p + p^2 + \cdots.$$

In fact,

$$1 = (1 + p + \cdots + p^{n-1})(1-p) + p^n,$$

$$\text{hence } \frac{1}{1-p} \equiv 1 + p + \cdots + p^{n-1} \pmod{p^n}.$$

One can define addition and multiplication of p -adic numbers which turn $\mathbb{Z}_p$ into a ring, and $\mathbb{Q}_p$ into its field of fractions. However, the direct approach, defining sum and product via the usual carry-over rules for digits, as one does it when dealing with real numbers as decimal fractions, leads into complications. They disappear once we use another representation of the p -adic numbers $f = \sum_{v=0}^{\infty} a_v p^v$, viewing them not as sequences of sums of integers

$$s_n = \sum_{v=0}^{n-1} a_v p^v \in \mathbb{Z},$$

but rather as sequences of **residue classes**

$$\bar{s}_n = s_n \bmod p^n \in \mathbb{Z}/p^n\mathbb{Z}.$$

The terms of such a sequence lie in different rings $\mathbb{Z}/p^n\mathbb{Z}$, but these are related by the canonical projections

$$\mathbb{Z}/p\mathbb{Z} \xleftarrow{\lambda_1} \mathbb{Z}/p^2\mathbb{Z} \xleftarrow{\lambda_2} \mathbb{Z}/p^3\mathbb{Z} \xleftarrow{\lambda_3} \dots$$

and we find

$$\lambda_n(\bar{s}_{n+1}) = \bar{s}_n.$$

In the direct product

$$\prod_{n=1}^{\infty} \mathbb{Z}/p^n\mathbb{Z} = \{ (x_n)_{n \in \mathbb{N}} \mid x_n \in \mathbb{Z}/p^n\mathbb{Z} \},$$

we now consider all elements $(x_n)_{n \in \mathbb{N}}$ with the property that

$$\lambda_n(x_{n+1}) = x_n \quad \text{for all } n = 1, 2, \dots$$

This set is called the **projective limit** of the rings $\mathbb{Z}/p^n\mathbb{Z}$ and is denoted by $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$. In other words, we have

$$\varprojlim_n \mathbb{Z}/p^n\mathbb{Z} = \{ (x_n)_{n \in \mathbb{N}} \in \prod_{n=1}^{\infty} \mathbb{Z}/p^n\mathbb{Z} \mid \lambda_n(x_{n+1}) = x_n, n = 1, 2, \dots \}.$$

The modified representation of the p -adic numbers alluded to above now follows from the

(1.3) Proposition. *Associating to every p -adic integer*

$$f = \sum_{v=0}^{\infty} a_v p^v$$

the sequence $(\bar{s}_n)_{n \in \mathbb{N}}$ of residue classes

$$\bar{s}_n = \sum_{v=0}^{n-1} a_v p^v \bmod p^n \in \mathbb{Z}/p^n\mathbb{Z},$$

yields a bijection

$$\mathbb{Z}_p \xrightarrow{\sim} \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}.$$

The proof is an immediate consequence of proposition (1.2). The projective limit $\varprojlim \mathbb{Z}/p^n\mathbb{Z}$ offers the advantage of being clearly a ring. In fact, it is a subring of the direct product $\prod_{n=1}^{\infty} \mathbb{Z}/p^n\mathbb{Z}$ where addition and multiplication are defined componentwise. We identify $\mathbb{Z}_p$ with $\varprojlim \mathbb{Z}/p^n\mathbb{Z}$ and obtain the **ring of p -adic integers $\mathbb{Z}_p$** .

Since every element $f \in \mathbb{Q}_p$ admits a representation

$$f = p^{-m}g$$

with $g \in \mathbb{Z}_p$, addition and multiplication extend from $\mathbb{Z}_p$ to $\mathbb{Q}_p$ and $\mathbb{Q}_p$ becomes the field of fractions of $\mathbb{Z}_p$.

In $\mathbb{Z}_p$, we found the rational integers $a \in \mathbb{Z}$ which were determined by the congruences

$$a \equiv a_0 + a_1p + \cdots + a_{n-1}p^{n-1} \pmod{p^n},$$

$0 \leq a_i < p$. Making the identification

$$\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

the subset $\mathbb{Z}$ is taken to the set of tuples

$$(a \bmod p, a \bmod p^2, a \bmod p^3, \dots) \in \prod_{n=1}^{\infty} \mathbb{Z}/p^n\mathbb{Z}$$

and thereby is realized as a subring of $\mathbb{Z}_p$. We obtain $\mathbb{Q}$ as a subfield of the field $\mathbb{Q}_p$ of p -adic numbers in the same way.

Despite their origin in function-theoretic ideas, the p -adic numbers live up to their destiny entirely within arithmetic, more precisely at its classical heart, the **Diophantine equations**. Such an equation

$$F(x_1, \dots, x_n) = 0$$

is given by a polynomial $F \in \mathbb{Z}[x_1, \dots, x_n]$, and the question is whether it admits solutions in integers. This difficult problem can be weakened by considering, instead of the equation, all the congruences

$$F(x_1, \dots, x_n) \equiv 0 \pmod{m}.$$

By the Chinese remainder theorem, this amounts to considering the congruences

$$F(x_1, \dots, x_n) \equiv 0 \pmod{p^v}$$

modulo all prime powers. The hope is to obtain in this way information about the original equation. This plethora of congruences is now synthesized again into a single equation by means of the p -adic numbers. In fact, one has the

(1.4) Proposition. *Let $F(x_1, \dots, x_n)$ be a polynomial with integer coefficients, and fix a prime number p . The congruence*

$$F(x_1, \dots, x_n) \equiv 0 \pmod{p^\nu}$$

is solvable for arbitrary $\nu \geq 1$ if and only if the equation

$$F(x_1, \dots, x_n) = 0$$

is solvable in p -adic integers.

Proof: As established above, we view the ring $\mathbb{Z}_p$ as the projective limit

$$\mathbb{Z}_p = \varprojlim_{\nu} \mathbb{Z}/p^\nu \mathbb{Z} \subseteq \prod_{\nu=1}^{\infty} \mathbb{Z}/p^\nu \mathbb{Z}.$$

Viewed over the ring on the right, the equation $F = 0$ splits up into components over the individual rings $\mathbb{Z}/p^\nu \mathbb{Z}$, namely, the congruences

$$F(x_1, \dots, x_n) \equiv 0 \pmod{p^\nu}.$$

If now

$$(x_1, \dots, x_n) = (x_1^{(\nu)}, \dots, x_n^{(\nu)})_{\nu \in \mathbb{N}} \in \mathbb{Z}_p^n,$$

with $(x_i^{(\nu)})_{\nu \in \mathbb{N}} \in \mathbb{Z}_p = \varprojlim_{\nu} \mathbb{Z}/p^\nu \mathbb{Z}$, is a p -adic solution of the equation $F(x_1, \dots, x_n) = 0$, then the congruences are solved by

$$F(x_1^{(\nu)}, \dots, x_n^{(\nu)}) \equiv 0 \pmod{p^\nu}, \quad \nu = 1, 2, \dots$$

Conversely, let a solution $(x_1^{(\nu)}, \dots, x_n^{(\nu)})$ of the congruence

$$F(x_1, \dots, x_n) \equiv 0 \pmod{p^\nu}$$

be given for every $\nu \geq 1$. If the elements $(x_i^{(\nu)})_{\nu \in \mathbb{N}} \in \prod_{\nu=1}^{\infty} \mathbb{Z}/p^\nu \mathbb{Z}$ are already in $\varprojlim_{\nu} \mathbb{Z}/p^\nu \mathbb{Z}$, for all $i = 1, \dots, n$, then we have a p -adic solution of the equation $F = 0$. But this is not automatically the case. We will therefore extract a subsequence from the sequence $(x_1^{(\nu)}, \dots, x_n^{(\nu)})$ which fits our needs. For simplicity of notation we only carry this out in the case $n = 1$, writing $x_\nu = x_1^{(\nu)}$. The general case follows exactly the same pattern.

In what follows, we view (x_ν) as a sequence in $\mathbb{Z}$. Since $\mathbb{Z}/p\mathbb{Z}$ is finite, there are infinitely many terms x_ν which mod p are congruent to the same element $y_1 \in \mathbb{Z}/p\mathbb{Z}$. Hence we may choose a subsequence $\{x_\nu^{(1)}\}$ of $\{x_\nu\}$ such that

$$x_\nu^{(1)} \equiv y_1 \pmod{p} \quad \text{and} \quad F(x_\nu^{(1)}) \equiv 0 \pmod{p}.$$

Likewise, we may extract from $\{x_v^{(1)}\}$ a subsequence $\{x_v^{(2)}\}$ such that

$$x_v^{(2)} \equiv y_2 \pmod{p^2} \quad \text{and} \quad F(x_v^{(2)}) \equiv 0 \pmod{p^2},$$

where $y_2 \in \mathbb{Z}/p^2\mathbb{Z}$ evidently satisfies $y_2 \equiv y_1 \pmod{p}$. Continuing in this way, we obtain for each $k \geq 1$ a subsequence $\{x_v^{(k)}\}$ of $\{x_v^{(k-1)}\}$ the terms of which satisfy the congruences

$$x_v^{(k)} \equiv y_k \pmod{p^k} \quad \text{and} \quad F(x_v^{(k)}) \equiv 0 \pmod{p^k}$$

for some $y_k \in \mathbb{Z}/p^k\mathbb{Z}$ such that

$$y_k \equiv y_{k-1} \pmod{p^{k-1}}.$$

The y_k define a p -adic integer $y = (y_k)_{k \in \mathbb{N}} \in \varprojlim_k \mathbb{Z}/p^k\mathbb{Z} = \mathbb{Z}_p$ satisfying

$$F(y_k) \equiv 0 \pmod{p^k}$$

for all $k \geq 1$. In other words, $F(y) = 0$. □

Exercise 1. A p -adic number $a = \sum_{v=-m}^{\infty} a_v p^v \in \mathbb{Q}_p$ is a rational number if and only if the sequence of digits is periodic (possibly with a finite string before the first period).

Hint: Write $p^m a = b + c \frac{p^\ell}{1 - p^n}$, $0 \leq b < p^\ell$, $0 \leq c < p^n$.

Exercise 2. A p -adic integer $a = a_0 + a_1 p + a_2 p^2 + \cdots$ is a unit in the ring $\mathbb{Z}_p$ if and only if $a_0 \neq 0$.

Exercise 3. Show that the equation $x^2 = 2$ has a solution in $\mathbb{Z}_7$.

Exercise 4. Write the numbers $\frac{2}{3}$ and $-\frac{2}{3}$ as 5-adic numbers.

Exercise 5. The field $\mathbb{Q}_p$ of p -adic numbers has no automorphisms except the identity.

Exercise 6. How is the addition, subtraction, multiplication and division of rational numbers reflected in the representation by p -adic digits?

§ 2. The p -adic Absolute Value

The representation of a p -adic integer

$$(1) \quad a_0 + a_1 p + a_2 p^2 + \cdots, \quad 0 \leq a_i < p,$$

resembles very much the decimal fraction representation

$$a_0 + a_1\left(\frac{1}{10}\right) + a_2\left(\frac{1}{10}\right)^2 + \cdots, \quad 0 \leq a_i < 10,$$

of a real number between 0 and 10. But it does not converge as the decimal fraction does. Nonetheless, the field $\mathbb{Q}_p$ of p -adic numbers can be constructed from the field $\mathbb{Q}$ in the same fashion as the field of real numbers $\mathbb{R}$. The key to this is to replace the ordinary absolute value by a new “ p -adic” absolute value $|\cdot|_p$ with respect to which the series (1) converge so that the p -adic numbers appear in the usual manner as limits of Cauchy sequences of rational numbers. This approach was proposed by the Hungarian mathematician *J. KÜRSCHAK*. The p -adic absolute value $|\cdot|_p$ is defined as follows.

Let $a = \frac{b}{c}$, $b, c \in \mathbb{Z}$ be a nonzero rational number. We extract from b and from c as high a power of the prime number p as possible,

$$(2) \quad a = p^m \frac{b'}{c'}, \quad (b'c', p) = 1,$$

and we put

$$|a|_p = \frac{1}{p^m}.$$

Thus the p -adic value no longer measures the size of a number $a \in \mathbb{N}$. Instead it becomes small if the number is divisible by a high power of p . This elaborates on the idea suggested in (1.4) that an integer has to be 0 if it is infinitely divisible by p . In particular, the summands of a p -adic series $a_0 + a_1 p + a_2 p^2 + \cdots$ form a sequence converging to 0 with respect to $|\cdot|_p$.

The exponent m in the representation (2) of the number a is denoted by $v_p(a)$, and one puts formally $v_p(0) = \infty$. This gives the function

$$v_p : \mathbb{Q} \longrightarrow \mathbb{Z} \cup \{\infty\},$$

which is easily checked to satisfy the properties

- 1) $v_p(a) = \infty \iff a = 0$,
- 2) $v_p(ab) = v_p(a) + v_p(b)$,
- 3) $v_p(a + b) \geq \min\{v_p(a), v_p(b)\}$,

where $x + \infty = \infty$, $\infty + \infty = \infty$ and $\infty > x$, for all $x \in \mathbb{Z}$. The function v_p is called the **p -adic exponential valuation** of $\mathbb{Q}$. The **p -adic absolute value** is given by

$$|\cdot|_p : \mathbb{Q} \longrightarrow \mathbb{R}, \quad a \longmapsto |a|_p = p^{-v_p(a)}.$$

In view of 1), 2), 3), it satisfies the conditions of a *norm* on $\mathbb{Q}$:

- 1) $|a|_p = 0 \iff a = 0$,
- 2) $|ab|_p = |a|_p |b|_p$,
- 3) $|a + b|_p \leq \max\{|a|_p, |b|_p\} \leq |a|_p + |b|_p$.

One can show that the absolute values $|\cdot|_p$ and $|\cdot|$ essentially exhaust all norms on $\mathbb{Q}$: any further norm is a power $|\cdot|_p^s$ or $|\cdot|^s$, for some real number $s > 0$ (see (3.7)). The usual absolute value $|\cdot|$ is denoted in this context by $|\cdot|_\infty$. The good reason for this will be explained in due course. In conjunction with the absolute values $|\cdot|_p$, it satisfies the following important **product formula**:

(2.1) Proposition. *For every rational number $a \neq 0$, one has*

$$\prod_p |a|_p = 1,$$

where p varies over all prime numbers as well as the symbol ∞ .

Proof: In the prime factorization

$$a = \pm \prod_{p \neq \infty} p^{\nu_p}$$

of a , the exponent ν_p of p is precisely the exponential valuation $v_p(a)$ and the sign equals $\frac{a}{|a|_\infty}$. The equation therefore reads

$$a = \frac{a}{|a|_\infty} \prod_{p \neq \infty} \frac{1}{|a|_p},$$

so that one has indeed $\prod_p |a|_p = 1$. □

The notation $|\cdot|_\infty$ for the ordinary absolute value is motivated by the analogy of the field of rational numbers $\mathbb{Q}$ with the rational function field $k(t)$ over a finite field k , with which we started our considerations. Instead of $\mathbb{Z}$, we have inside $k(t)$ the polynomial ring $k[t]$, the prime ideals $\mathfrak{p} \neq 0$ of which are given by the monic irreducible polynomials $p(t) \in k[t]$. For every such $\mathfrak{p}$, one defines an absolute value

$$|\cdot|_{\mathfrak{p}} : k(t) \longrightarrow \mathbb{R}$$

as follows. Let $f(t) = \frac{g(t)}{h(t)}$, $g(t), h(t) \in k[t]$ be a nonzero rational function. We extract from $g(t)$ and $h(t)$ the highest possible power of the irreducible polynomial $p(t)$,

$$f(t) = p(t)^m \frac{\tilde{g}(t)}{\tilde{h}(t)}, \quad (\tilde{g}, \tilde{h}, p) = 1,$$

and put

$$v_p(f) = m, \quad |f|_p = q_p^{-v_p(f)},$$

where $q_p = q^{d_p}$, d_p being the degree of the residue class field of p over k and q a fixed real number > 1 . Furthermore we put $v_p(0) = \infty$ and $|0|_p = 0$, and obtain for v_p and $|\cdot|_p$ the same conditions 1), 2), 3) as for v_p and $|\cdot|_p$ above. In the case $p = (t - a)$ for $a \in k$, the valuation $v_p(f)$ is clearly the order of the zero, resp. pole, of the function $f = f(t)$ at $t = a$.

But for the function field $k(t)$, there is one more exponential valuation

$$v_\infty : k(t) \longrightarrow \mathbb{Z} \cup \{\infty\},$$

namely

$$v_\infty(f) = \deg(h) - \deg(g),$$

where $f = \frac{g}{h} \neq 0$, $g, h \in k[t]$. It describes the order of zero, resp. pole, of $f(t)$ at the point at infinity ∞ , i.e., the order of zero, resp. pole, of the function $f(1/t)$ at the point $t = 0$. It is associated to the prime ideal $p = (1/t)$ of the ring $k[1/t] \subseteq k(t)$ in the same way as the exponential valuations v_p are associated to the prime ideals p of $k[t]$. Putting

$$|f|_\infty = q^{-v_\infty(f)},$$

the unique factorization in $k(t)$ yields, as in (2.1) above, the formula

$$\prod_p |f|_p = 1,$$

where p varies over the prime ideals of $k[t]$ as well as the symbol ∞ , which now denotes the point at infinity (see chap. I, § 14, p. 95).

In view of the product formula (2.1), the above consideration shows that the ordinary absolute value $|\cdot|$ of $\mathbb{Q}$ should be thought of as being associated to a virtual point at infinity. This point of view justifies the notation $|\cdot|_\infty$, obeys our constant *leitmotiv* to study numbers as functions from a geometric perspective, and it will fulfill the expectations thus raised in an ever growing and amazing manner. The decisive difference between the absolute value $|\cdot|_\infty$ of $\mathbb{Q}$ and the absolute value $|\cdot|_\infty$ of $k(t)$ is, however, that the former is not derived from any exponential valuation v_p attached to a prime ideal.

Having introduced the p -adic absolute value $|\cdot|_p$ on the field $\mathbb{Q}$, let us now give a new definition of the field $\mathbb{Q}_p$ of p -adic numbers, imitating the construction of the field of real numbers. We will verify afterwards that this new, analytic construction does agree with Hensel's definition, which was motivated by function theory.

A **Cauchy sequence** with respect to $|\cdot|_p$ is by definition a sequence $\{x_n\}$ of rational numbers such that for every $\varepsilon > 0$, there exists a positive integer n_0 satisfying

$$|x_n - x_m|_p < \varepsilon \quad \text{for all } n, m \geq n_0.$$

Example: Every formal series

$$\sum_{v=0}^{\infty} a_v p^v, \quad 0 \leq a_v < p,$$

provides a Cauchy sequence via its partial sums

$$x_n = \sum_{v=0}^{n-1} a_v p^v,$$

because for $n > m$ one has

$$|x_n - x_m|_p = \left| \sum_{v=m}^{n-1} a_v p^v \right|_p \leq \max_{m \leq v < n} \{ |a_v p^v|_p \} \leq \frac{1}{p^m}.$$

A sequence $\{x_n\}$ in $\mathbb{Q}$ is called a **nullsequence** with respect to $|\cdot|_p$ if $|x_n|_p$ is a sequence converging to 0 in the usual sense.

Example: $1, p, p^2, p^3, \dots$

The Cauchy sequences form a ring R , the nullsequences form a maximal ideal $\mathfrak{m}$, and we define afresh the field of p -adic numbers to be the residue class field

$$\mathbb{Q}_p := R/\mathfrak{m}.$$

We embed $\mathbb{Q}$ in $\mathbb{Q}_p$ by associating to every element $a \in \mathbb{Q}$ the residue class of the constant sequence $(a, a, a, \dots)$. The p -adic absolute value $|\cdot|_p$ on $\mathbb{Q}$ is extended to $\mathbb{Q}_p$ by giving the element $x = \{x_n\} \bmod \mathfrak{m} \in R/\mathfrak{m}$ the absolute value

$$|x|_p := \lim_{n \rightarrow \infty} |x_n|_p \in \mathbb{R}.$$

This limit exists because $\{|x_n|_p\}$ is a Cauchy sequence in $\mathbb{R}$, and it is independent of the choice of the sequence $\{x_n\}$ within its class $\bmod \mathfrak{m}$ because any p -adic nullsequence $\{y_n\} \in \mathfrak{m}$ satisfies of course $\lim_{n \rightarrow \infty} |y_n|_p = 0$.

The p -adic exponential valuation v_p on $\mathbb{Q}$ extends to an exponential valuation

$$v_p : \mathbb{Q}_p \longrightarrow \mathbb{Z} \cup \{\infty\}.$$

In fact, if $x \in \mathbb{Q}_p$ is the class of the Cauchy sequence $\{x_n\}$ where $x_n \neq 0$, then

$$v_p(x_n) = -\log_p |x_n|_p$$

either diverges to ∞ or is a Cauchy sequence in $\mathbb{Z}$ which eventually must become constant for large n because $\mathbb{Z}$ is discrete. We put

$$v_p(x) = \lim_{n \rightarrow \infty} v_p(x_n) = v_p(x_n) \quad \text{for } n \geq n_0.$$

Again we find for all $x \in \mathbb{Q}_p$ that

$$|x|_p = p^{-v_p(x)}.$$

As for the field of real numbers one proves the

(2.2) Proposition. *The field $\mathbb{Q}_p$ of p -adic numbers is **complete** with respect to the absolute value $|\cdot|_p$, i.e., every Cauchy sequence in $\mathbb{Q}_p$ converges with respect to $|\cdot|_p$.*

As well as the field $\mathbb{R}$, we thus obtain for each prime number p a new field $\mathbb{Q}_p$ with equal rights and standing, so that $\mathbb{Q}$ has given rise to the infinite family of fields

$$\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7, \mathbb{Q}_{11}, \dots, \mathbb{Q}_\infty = \mathbb{R}.$$

An important special property of the p -adic absolute values $|\cdot|_p$ lies in the fact that they do not only satisfy the usual triangle inequality, but also the stronger version

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

This yields the following remarkable proposition, which gives us a new definition of the p -adic integers.

(2.3) Proposition. *The set*

$$\mathbb{Z}_p := \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}$$

is a subring of $\mathbb{Q}_p$. It is the closure with respect to $|\cdot|_p$ of the ring $\mathbb{Z}$ in the field $\mathbb{Q}_p$.

Proof: That $\mathbb{Z}_p$ is closed under addition and multiplication follows from

$$|x + y|_p \leq \max\{|x|_p, |y|_p\} \quad \text{and} \quad |xy|_p = |x|_p |y|_p.$$

If $\{x_n\}$ is a Cauchy sequence in $\mathbb{Z}$ and $x = \lim_{n \rightarrow \infty} x_n$, then $|x_n|_p \leq 1$ implies also $|x|_p \leq 1$, hence $x \in \mathbb{Z}_p$. Conversely, let $x = \lim_{n \rightarrow \infty} x_n \in \mathbb{Z}_p$, for a Cauchy sequence $\{x_n\}$ in $\mathbb{Q}$. We saw above that one has $|x|_p = |x_n|_p \leq 1$ for $n \geq n_0$, i.e., $x_n = \frac{a_n}{b_n}$, with $a_n, b_n \in \mathbb{Z}$, $(b_n, p) = 1$. Choosing for each $n \geq n_0$ a solution $y_n \in \mathbb{Z}$ of the congruence $b_n y_n \equiv a_n \pmod{p^n}$ yields $|x_n - y_n|_p \leq \frac{1}{p^n}$ and hence $x = \lim_{n \rightarrow \infty} y_n$, so that x belongs to the closure of $\mathbb{Z}$. $\square$

The group of units of $\mathbb{Z}_p$ is obviously

$$\mathbb{Z}_p^* = \{x \in \mathbb{Z}_p \mid |x|_p = 1\}.$$

Every element $x \in \mathbb{Q}_p^*$ admits a unique representation

$$x = p^m u \quad \text{with } m \in \mathbb{Z} \text{ and } u \in \mathbb{Z}_p^*.$$

For if $v_p(x) = m \in \mathbb{Z}$, then $v_p(xp^{-m}) = 0$, hence $|xp^{-m}|_p = 1$, i.e., $u = xp^{-m} \in \mathbb{Z}_p^*$. Furthermore we have the

(2.4) Proposition. *The nonzero ideals of the ring $\mathbb{Z}_p$ are the principal ideals*

$$p^n \mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid v_p(x) \geq n\},$$

with $n \geq 0$, and one has

$$\mathbb{Z}_p / p^n \mathbb{Z}_p \cong \mathbb{Z} / p^n \mathbb{Z}.$$

Proof: Let $\mathfrak{a} \neq (0)$ be an ideal of $\mathbb{Z}_p$ and $x = p^m u$, $u \in \mathbb{Z}_p^*$, an element of $\mathfrak{a}$ with smallest possible m (since $|x|_p \leq 1$, one has $m \geq 0$). Then $\mathfrak{a} = p^m \mathbb{Z}_p$ because $y = p^n u' \in \mathfrak{a}$, $u' \in \mathbb{Z}_p^*$, implies $n \geq m$, hence $y = (p^{n-m} u') p^m \in p^m \mathbb{Z}_p$. The homomorphism

$$\mathbb{Z} \longrightarrow \mathbb{Z}_p / p^n \mathbb{Z}_p, \quad a \longmapsto a \bmod p^n \mathbb{Z}_p,$$

has kernel $p^n \mathbb{Z}$ and is surjective. Indeed, for every $x \in \mathbb{Z}_p$, there exists by (2.3) an $a \in \mathbb{Z}$ such that

$$|x - a|_p \leq \frac{1}{p^n},$$

i.e., $v_p(x - a) \geq n$, therefore $x - a \in p^n \mathbb{Z}_p$ and hence $x \equiv a \pmod{p^n \mathbb{Z}_p}$. So we obtain an isomorphism

$$\mathbb{Z}_p / p^n \mathbb{Z}_p \cong \mathbb{Z} / p^n \mathbb{Z}. \quad \square$$

We now want to establish the link with Hensel's definition of the ring $\mathbb{Z}_p$ and the field $\mathbb{Q}_p$ which was given in §1. There we defined the p -adic integers as formal series

$$\sum_{v=0}^{\infty} a_v p^v, \quad 0 \leq a_v < p,$$

which we identified with sequences

$$\bar{s}_n = s_n \pmod{p^n} \in \mathbb{Z} / p^n \mathbb{Z}, \quad n = 1, 2, \dots,$$

where s_n was the partial sum

$$s_n = \sum_{v=0}^{n-1} a_v p^v.$$

These sequences constituted the projective limit

$$\varprojlim_n \mathbb{Z} / p^n \mathbb{Z} = \left\{ (x_n)_{n \in \mathbb{N}} \in \prod_{n=1}^{\infty} \mathbb{Z} / p^n \mathbb{Z} \mid x_{n+1} \mapsto x_n \right\}.$$

We viewed the p -adic integers as elements of this ring. Since

$$\mathbb{Z}_p / p^n \mathbb{Z}_p \cong \mathbb{Z} / p^n \mathbb{Z},$$

we obtain, for every $n \geq 1$, a surjective homomorphism

$$\mathbb{Z}_p \longrightarrow \mathbb{Z} / p^n \mathbb{Z}.$$

It is clear that the family of these homomorphisms yields a homomorphism

$$\mathbb{Z}_p \longrightarrow \varprojlim_n \mathbb{Z} / p^n \mathbb{Z}.$$

It is now possible to identify both definitions given for $\mathbb{Z}_p$ (and therefore also for $\mathbb{Q}_p$) via the

(2.5) Proposition. *The homomorphism*

$$\mathbb{Z}_p \longrightarrow \varprojlim_n \mathbb{Z} / p^n \mathbb{Z}$$

is an isomorphism.

Proof: If $x \in \mathbb{Z}_p$ is mapped to zero, this means that $x \in p^n \mathbb{Z}_p$ for all $n \geq 1$, i.e., $|x|_p \leq \frac{1}{p^n}$ for all $n \geq 1$, so that $|x|_p = 0$ and thus $x = 0$. This shows injectivity.

An element of $\varprojlim_n \mathbb{Z}/p^n \mathbb{Z}$ is given by a sequence of partial sums

$$s_n = \sum_{v=0}^{n-1} a_v p^v, \quad 0 \leq a_v < p.$$

We saw above that this sequence is a Cauchy sequence in $\mathbb{Z}_p$, and thus converges to an element

$$x = \sum_{v=0}^{\infty} a_v p^v \in \mathbb{Z}_p.$$

Since

$$x - s_n = \sum_{v=n}^{\infty} a_v p^v \in p^n \mathbb{Z}_p,$$

one has $x \equiv s_n \pmod{p^n}$ for all n , i.e., x is mapped to the element of $\varprojlim_n \mathbb{Z}/p^n \mathbb{Z}$ which is defined by the given sequence $(s_n)_{n \in \mathbb{N}}$. This shows surjectivity. $\square$

We emphasize that the elements on the right hand side of the isomorphism

$$\mathbb{Z}_p \xrightarrow{\sim} \varprojlim_n \mathbb{Z}/p^n \mathbb{Z}$$

are given formally by sequences of partial sums

$$s_n = \sum_{v=0}^{n-1} a_v p^v, \quad n = 1, 2, \dots$$

On the left, however, these sequences converge with respect to the absolute value and yield the elements of $\mathbb{Z}_p$ in the familiar way, as convergent infinite series

$$x = \sum_{v=0}^{\infty} a_v p^v.$$

Yet another, very elegant method to introduce the p -adic numbers comes about as follows. Let $\mathbb{Z}[[X]]$ denote the ring of all formal power series $\sum_{i=0}^{\infty} a_i X^i$ with integer coefficients. Then one has the

(2.6) Proposition. *There is a canonical isomorphism*

$$\mathbb{Z}_p \cong \mathbb{Z}[[X]]/(X - p).$$

Proof: Consider the visibly surjective homomorphism $\mathbb{Z}[[X]] \rightarrow \mathbb{Z}_p$ which to every formal power series $\sum_{v=0}^{\infty} a_v X^v$ associates the convergent series $\sum_{v=0}^{\infty} a_v p^v$. The principal ideal $(X - p)$ clearly belongs to the kernel of this mapping. In order to show that it is the whole kernel, let $f(X) = \sum_{v=0}^{\infty} a_v X^v$ be a power series such that $f(p) = \sum_{v=0}^{\infty} a_v p^v = 0$. Since $\mathbb{Z}_p/p^n \mathbb{Z}_p \cong \mathbb{Z}/p^n \mathbb{Z}$, this means that

$$a_0 + a_1 p + \cdots + a_{n-1} p^{n-1} \equiv 0 \pmod{p^n}$$

for all n . We put, for $n \geq 1$,

$$b_{n-1} = -\frac{1}{p^n} (a_0 + a_1 p + \cdots + a_{n-1} p^{n-1}).$$

Then we obtain successively

$$\begin{aligned} a_0 &= -pb_0, \\ a_1 &= b_0 - pb_1, \\ a_2 &= b_1 - pb_2, \quad \text{etc.} \end{aligned}$$

But this amounts to the equality

$$(a_0 + a_1 X + a_2 X^2 + \cdots) = (X - p)(b_0 + b_1 X + b_2 X^2 + \cdots),$$

i.e., $f(X)$ belongs to the principal ideal $(X - p)$. $\square$

Exercise 1. $|x - y|_p \geq ||x|_p - |y|_p|$.

Exercise 2. Let n be a natural number, $n = a_0 + a_1 p + \cdots + a_{r-1} p^{r-1}$ its p -adic expansion, with $0 \leq a_i < p$, and $s = a_0 + a_1 + \cdots + a_{r-1}$. Show that $v_p(n!) = \frac{n-s}{p-1}$.

Exercise 3. The sequence $1, \frac{1}{10}, \frac{1}{10^2}, \frac{1}{10^3}, \dots$ does not converge in $\mathbb{Q}_p$, for any p .

Exercise 4. Let $\varepsilon \in 1 + p\mathbb{Z}_p$, and let $\alpha = a_0 + a_1 p + a_2 p^2 + \cdots$ be a p -adic integer, and write $s_n = a_0 + a_1 p + \cdots + a_{n-1} p^{n-1}$. Show that the sequence ε^{s_n} converges to a number ε^α in $1 + p\mathbb{Z}_p$. Show furthermore that $1 + p\mathbb{Z}_p$ is thus turned into a multiplicative $\mathbb{Z}_p$ -module.

Exercise 5. For every $a \in \mathbb{Z}$, $(a, p) = 1$, the sequence $\{a^{p^n}\}_{n \in \mathbb{N}}$ converges in $\mathbb{Q}_p$.

Exercise 6. The fields $\mathbb{Q}_p$ and $\mathbb{Q}_q$ are not isomorphic, unless $p = q$.

Exercise 7. The algebraic closure of $\mathbb{Q}_p$ has infinite degree.

Exercise 8. In the ring $\mathbb{Z}_p[[X]]$ of formal power series $\sum_{v=0}^{\infty} a_v X^v$ over $\mathbb{Z}_p$, one has the following **division with remainder**. Let $f, g \in \mathbb{Z}_p[[X]]$ and let $f(X) = a_0 + a_1 X + \cdots$ such that $p|a_v$ for $v = 0, \dots, n-1$, but $p \nmid a_n$. Then one may write in a unique way

$$g = qf + r,$$

where $q \in \mathbb{Z}_p[[X]]$, and $r \in \mathbb{Z}_p[X]$ is a polynomial of degree $\leq n-1$.

Hint: Let τ be the operator $\tau(\sum_{v=0}^{\infty} b_v X^v) = \sum_{v=n}^{\infty} b_v X^{v-n}$. Show that $U(X) = a_n + a_{n+1}X + \cdots = \tau(f(X))$ is a unit in $\mathbb{Z}_p[[X]]$ and write $f(X) = pP(X) + X^n U(X)$ with a polynomial $P(X)$ of degree $\leq n-1$. Show that

$$q(X) = \frac{1}{U(X)} \sum_{i=0}^{\infty} (-1)^i p^i \left(\tau \circ \frac{P}{U} \right)^i \circ \tau(g)$$

is a well-defined power series in $\mathbb{Z}_p[[X]]$ such that $\tau(qf) = \tau(g)$.

Exercise 9 (p -adic Weierstrass Preparation Theorem). Every nonzero power series

$$f(X) = \sum_{v=0}^{\infty} a_v X^v \in \mathbb{Z}_p[[X]]$$

admits a unique representation

$$f(X) = p^\mu P(X) U(X),$$

where $U(X)$ is a unit in $\mathbb{Z}_p[[X]]$ and $P(X) \in \mathbb{Z}_p[X]$ is a monic polynomial satisfying $P(X) \equiv X^n \pmod{p}$.

§ 3. Valuations

The procedure we performed in the previous section with the field $\mathbb{Q}$ in order to obtain the p -adic numbers can be generalized to arbitrary fields using the concept of (multiplicative) valuation.

(3.1) Definition. A **valuation** of a field K is a function

$$| \cdot | : K \rightarrow \mathbb{R}$$

enjoying the properties

- (i) $|x| \geq 0$, and $|x| = 0 \iff x = 0$,
- (ii) $|xy| = |x||y|$,
- (iii) $|x + y| \leq |x| + |y|$ “triangle inequality”.

We tacitly exclude in the sequel the case where $| \cdot |$ is the trivial valuation of K which satisfies $|x| = 1$ for all $x \neq 0$. Defining the distance between two points $x, y \in K$ by

$$d(x, y) = |x - y|$$

makes K into a metric space, and hence in particular a topological space.

(3.2) Definition. Two valuations of K are called **equivalent** if they define the same topology on K .

(3.3) Proposition. *Two valuations $| \cdot |_1$ and $| \cdot |_2$ on K are equivalent if and only if there exists a real number $s > 0$ such that one has*

$$|x|_1 = |x|_2^s$$

for all $x \in K$.

Proof: If $| \cdot |_1 = | \cdot |_2^s$, with $s > 0$, then $| \cdot |_1$ and $| \cdot |_2$ are obviously equivalent. For an arbitrary valuation $| \cdot |$ on K , the inequality $|x| < 1$ is tantamount to the condition that $\{x^n\}_{n \in \mathbb{N}}$ converges to zero in the topology defined by $| \cdot |$. Therefore if $| \cdot |_1$ and $| \cdot |_2$ are equivalent, one has the implication

$$(*) \quad |x|_1 < 1 \implies |x|_2 < 1.$$

Now let $y \in K$ be a fixed element satisfying $|y|_1 > 1$. Let $x \in K$, $x \neq 0$. Then $|x|_1 = |y|_1^\alpha$ for some $\alpha \in \mathbb{R}$. Let m_i/n_i be a sequence of rational numbers (with $n_i > 0$) which converges to α from above. Then we have $|x|_1 = |y|_1^\alpha < |y|_1^{m_i/n_i}$, hence

$$\left| \frac{x^{n_i}}{y^{m_i}} \right|_1 < 1 \implies \left| \frac{x^{n_i}}{y^{m_i}} \right|_2 < 1,$$

so that $|x|_2 \leq |y|_2^{m_i/n_i}$, and thus $|x|_2 \leq |y|_2^\alpha$. Using a sequence m_i/n_i which converges to α from below $(*)$ tells us that $|x|_2 \geq |y|_2^\alpha$. So we have $|x|_2 = |y|_2^\alpha$. For all $x \in K$, $x \neq 0$, we therefore get

$$\frac{\log |x|_1}{\log |x|_2} = \frac{\log |y|_1}{\log |y|_2} =: s,$$

hence $|x|_1 = |x|_2^s$. But $|y|_1 > 1$ implies $|y|_2 > 1$, hence $s > 0$. $\square$

The proof shows that the equivalence of $| \cdot |_1$ and $| \cdot |_2$ is also equivalent to the condition

$$|x|_1 < 1 \implies |x|_2 < 1.$$

We use this for the proof of the following approximation theorem, which may be considered a variant of the Chinese remainder theorem.

(3.4) Approximation Theorem. *Let $| \cdot |_1, \dots, | \cdot |_n$ be pairwise inequivalent valuations of the field K and let $a_1, \dots, a_n \in K$ be given elements. Then for every $\varepsilon > 0$ there exists an $x \in K$ such that*

$$|x - a_i|_i < \varepsilon \quad \text{for all } i = 1, \dots, n.$$

Proof: By the above remark, since $|\cdot|_1$ and $|\cdot|_n$ are inequivalent, there exists $\alpha \in K$ such that $|\alpha|_1 < 1$ and $|\alpha|_n \geq 1$. By the same token, there exists $\beta \in K$ such that $|\beta|_n < 1$ and $|\beta|_1 \geq 1$. Putting $y = \beta/\alpha$, one finds $|y|_1 > 1$ and $|y|_n < 1$.

We now prove by induction on n that there exists $z \in K$ such that

$$|z|_1 > 1 \quad \text{and} \quad |z|_j < 1 \quad \text{for } j = 2, \dots, n.$$

We have just done this for $n = 2$. Assume we have found $z \in K$ satisfying

$$|z|_1 > 1 \quad \text{and} \quad |z|_j < 1 \quad \text{for } j = 2, \dots, n-1.$$

If $|z|_n \leq 1$, then $z^m y$ will do, for m large. If however $|z|_n > 1$, the sequence $t_m = z^m/(1+z^m)$ will converge to 1 with respect to $|\cdot|_1$ and $|\cdot|_n$, and to 0 with respect to $|\cdot|_2, \dots, |\cdot|_{n-1}$. Hence, for m large, $t_m y$ will suffice.

The sequence $z^m/(1+z^m)$ converges to 1 with respect to $|\cdot|_1$ and to 0 with respect to $|\cdot|_2, \dots, |\cdot|_n$. For every i we may construct in this way a z_i which is very close to 1 with respect to $|\cdot|_i$, and very close to 0 with respect to $|\cdot|_j$ for $j \neq i$. The element

$$x = a_1 z_1 + \dots + a_n z_n$$

then satisfies the statement of the approximation theorem. $\square$

(3.5) Definition. The valuation $|\cdot|$ is called **nonarchimedean** if $|n|$ stays bounded, for all $n \in \mathbb{N}$. Otherwise it is called **archimedean**.

(3.6) Proposition. The valuation $|\cdot|$ is nonarchimedean if and only if it satisfies the **strong triangle inequality**

$$|x + y| \leq \max\{|x|, |y|\}.$$

Proof: If the strong triangle inequality holds, then one has

$$|n| = |1 + \dots + 1| \leq 1.$$

Conversely, let $|n| \leq N$ for all $n \in \mathbb{N}$. Let $x, y \in K$ and suppose $|x| \geq |y|$. Then $|x|^\nu |y|^{n-\nu} \leq |x|^n$ for $\nu \geq 0$ and one gets

$$|x + y|^n \leq \sum_{\nu=0}^n \binom{n}{\nu} |x|^\nu |y|^{n-\nu} \leq N(n+1)|x|^n,$$

hence

$$|x + y| \leq N^{1/n} (1+n)^{1/n} |x| = N^{1/n} (1+n)^{1/n} \max\{|x|, |y|\},$$

and thus $|x + y| \leq \max\{|x|, |y|\}$ by letting $n \rightarrow \infty$. $\square$

Remark: The strong triangle inequality immediately implies that

$$|x| \neq |y| \implies |x + y| = \max\{|x|, |y|\}.$$

One may extend the nonarchimedean valuation $|\cdot|$ of K to a valuation of the function field $K(t)$ in a canonical way by setting, for a polynomial $f(t) = a_0 + a_1t + \cdots + a_nt^n$,

$$|f| = \max\{|a_0|, \dots, |a_n|\}.$$

The triangle inequality $|f + g| \leq \max\{|f|, |g|\}$ is immediate. The proof that $|fg| = |f||g|$ is the same as the proof of Gauss's lemma for polynomials over factorial rings once we replace the **content** of f in this lemma by the absolute value $|f|$.

For the field $\mathbb{Q}$, we have the usual absolute value $|\cdot|_\infty = |\cdot|$, this being the archimedean valuation, and for each prime number p the nonarchimedean valuation $|\cdot|_p$. As a matter of fact:

(3.7) Proposition. *Every valuation of $\mathbb{Q}$ is equivalent to one of the valuations $|\cdot|_p$ or $|\cdot|_\infty$.*

Proof: Let $\|\cdot\|$ be a nonarchimedean valuation of $\mathbb{Q}$. Then $\|n\| = \|1 + \cdots + 1\| \leq 1$, and there is a prime number p such that $\|p\| < 1$ because, if not, unique prime factorization would imply $\|x\| = 1$ for all $x \in \mathbb{Q}^*$. The set

$$\mathfrak{a} = \{a \in \mathbb{Z} \mid \|a\| < 1\}$$

is an ideal of $\mathbb{Z}$ satisfying $p\mathbb{Z} \subseteq \mathfrak{a} \neq \mathbb{Z}$, and since $p\mathbb{Z}$ is a maximal ideal, we have $\mathfrak{a} = p\mathbb{Z}$. If now $a \in \mathbb{Z}$ and $a = bp^m$ with $p \nmid b$, so that $b \notin \mathfrak{a}$, then $\|b\| = 1$ and hence

$$\|a\| = \|p\|^m = |a|_p^s$$

where $s = -\log \|p\| / \log p$. Consequently $\|\cdot\|$ is equivalent to $|\cdot|_p$.

Now let $\|\cdot\|$ be archimedean. Then one has, for every two natural numbers $n, m > 1$,

$$(*) \quad \|m\|^{1/\log m} = \|n\|^{1/\log n}.$$

In fact, we may write

$$m = a_0 + a_1n + \cdots + a_r n^r$$

where $a_i \in \{0, 1, \dots, n-1\}$ and $n^r \leq m$. Hence, observing that $r \leq \log m / \log n$ and $\|a_i\| = \|1 + \cdots + 1\| \leq a_i \|1\| \leq n$, one gets the inequality

$$\|m\| \leq \sum \|a_i\| \cdot \|n\|^i \leq \sum \|a_i\| \cdot \|n\|^r \leq \left(1 + \frac{\log m}{\log n}\right) n \cdot \|n\|^{\log m / \log n}.$$

Substituting here m^k for m , taking k -th roots on both sides, and letting k tend to ∞ , one finally obtains

$$\|m\| \leq \|n\|^{\log m / \log n}, \quad \text{or} \quad \|m\|^{1/\log m} \leq \|n\|^{1/\log n}.$$

Swapping m with n gives the identity (*). Putting $c = \|n\|^{1/\log n}$ we have $\|n\| = c^{\log n}$, and putting $c = e^s$ yields, for every positive rational number $x = a/b$,

$$\|x\| = e^{s \log x} = |x|^s.$$

Therefore $\| \cdot \|$ is equivalent to the usual absolute value $| \cdot |$ on $\mathbb{Q}$. □

Let $| \cdot |$ be a nonarchimedean valuation of the field K . Putting

$$v(x) = -\log |x| \quad \text{for } x \neq 0, \quad \text{and } v(0) = \infty,$$

we obtain a function

$$v : K \longrightarrow \mathbb{R} \cup \{\infty\}$$

verifying the properties

- (i) $v(x) = \infty \iff x = 0$,
- (ii) $v(xy) = v(x) + v(y)$,
- (iii) $v(x + y) \geq \min\{v(x), v(y)\}$,

where we fix the following conventions regarding elements $a \in \mathbb{R}$ and the symbol ∞ : $a < \infty$, $a + \infty = \infty$, $\infty + \infty = \infty$.

A function v on K with these properties is called an **exponential valuation** of K . We exclude the case of the trivial function $v(x) = 0$ for $x \neq 0$, $v(0) = \infty$. Two exponential valuations v_1 and v_2 of K are called **equivalent** if $v_1 = sv_2$, for some real number $s > 0$. For every exponential valuation v we obtain a valuation in the sense of (3.1) by putting

$$|x| = q^{-v(x)},$$

for some fixed real number $q > 1$. To distinguish it from v , we call $| \cdot |$ an associated **multiplicative valuation**, or **absolute value**. Replacing v by an equivalent valuation sv (i.e., replacing q by $q' = q^s$) changes $| \cdot |$ into the equivalent multiplicative valuation $| \cdot |^s$. The conditions (i), (ii), (iii) immediately imply the

(3.8) Proposition. *The subset*

$$\mathcal{O} = \{x \in K \mid v(x) \geq 0\} = \{x \in K \mid |x| \leq 1\}$$

is a ring with group of units

$$\mathcal{O}^* = \{x \in K \mid v(x) = 0\} = \{x \in K \mid |x| = 1\}$$

and the unique maximal ideal

$$\mathfrak{p} = \{x \in K \mid v(x) > 0\} = \{x \in K \mid |x| < 1\}.$$

$\mathcal{O}$ is an integral domain with field of fractions K and has the property that, for every $x \in K$, either $x \in \mathcal{O}$ or $x^{-1} \in \mathcal{O}$. Such a ring is called a **valuation ring**. Its only maximal ideal is $\mathfrak{p} = \{x \in \mathcal{O} \mid x^{-1} \notin \mathcal{O}\}$. The field $\mathcal{O}/\mathfrak{p}$ is called the **residue class field** of $\mathcal{O}$. A valuation ring is always integrally closed. For if $x \in K$ is integral over $\mathcal{O}$, then there is an equation

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0$$

with $a_i \in \mathcal{O}$ and the hypothesis $x \notin \mathcal{O}$, so that $x^{-1} \in \mathcal{O}$, would imply the contradiction $x = -a_1 - a_2 x^{-1} - \cdots - a_n (x^{-1})^{n-1} \in \mathcal{O}$.

An exponential valuation v is called **discrete** if it admits a smallest positive value s . In this case, one finds

$$v(K^*) = s\mathbb{Z}.$$

It is called **normalized** if $s = 1$. Dividing by s we may always pass to a normalized valuation without changing the invariants $\mathcal{O}, \mathcal{O}^*, \mathfrak{p}$. Having done so, an element

$$\pi \in \mathcal{O} \quad \text{such that} \quad v(\pi) = 1$$

is a **prime element**, and every element $x \in K^*$ admits a unique representation

$$x = u \pi^m$$

with $m \in \mathbb{Z}$ and $u \in \mathcal{O}^*$. For if $v(x) = m$, then $v(x \pi^{-m}) = 0$, hence $u = x \pi^{-m} \in \mathcal{O}^*$.

(3.9) Proposition. *If v is a discrete exponential valuation of K , then*

$$\mathcal{O} = \{x \in K \mid v(x) \geq 0\}$$

is a principal ideal domain, hence a discrete valuation ring (see I, (11.3)).

Suppose v is normalized. Then the nonzero ideals of $\mathcal{O}$ are given by

$$\mathfrak{p}^n = \pi^n \mathcal{O} = \{x \in K \mid v(x) \geq n\}, \quad n \geq 0,$$

where π is a prime element, i.e., $v(\pi) = 1$. One has

$$\mathfrak{p}^n / \mathfrak{p}^{n+1} \cong \mathcal{O}/\mathfrak{p}.$$

Proof: Let $\mathfrak{a} \neq 0$ be an ideal of $\mathcal{O}$ and $x \neq 0$ an element in $\mathfrak{a}$ with smallest possible value $v(x) = n$. Then $x = u \pi^n$, $u \in \mathcal{O}^*$, so that $\pi^n \mathcal{O} \subseteq \mathfrak{a}$. If $y = \varepsilon \pi^m \in \mathfrak{a}$ is arbitrary with $\varepsilon \in \mathcal{O}^*$, then $m = v(y) \geq n$, hence $y = (\varepsilon \pi^{m-n}) \pi^n \in \pi^n \mathcal{O}$, so that $\mathfrak{a} = \pi^n \mathcal{O}$. The isomorphism

$$\mathfrak{p}^n / \mathfrak{p}^{n+1} \cong \mathcal{O}/\mathfrak{p}$$

results from the correspondence $a \pi^n \mapsto a \bmod \mathfrak{p}$. □

In a discretely valued field K the chain

$$\mathcal{O} \supseteq \mathfrak{p} \supseteq \mathfrak{p}^2 \supseteq \mathfrak{p}^3 \supseteq \dots$$

consisting of the ideals of the valuation ring $\mathcal{O}$ forms a basis of neighbourhoods of the zero element. Indeed, if v is a normalized exponential valuation and $|\cdot| = q^{-v}$ ($q > 1$) an associated multiplicative valuation, then

$$\mathfrak{p}^n = \left\{ x \in K \mid |x| < \frac{1}{q^{n-1}} \right\}.$$

As a basis of neighbourhoods of the element 1 of K^* , we obtain in the same way the descending chain

$$\mathcal{O}^* = U^{(0)} \supseteq U^{(1)} \supseteq U^{(2)} \supseteq \dots$$

of subgroups

$$U^{(n)} = 1 + \mathfrak{p}^n = \left\{ x \in K^* \mid |1 - x| < \frac{1}{q^{n-1}} \right\}, \quad n > 0,$$

of $\mathcal{O}^*$. (Observe that $1 + \mathfrak{p}^n$ is closed under multiplication and that, if $x \in U^{(n)}$, then so is x^{-1} because $|1 - x^{-1}| = |x|^{-1}|x - 1| = |1 - x| < \frac{1}{q^{n-1}}$.) $U^{(n)}$ is called the n -th **higher unit group** and $U^{(1)}$ the group of **principal units**. Regarding the successive quotients of the chain of higher unit groups, we have the

(3.10) Proposition. $\mathcal{O}^*/U^{(n)} \cong (\mathcal{O}/\mathfrak{p}^n)^*$ and $U^{(n)}/U^{(n+1)} \cong \mathcal{O}/\mathfrak{p}$, for $n \geq 1$.

Proof: The first isomorphism is induced by the canonical and obviously surjective homomorphism

$$\mathcal{O}^* \longrightarrow (\mathcal{O}/\mathfrak{p}^n)^*, \quad u \longmapsto u \bmod \mathfrak{p}^n,$$

the kernel of which is $U^{(n)}$. The second isomorphism is given, once we choose a prime element π , by the surjective homomorphism

$$U^{(n)} = 1 + \pi^n \mathcal{O} \longrightarrow \mathcal{O}/\mathfrak{p}, \quad 1 + \pi^n a \longmapsto a \bmod \mathfrak{p},$$

which has kernel $U^{(n+1)}$. □

Exercise 1. Show that $|z| = (z\bar{z})^{1/2} = \sqrt{|N_{\mathbb{C}|\mathbb{R}}(z)|}$ is the only valuation of $\mathbb{C}$ which extends the absolute value $|\cdot|$ of $\mathbb{R}$.

Exercise 2. What is the relation between the Chinese remainder theorem and the approximation theorem (3.4)?

Exercise 3. Let k be a field and $K = k(t)$ the function field in one variable. Show that the valuations v_p associated to the prime ideals $\mathfrak{p} = (p(t))$ of $k[t]$, together with the degree valuation v_∞ , are the only valuations of K , up to equivalence. What are the residue class fields?

Exercise 4. Let $\mathcal{O}$ be an arbitrary valuation ring with field of fractions K , and let $\Gamma = K^*/\mathcal{O}^*$. Then Γ becomes a totally ordered group if we define $x \bmod \mathcal{O}^* \geq y \bmod \mathcal{O}^*$ to mean $x/y \in \mathcal{O}$.

Write Γ additively and show that the function

$$v : K \longrightarrow \Gamma \cup \{\infty\},$$

$v(0) = \infty$, $v(x) = x \bmod \mathcal{O}^*$ for $x \in K^*$, satisfies the conditions

$$1) \ v(x) = \infty \implies x = 0,$$

$$2) \ v(xy) = v(x) + v(y),$$

$$3) \ v(x + y) \geq \min\{v(x), v(y)\}.$$

v is called a **Krull valuation**.

§ 4. Completions

(4.1) Definition. A valued field $(K, |\cdot|)$ is called **complete** if every Cauchy sequence $\{a_n\}_{n \in \mathbb{N}}$ in K converges to an element $a \in K$, i.e.,

$$\lim_{n \rightarrow \infty} |a_n - a| = 0.$$

Here, as usual, we call $\{a_n\}_{n \in \mathbb{N}}$ a **Cauchy sequence** if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_n - a_m| < \varepsilon \quad \text{for all } n, m \geq N.$$

From any valued field $(K, |\cdot|)$ we get a complete valued field $(\widehat{K}, |\cdot|)$ by the process of **completion**. This completion is obtained in the same way as the field of real numbers is constructed from the field of rational numbers.

Take the ring R of all Cauchy sequences of $(K, |\cdot|)$, consider therein the maximal ideal $\mathfrak{m}$ of all nullsequences with respect to $|\cdot|$, and define

$$\widehat{K} = R/\mathfrak{m}.$$

One embeds the field K into $\widehat{K}$ by sending every $a \in K$ to the class of the constant Cauchy sequence $(a, a, a, \dots)$. The valuation $|\cdot|$ is extended from K to $\widehat{K}$ by giving the element $a \in \widehat{K}$ which is represented by the Cauchy sequence $\{a_n\}_{n \in \mathbb{N}}$ the absolute value

$$|a| = \lim_{n \rightarrow \infty} |a_n|.$$

This limit exists because $||a_n| - |a_m|| \leq |a_n - a_m|$ implies that $|a_n|$ is a Cauchy sequence of real numbers. As in the case of the field of real numbers, one proves that $\widehat{K}$ is complete with respect to the extended $|\cdot|$, and that each $a \in \widehat{K}$ is a limit of a sequence $\{a_n\}$ in K . Finally one proves the uniqueness of the completion $(\widehat{K}, |\cdot|)$: if $(\widehat{K}', |\cdot'|)$ is another complete valued field that contains $(K, |\cdot|)$ as a dense subfield, then mapping

$$|\cdot| - \lim_{n \rightarrow \infty} a_n \longmapsto |\cdot'| - \lim_{n \rightarrow \infty} a_n$$

gives a K -isomorphism $\sigma : \widehat{K} \rightarrow \widehat{K}'$ such that $|a| = |\sigma a|'$.

The fields $\mathbb{R}$ and $\mathbb{C}$ are the most familiar examples of complete fields. They are complete with respect to an archimedean valuation. Amazingly enough, there are no others of this type. More precisely we have the

(4.2) Theorem (OSTROWSKI). *Let K be a field which is complete with respect to an archimedean valuation $|\cdot|$. Then there is an isomorphism σ from K onto $\mathbb{R}$ or $\mathbb{C}$ satisfying*

$$|a| = |\sigma a|^s \quad \text{for all } a \in K,$$

for some fixed $s \in (0, 1]$.

Proof: We may assume without loss of generality that $\mathbb{R} \subseteq K$ and that the valuation $|\cdot|$ of K is an extension of the usual absolute value of $\mathbb{R}$. In fact, replacing $|\cdot|$ by $|\cdot|^{s^{-1}}$ for a suitable $s > 0$, we may assume by (3.7) that the restriction of $|\cdot|$ to $\mathbb{Q}$ is equal to the usual absolute value. Then taking the closure $\widehat{\mathbb{Q}}$ in K we find that $\widehat{\mathbb{Q}}$ is complete with respect to the restriction of $|\cdot|$ to $\widehat{\mathbb{Q}}$, in other words, it is a completion of $(\mathbb{Q}, |\cdot|)$. In view of the uniqueness of completions, there is an isomorphism $\sigma : \mathbb{R} \rightarrow \widehat{\mathbb{Q}}$ such that $|a| = |\sigma a|$ as required.

In order to prove that $K = \mathbb{R}$ or $= \mathbb{C}$ we show that each $\xi \in K$ satisfies a quadratic equation over $\mathbb{R}$. For this, consider the continuous function $f : \mathbb{C} \rightarrow \mathbb{R}$ defined by

$$f(z) = |\xi^2 - (z + \bar{z})\xi + z\bar{z}|.$$

Note here that $z + \bar{z}$, $z\bar{z} \in \mathbb{R} \subseteq K$. Since $\lim_{z \rightarrow \infty} f(z) = \infty$, $f(z)$ has a minimum m . The set

$$S = \{z \in \mathbb{C} \mid f(z) = m\}$$

is therefore nonempty, bounded, and closed, and there is a $z_0 \in S$ such that $|z_0| \geq |z|$ for all $z \in S$. It suffices to show that $m = 0$, because then one has the equation $\xi^2 - (z_0 + \bar{z}_0)\xi + z_0\bar{z}_0 = 0$.

Assume $m > 0$. Consider the real polynomial

$$g(x) = x^2 - (z_0 + \bar{z}_0)x + z_0\bar{z}_0 + \varepsilon,$$

where $0 < \varepsilon < m$, with the roots $z_1, \bar{z}_1 \in \mathbb{C}$. We have $z_1\bar{z}_1 = z_0\bar{z}_0 + \varepsilon$, hence $|z_1| > |z_0|$ and thus

$$f(z_1) > m.$$

For fixed $n \in \mathbb{N}$, consider on the other hand the real polynomial

$$G(x) = [g(x) - \varepsilon]^n - (-\varepsilon)^n = \prod_{i=1}^{2n} (x - \alpha_i) = \prod_{i=1}^{2n} (x - \bar{\alpha}_i)$$

with roots $\alpha_1, \dots, \alpha_{2n} \in \mathbb{C}$. It follows that $G(z_1) = 0$; say, $z_1 = \alpha_1$. We may substitute $\xi \in K$ into the polynomial

$$G(x)^2 = \prod_{i=1}^{2n} (x^2 - (\alpha_i + \bar{\alpha}_i)x + \alpha_i\bar{\alpha}_i)$$

and get

$$|G(\xi)|^2 = \prod_{i=1}^{2n} f(\alpha_i) \geq f(\alpha_1)m^{2n-1}.$$

From this and the inequality

$$|G(\xi)| \leq |\xi^2 - (z_0 + \bar{z}_0)\xi + z_0\bar{z}_0|^n + |-\varepsilon|^n = f(z_0)^n + \varepsilon^n = m^n + \varepsilon^n,$$

it follows that $f(\alpha_1)m^{2n-1} \leq (m^n + \varepsilon^n)^2$ and hence

$$\frac{f(\alpha_1)}{m} \leq \left(1 + \left(\frac{\varepsilon}{m}\right)^n\right)^2.$$

For $n \rightarrow \infty$ we have $f(\alpha_1) \leq m$, which contradicts the inequality $f(\alpha_1) > m$ proved before. $\square$

In view of *OSTROWSKI*'s theorem, we will henceforth restrict attention to the case of nonarchimedean valuations. In this case it is usually expedient — both with regard to the substance and to practical technique — to work with the exponential valuations v rather than the multiplicative valuations. So let v

be an exponential valuation of the field K . It is canonically continued to an exponential valuation $\hat{v}$ of the completion $\widehat{K}$ by setting

$$\hat{v}(a) = \lim_{n \rightarrow \infty} v(a_n),$$

where $a = \lim_{n \rightarrow \infty} a_n \in \widehat{K}$, $a_n \in K$. Observe here that the sequence $v(a_n)$ has to become stationary (provided $a \neq 0$) because, for $n \geq n_0$, one has $\hat{v}(a - a_n) > \hat{v}(a)$, so that it follows from the remark on p. 119

$$v(a_n) = \hat{v}(a_n - a + a) = \min\{\hat{v}(a_n - a), \hat{v}(a)\} = \hat{v}(a).$$

Therefore it follows that

$$v(K^*) = \hat{v}(\widehat{K}^*),$$

and if v is discrete and normalized, then so is the extension $\hat{v}$. In the nonarchimedean case, for a sequence $\{a_n\}_{n \in \mathbb{N}}$ to be a Cauchy sequence, it suffices that $a_{n+1} - a_n$ be a nullsequence. In fact, $v(a_n - a_m) \geq \min_{m \leq i < n} \{v(a_{i+1} - a_i)\}$. By the same token an infinite series $\sum_{v=0}^{\infty} a_v$ converges in $\widehat{K}$ if and only if the sequence of its terms a_v is a nullsequence. The following proposition is proved exactly as its analogue, proposition (2.4), in the special case $(\mathbb{Q}, v_p)$.

(4.3) Proposition. *If $\mathfrak{o} \subseteq K$, resp. $\widehat{\mathfrak{o}} \subseteq \widehat{K}$, is the valuation ring of v , resp. of $\hat{v}$, and $\mathfrak{p}$, resp. $\widehat{\mathfrak{p}}$, is the maximal ideal, then one has*

$$\widehat{\mathfrak{o}}/\widehat{\mathfrak{p}} \cong \mathfrak{o}/\mathfrak{p}$$

and, if v is discrete, one has furthermore

$$\widehat{\mathfrak{o}}/\widehat{\mathfrak{p}}^n \cong \mathfrak{o}/\mathfrak{p}^n \quad \text{for } n \geq 1.$$

Generalizing the p -adic expansion to the case of an arbitrary discrete valuation v of the field K , we have the

(4.4) Proposition. *Let $R \subseteq \mathfrak{o}$ be a system of representatives for $\kappa = \mathfrak{o}/\mathfrak{p}$ such that $0 \in R$, and let $\pi \in \mathfrak{o}$ be a prime element. Then every $x \neq 0$ in $\widehat{K}$ admits a unique representation as a convergent series*

$$x = \pi^m(a_0 + a_1\pi + a_2\pi^2 + \cdots)$$

where $a_i \in R$, $a_0 \neq 0$, $m \in \mathbb{Z}$.

Proof: Let $x = \pi^m u$ with $u \in \widehat{\mathcal{O}}^*$. Since $\widehat{\mathcal{O}}/\widehat{\mathfrak{p}} \cong \mathcal{O}/\mathfrak{p}$, the class $u \bmod \widehat{\mathfrak{p}}$ has a unique representative $a_0 \in R$, $a_0 \neq 0$. We thus have $u = a_0 + \pi b_1$, for some $b_1 \in \widehat{\mathcal{O}}$. Assume now that $a_0, \dots, a_{n-1} \in R$ have been found, satisfying

$$u = a_0 + a_1\pi + \dots + a_{n-1}\pi^{n-1} + \pi^n b_n$$

for some $b_n \in \widehat{\mathcal{O}}$, and that the a_i are uniquely determined by this equation. Then the representative $a_n \in R$ of $b_n \bmod \pi\widehat{\mathcal{O}} \in \widehat{\mathcal{O}}/\widehat{\mathfrak{p}} \cong \mathcal{O}/\mathfrak{p}$ is also uniquely determined by u and we have $b_n = a_n + \pi b_{n+1}$, for some $b_{n+1} \in \widehat{\mathcal{O}}$. Hence

$$u = a_0 + a_1\pi + \dots + a_{n-1}\pi^{n-1} + a_n\pi^n + \pi^{n+1}b_{n+1}.$$

In this way we find an infinite series $\sum_{v=0}^{\infty} a_v \pi^v$ which is uniquely determined by u . It converges to u because the remainder term $\pi^{n+1}b_{n+1}$ tends to zero. $\square$

In the case of the field of rational numbers $\mathbb{Q}$ and the p -adic valuation v_p with its completion $\mathbb{Q}_p$, the numbers $0, 1, \dots, p-1$ form a system of representatives R for the residue class field $\mathbb{Z}/p\mathbb{Z}$ of the valuation, and we get back the representation of p -adic numbers which has already been discussed in § 2:

$$x = p^m(a_0 + a_1p + a_2p^2 + \dots),$$

where $0 \leq a_i < p$ and $m \in \mathbb{Z}$.

In the case of the rational function field $k(t)$ and the valuation $v_{\mathfrak{p}}$ attached to a prime ideal $\mathfrak{p} = (t - a)$ of $k[t]$ (see § 2), we may take as a system of representatives R the field of coefficients k itself. The completion then turns out to be the **field of formal power series** $k((x))$, $x = t - a$, consisting of all formal Laurent series

$$f(t) = (t - a)^m(a_0 + a_1(t - a) + a_2(t - a)^2 + \dots),$$

with $a_i \in k$ and $m \in \mathbb{Z}$. The motivating analogy of the beginning of this chapter, between power series and p -adic numbers, thus appears as two special instances of the same concrete mathematical situation.

In § 1 we identified the ring $\mathbb{Z}_p$ of p -adic integers as being the projective limit $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$. We obtain a similar result in the general setting of valuation theory. To explain this, let K be complete with respect to a discrete valuation. Let $\mathcal{O}$ be the valuation ring with the maximal ideal $\mathfrak{p}$. We then have for every $n \geq 1$ the canonical homomorphisms

$$\mathcal{O} \longrightarrow \mathcal{O}/\mathfrak{p}^n$$

and

$$\mathcal{O}/\mathfrak{p} \xleftarrow{\lambda_1} \mathcal{O}/\mathfrak{p}^2 \xleftarrow{\lambda_2} \mathcal{O}/\mathfrak{p}^3 \xleftarrow{\lambda_3} \dots$$

This gives us a homomorphism

$$\mathcal{O} \longrightarrow \varprojlim_n \mathcal{O}/\mathfrak{p}^n$$

into the projective limit

$$\varprojlim_n \mathcal{O}/\mathfrak{p}^n = \left\{ (x_n) \in \prod_{n=1}^{\infty} \mathcal{O}/\mathfrak{p}^n \mid \lambda_n(x_{n+1}) = x_n \right\}.$$

Considering the rings $\mathcal{O}/\mathfrak{p}^n$ as topological rings, for the discrete topology, gives us the product topology on $\prod_{n=1}^{\infty} \mathcal{O}/\mathfrak{p}^n$, and the projective limit $\varprojlim_n \mathcal{O}/\mathfrak{p}^n$ becomes a topological ring in a canonical way, being a closed subset of the product (see chap. IV, §2).

(4.5) Proposition. *The canonical mapping*

$$\mathcal{O} \longrightarrow \varprojlim_n \mathcal{O}/\mathfrak{p}^n$$

is an isomorphism and a homeomorphism. The same is true for the mapping

$$\mathcal{O}^* \longrightarrow \varprojlim_n \mathcal{O}^*/U^{(n)}.$$

Proof: The map is injective since its kernel is $\bigcap_{n=1}^{\infty} \mathfrak{p}^n = (0)$. To prove surjectivity, let $\mathfrak{p} = \pi\mathcal{O}$ and let $R \subseteq \mathcal{O}$, $R \ni 0$, be a system of representatives of $\mathcal{O}/\mathfrak{p}$. We saw in the proof of (4.4) (and in fact already in (1.2)) that the elements $a \bmod \mathfrak{p}^n \in \mathcal{O}/\mathfrak{p}^n$ can be given uniquely in the form

$$a \equiv a_0 + a_1\pi + \dots + a_{n-1}\pi^{n-1} \bmod \mathfrak{p}^n,$$

where $a_i \in R$. Each element $s \in \varprojlim_n \mathcal{O}/\mathfrak{p}^n$ is therefore given by a sequence of sums

$$s_n = a_0 + a_1\pi + \dots + a_{n-1}\pi^{n-1}, \quad n = 1, 2, \dots,$$

with fixed coefficients $a_i \in R$, and it is thus the image of the element $x = \lim_{n \rightarrow \infty} s_n = \sum_{v=0}^{\infty} a_v \pi^v \in \mathcal{O}$.

The sets $P_n = \prod_{v \geq n} \mathcal{O}/\mathfrak{p}^v$ form a basis of neighbourhoods of the zero element of $\prod_{v=1}^{\infty} \mathcal{O}/\mathfrak{p}^v$. Under the bijection

$$\mathcal{O} \longrightarrow \varprojlim_v \mathcal{O}/\mathfrak{p}^v$$

the basis of neighbourhoods $\mathfrak{p}^n$ of zero in $\mathcal{O}$ is mapped onto the basis of neighbourhoods $P_n \cap \varprojlim_v \mathcal{O}/\mathfrak{p}^v$ of zero in $\varprojlim_v \mathcal{O}/\mathfrak{p}^v$. Thus the bijection is a homeomorphism. It induces an isomorphism and homeomorphism on the group of units

$$\mathcal{O}^* \cong (\varprojlim \mathcal{O}/\mathfrak{p}^n)^* \cong \varprojlim (\mathcal{O}/\mathfrak{p}^n)^* \cong \varprojlim \mathcal{O}^*/U^{(n)}. \quad \square$$

One of our chief concerns will be to study the finite extensions $L|K$ of a complete valued field K . This means that we have to turn to the question of factoring algebraic equations

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = 0$$

over complete valued fields. For this, Hensel's seminal "lemma" is of fundamental importance. Let K again be a field which is complete with respect to a nonarchimedean valuation $|\cdot|$. Let $\mathcal{O}$ be the corresponding valuation ring with maximal ideal $\mathfrak{p}$ and residue class field $\kappa = \mathcal{O}/\mathfrak{p}$. We call a polynomial $f(x) = a_0 + a_1 x + \cdots + a_n x^n \in \mathcal{O}[x]$ **primitive** if $f(x) \not\equiv 0 \pmod{\mathfrak{p}}$, i.e., if

$$|f| = \max\{|a_0|, \dots, |a_n|\} = 1.$$

(4.6) Hensel's Lemma. *If a primitive polynomial $f(x) \in \mathcal{O}[x]$ admits modulo $\mathfrak{p}$ a factorization*

$$f(x) \equiv \bar{g}(x)\bar{h}(x) \pmod{\mathfrak{p}}$$

into relatively prime polynomials $\bar{g}, \bar{h} \in \kappa[x]$, then $f(x)$ admits a factorization

$$f(x) = g(x)h(x)$$

into polynomials $g, h \in \mathcal{O}[x]$ such that $\deg(g) = \deg(\bar{g})$ and

$$g(x) \equiv \bar{g}(x) \pmod{\mathfrak{p}} \quad \text{and} \quad h(x) \equiv \bar{h}(x) \pmod{\mathfrak{p}}.$$

Proof: Let $d = \deg(f)$, $m = \deg(\bar{g})$, hence $d - m \geq \deg(\bar{h})$. Let $g_0, h_0 \in \mathcal{O}[x]$ be polynomials such that $g_0 \equiv \bar{g} \pmod{\mathfrak{p}}$, $h_0 \equiv \bar{h} \pmod{\mathfrak{p}}$ and $\deg(g_0) = m$, $\deg(h_0) \leq d - m$. Since $(\bar{g}, \bar{h}) = 1$, there exist polynomials $a(x), b(x) \in \mathcal{O}[x]$ satisfying $ag_0 + bh_0 \equiv 1 \pmod{\mathfrak{p}}$. Among the coefficients of the two polynomials $f - g_0 h_0$ and $ag_0 + bh_0 - 1 \in \mathfrak{p}[x]$ we pick one with minimum value and call it π .

Let us look for the polynomials g and h in the following form:

$$g = g_0 + p_1\pi + p_2\pi^2 + \cdots,$$

$$h = h_0 + q_1\pi + q_2\pi^2 + \cdots,$$

where $p_i, q_i \in \mathcal{O}[x]$ are polynomials of degree $< m$, resp. $\leq d - m$. We then determine successively the polynomials

$$g_{n-1} = g_0 + p_1\pi + \cdots + p_{n-1}\pi^{n-1},$$

$$h_{n-1} = h_0 + q_1\pi + \cdots + q_{n-1}\pi^{n-1},$$

in such a way that one has

$$f \equiv g_{n-1}h_{n-1} \pmod{\pi^n}.$$

Passing to the limit as $n \rightarrow \infty$, we will finally obtain the identity $f = gh$. For $n = 1$ the congruence is satisfied in view of our choice of π . Let us assume that it is already established for some $n \geq 1$. Then, in view of the relation

$$g_n = g_{n-1} + p_n\pi^n, \quad h_n = h_{n-1} + q_n\pi^n,$$

the condition on g_n, h_n reduces to

$$f - g_{n-1}h_{n-1} \equiv (g_{n-1}q_n + h_{n-1}p_n)\pi^n \pmod{\pi^{n+1}}.$$

Dividing by π^n , this means

$$g_{n-1}q_n + h_{n-1}p_n \equiv g_0q_n + h_0p_n \equiv f_n \pmod{\pi},$$

where $f_n = \pi^{-n}(f - g_{n-1}h_{n-1}) \in \mathcal{O}[x]$. Since $g_0a + h_0b \equiv 1 \pmod{\pi}$, one has

$$g_0af_n + h_0bf_n \equiv f_n \pmod{\pi}.$$

At this point we would like to put $q_n = af_n$ and $p_n = bf_n$, but the degrees might be too big. For this reason, we write

$$b(x)f_n(x) = q(x)g_0(x) + p_n(x),$$

where $\deg(p_n) < \deg(g_0) = m$. Since $g_0 \equiv \bar{g} \pmod{\mathfrak{p}}$ and $\deg(g_0) = \deg(\bar{g})$, the highest coefficient of g_0 is a unit; hence $q(x) \in \mathcal{O}[x]$ and we obtain the congruence

$$g_0(af_n + h_0q) + h_0p_n \equiv f_n \pmod{\pi}.$$

Omitting now from the polynomial $af_n + h_0q$ all coefficients divisible by π , we get a polynomial q_n such that $g_0q_n + h_0p_n \equiv f_n \pmod{\pi}$ and which, in view of $\deg(f_n) \leq d$, $\deg(g_0) = m$ and $\deg(h_0p_n) < (d - m) + m = d$, has degree $\leq d - m$ as required. $\square$

Example: The polynomial $x^{p-1} - 1 \in \mathbb{Z}_p[x]$ splits over the residue class field $\mathbb{Z}_p/p\mathbb{Z}_p = \mathbb{F}_p$ into distinct linear factors. Applying (repeatedly) Hensel's lemma, we see that it also splits into linear factors over $\mathbb{Z}_p$. We thus obtain the astonishing result that the field $\mathbb{Q}_p$ of p -adic numbers contains the $(p-1)$ -th roots of unity. These, together with 0, even form a system of representatives for the residue class field, which is closed under multiplication.

(4.7) Corollary. *Let the field K be complete with respect to the nonarchimedean valuation $|\cdot|$. Then, for every irreducible polynomial $f(x) = a_0 + a_1x + \cdots + a_nx^n \in K[x]$ such that $a_0a_n \neq 0$, one has*

$$|f| = \max\{|a_0|, |a_n|\}.$$

In particular, $a_n = 1$ and $a_0 \in \mathfrak{o}$ imply that $f \in \mathfrak{o}[x]$.

Proof: After multiplying by a suitable element of K we may assume that $f \in \mathfrak{o}[x]$ and $|f| = 1$. Let a_r be the first one among the coefficients $a_0, \dots, a_n$ such that $|a_r| = 1$. In other words, we have

$$f(x) \equiv x^r(a_r + a_{r+1}x + \cdots + a_nx^{n-r}) \pmod{\mathfrak{p}}.$$

If one had $\max\{|a_0|, |a_n|\} < 1$, then $0 < r < n$ and the congruence would contradict Hensel's lemma. $\square$

From this corollary we can now deduce the following theorem on extensions of valuations.

(4.8) Theorem. *Let K be complete with respect to the valuation $|\cdot|$. Then $|\cdot|$ may be extended in a unique way to a valuation of any given algebraic extension $L|K$. This extension is given by the formula*

$$|\alpha| = \sqrt[n]{|N_{L|K}(\alpha)|},$$

when $L|K$ has finite degree n . In this case L is again complete.

Proof: If the valuation $|\cdot|$ is archimedean, then by Ostrowski's theorem, $K = \mathbb{R}$ or $\mathbb{C}$. We have $N_{\mathbb{C}|\mathbb{R}}(z) = z\bar{z} = |z|^2$ and the theorem is part of classical analysis. So let $|\cdot|$ be nonarchimedean. Since every algebraic extension $L|K$ is the union of its finite subextensions, we may assume that the degree $n = [L : K]$ is finite.

Existence of the extended valuation: let $\mathcal{O}$ be the valuation ring of K and $\mathcal{O}$ its integral closure in L . Then one has

$$(*) \quad \mathcal{O} = \{ \alpha \in L \mid N_{L|K}(\alpha) \in \mathcal{O} \}.$$

The implication $\alpha \in \mathcal{O} \Rightarrow N_{L|K}(\alpha) \in \mathcal{O}$ is evident (see chap. I, § 2, p. 12). Conversely, let $\alpha \in L^*$ and $N_{L|K}(\alpha) \in \mathcal{O}$. Let

$$f(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_0 \in K[x]$$

be the minimal polynomial of α over K . Then $N_{L|K}(\alpha) = \pm a_0^m \in \mathcal{O}$, so that $|a_0| \leq 1$, i.e., $a_0 \in \mathcal{O}$. By (4.7) this gives $f(x) \in \mathcal{O}[x]$, i.e., $\alpha \in \mathcal{O}$.

For the function $|\alpha| = \sqrt[n]{|N_{L|K}(\alpha)|}$, the conditions $|\alpha| = 0 \iff \alpha = 0$ and $|\alpha\beta| = |\alpha||\beta|$ are obvious. The strong triangle inequality

$$|\alpha + \beta| \leq \max\{|\alpha|, |\beta|\}$$

reduces, after dividing by α or β , to the implication

$$|\alpha| \leq 1 \implies |\alpha + 1| \leq 1,$$

and then, by (*), to $\alpha \in \mathcal{O} \Rightarrow \alpha + 1 \in \mathcal{O}$, which is trivially true. Thus the formula $|\alpha| = \sqrt[n]{|N_{L|K}(\alpha)|}$ does define a valuation of L and, restricted to K , it clearly gives back the given valuation. Equally obviously it has $\mathcal{O}$ as its valuation ring.

Uniqueness of the extended valuation: let $|\cdot|'$ be another extension with valuation ring $\mathcal{O}'$. Let $\mathfrak{P}$, resp. $\mathfrak{P}'$, be the maximal ideal of $\mathcal{O}$, resp. $\mathcal{O}'$. We show that $\mathcal{O} \subseteq \mathcal{O}'$. Let $\alpha \in \mathcal{O} \setminus \mathcal{O}'$ and let

$$f(x) = x^d + a_1x^{d-1} + \cdots + a_d$$

be the minimal polynomial of α over K . Then one has $a_1, \dots, a_d \in \mathcal{O}$ and $\alpha^{-1} \in \mathfrak{P}'$, hence $1 = -a_1\alpha^{-1} - \cdots - a_d(\alpha^{-1})^d \in \mathfrak{P}'$, a contradiction. This shows the inclusion $\mathcal{O} \subseteq \mathcal{O}'$. In other words, we have that $|\alpha| \leq 1 \Rightarrow |\alpha|' \leq 1$ and this implies that the valuations $|\cdot|$ and $|\cdot|'$ are equivalent. For if they were not, then the approximation theorem (3.4) would allow us to find an $\alpha \in L$ such that $|\alpha| \leq 1 \Rightarrow |\alpha|' > 1$. Thus $|\cdot|$ and $|\cdot|'$ are equal because they agree on K .

The fact that L is again complete with respect to the extended valuation is deduced from the following general result. $\square$

(4.9) Proposition. *Let K be complete with respect to the valuation $|\cdot|$ and let V be an n -dimensional normed vector space over K . Then, for any basis $v_1, \dots, v_n$ of V the maximum norm*

$$\|x_1v_1 + \cdots + x_nv_n\| = \max\{|x_1|, \dots, |x_n|\}$$

is equivalent to the given norm on V . In particular, V is complete and the isomorphism

$$K^n \longrightarrow V, \quad (x_1, \dots, x_n) \longmapsto x_1 v_1 + \dots + x_n v_n,$$

is a homeomorphism.

Proof: Let $v_1, \dots, v_n$ be a basis and $\| \cdot \|$ be the corresponding maximum norm on V . It suffices to show that, for every norm $| \cdot |$ on V , there exist constants $\rho, \rho' > 0$ such that

$$\rho \|x\| \leq |x| \leq \rho' \|x\| \quad \text{for all } x \in V.$$

Then the norm $| \cdot |$ defines the same topology on V as the norm $\| \cdot \|$, and we obtain the topological isomorphism $K^n \rightarrow V$, $(x_1, \dots, x_n) \mapsto x_1 v_1 + \dots + x_n v_n$. In fact, $\| \cdot \|$ is transformed into the maximum norm on K^n .

For ρ' we may obviously take $|v_1| + \dots + |v_n|$. The existence of ρ is proved by induction on n . For $n = 1$ we may take $\rho = |v_1|$. Suppose that everything is proved for $(n - 1)$ -dimensional vector spaces. Let

$$V_i = K v_1 + \dots + K v_{i-1} + K v_{i+1} + \dots + K v_n,$$

so that $V = V_i + K v_i$. Then V_i is complete with respect to the restriction of $| \cdot |$ by induction, hence it is closed in V . Thus $V_i + v_i$ is also closed. Since $0 \notin \bigcup_{i=1}^n (V_i + v_i)$, there exists a neighbourhood of 0 which is disjoint from $\bigcup_{i=1}^n (V_i + v_i)$, i.e., there exists $\rho > 0$ such that

$$|w_i + v_i| \geq \rho \quad \text{for all } w_i \in V_i \quad \text{and all } i = 1, \dots, n.$$

For $x = x_1 v_1 + \dots + x_n v_n \neq 0$ and $|x_r| = \max\{|x_i|\}$, one finds

$$|x_r^{-1} x| = \left| \frac{x_1}{x_r} v_1 + \dots + v_r + \dots + \frac{x_n}{x_r} v_n \right| \geq \rho,$$

so that one has $|x| \geq \rho |x_r| = \rho \|x\|$. □

The fact that an exponential valuation v on K associated with $| \cdot |$ extends uniquely to L is a trivial consequence of theorem (4.8). The extension w is given by the formula

$$w(\alpha) = \frac{1}{n} v(N_{L|K}(\alpha))$$

if $n = [L : K] < \infty$.

Exercise 1. An infinite algebraic extension of a complete field K is never complete.

Exercise 2. Let $X_0, X_1, \dots$ be an infinite sequence of unknowns, p a fixed prime number and $W_n = X_0^{p^n} + pX_1^{p^{n-1}} + \dots + p^n X_n$, $n \geq 0$. Show that there exist polynomials $S_0, S_1, \dots; P_0, P_1, \dots \in \mathbb{Z}[X_0, X_1, \dots; Y_0, Y_1, \dots]$ such that

$$W_n(S_0, S_1, \dots) = W_n(X_0, X_1, \dots) + W_n(Y_0, Y_1, \dots),$$

$$W_n(P_0, P_1, \dots) = W_n(X_0, X_1, \dots) \cdot W_n(Y_0, Y_1, \dots).$$

Exercise 3. Let A be a commutative ring. For $a = (a_0, a_1, \dots)$, $b = (b_0, b_1, \dots)$, $a_i, b_i \in A$, put

$$a + b = (S_0(a, b), S_1(a, b), \dots), \quad a \cdot b = (P_0(a, b), P_1(a, b), \dots).$$

Show that with these operations the vectors $a = (a_0, a_1, \dots)$ form a commutative ring $W(A)$ with 1. It is called the **ring of Witt vectors** over A .

Exercise 4. Assume $pA = 0$. For every Witt vector $a = (a_0, a_1, \dots) \in W(A)$ consider the “ghost components”

$$a^{(n)} = W_n(a) = a_0^{p^n} + pa_1^{p^{n-1}} + \dots + p^n a_n$$

as well as the mappings $V, F : W(A) \rightarrow W(A)$ defined by

$$Va = (0, a_0, a_1, \dots) \quad \text{and} \quad Fa = (a_0^p, a_1^p, \dots),$$

called respectively “transfer” (“Verschiebung” in German) and “Frobenius”. Show that

$$(Va)^{(n)} = pa^{(n-1)} \quad \text{and} \quad a^{(n)} = (Fa)^{(n)} + p^n a_n.$$

Exercise 5. Let k be a field of characteristic p . Then V is a homomorphism of the additive group of $W(k)$ and F is a ring homomorphism, and one has

$$VF a = FVa = pa.$$

Exercise 6. If k is a perfect field of characteristic p , then $W(k)$ is a complete discrete valuation ring with residue class field k .

§ 5. Local Fields

Among all complete (nonarchimedean) valued fields, those arising as completions of a **global field**, i.e., of a finite extension of either $\mathbb{Q}$ or $\mathbb{F}_p(t)$, have the most eminent relevance for number theory. The valuation on such a completion is discrete and has a finite residue class field, as we shall see shortly. In contrast to the global fields, all fields which are complete with respect to a discrete valuation and have a finite residue class field are called **local fields**. For such a local field, the normalized exponential valuation is denoted by v_p , and $|\cdot|_p$ denotes the absolute value normalized by

$$|x|_p = q^{-v_p(x)},$$

where q is the cardinality of the residue class field.

(5.1) Proposition. *A local field K is locally compact. Its valuation ring $\mathcal{O}$ is compact.*

Proof: By (4.5) we have $\mathcal{O} \cong \varprojlim \mathcal{O}/\mathfrak{p}^n$, both algebraically and topologically. Since $\mathfrak{p}^v/\mathfrak{p}^{v+1} \cong \mathcal{O}/\mathfrak{p}$ (see (3.9)), the rings $\mathcal{O}/\mathfrak{p}^n$ are finite, hence compact. Being a closed subset of the compact product $\prod_{n=1}^{\infty} \mathcal{O}/\mathfrak{p}^n$, it follows that the projective limit $\varprojlim \mathcal{O}/\mathfrak{p}^n$, and thus $\mathcal{O}$, is also compact. For every $a \in K$, the set $a + \mathcal{O}$ is an open, and at the same time compact neighbourhood, so that K is locally compact. $\square$

In happy concord with the definition of global fields as the finite extensions of $\mathbb{Q}$ and $\mathbb{F}_p(t)$, we now obtain the following characterization of local fields.

(5.2) Proposition. *The local fields are precisely the finite extensions of the fields $\mathbb{Q}_p$ and $\mathbb{F}_p((t))$.*

Proof: A finite extension K of $k = \mathbb{Q}_p$ or $k = \mathbb{F}_p((t))$ is again complete, by (4.8), with respect to the extended valuation $|\alpha| = \sqrt[n]{|N_{K|k}(\alpha)|}$, which itself is obviously again discrete. Since $K|k$ is of finite degree, so is the residue class field extension $\kappa|\mathbb{F}_p$, for if $\bar{x}_1, \dots, \bar{x}_n \in \kappa$ are linearly independent, then any choice of preimages $x_1, \dots, x_n \in K$ is linearly independent over k . Indeed, dividing any nontrivial k -linear relation $\lambda_1 x_1 + \dots + \lambda_n x_n = 0$, $\lambda_i \in k$, by the coefficient λ_i with biggest absolute value, yields a linear combination with coefficients in the valuation ring of k with 1 as i -th coefficient, from which we obtain a nontrivial relation $\bar{\lambda}_1 \bar{x}_1 + \dots + \bar{\lambda}_n \bar{x}_n = 0$ by reducing to κ . Therefore K is a local field.

Conversely, let K be a local field, v its discrete exponential valuation, and p the characteristic of its residue class field κ . If K has characteristic 0, then the restriction of v to $\mathbb{Q}$ is equivalent to the p -adic valuation v_p of $\mathbb{Q}$ because $v(p) > 0$. In view of the completeness of K , the closure of $\mathbb{Q}$ in K is the completion of $\mathbb{Q}$ with respect to v_p , in other words $\mathbb{Q}_p \subseteq K$. The fact that $K|\mathbb{Q}_p$ is of finite degree results from the local compactness of the vector space K , by a general theorem of topological linear algebra (see [18], chap. I, § 2, n° 4, th. 3), but it also follows from (6.8) below. If on the other hand the characteristic of K is not equal to zero, then it has to equal p . In this case we find $K = \kappa((t))$, for a prime element t of K (see p. 127), hence $\mathbb{F}_p((t)) \subseteq K$. In fact, if $\kappa = \mathbb{F}_p(\alpha)$ and $p(X) \in \mathbb{F}_p[X] \subseteq K[X]$ is the minimal polynomial of α over $\mathbb{F}_p$, then, by Hensel's lemma, $p(X)$ splits over K into linear factors. We may therefore view κ as a subfield of K , and then the elements of K turn out to be, by (4.4), the Laurent series in t with coefficients in κ . $\square$

Remark: One can show that a field K which is locally compact with respect to a nondiscrete topology is isomorphic either to $\mathbb{R}$ or $\mathbb{C}$, or to a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_p((t))$, i.e., to a local field (see [137], chap. I, § 3).

We have just seen that the local fields of characteristic p are the **power series fields** $\mathbb{F}_q((t))$, with $q = p^f$. The local fields of characteristic 0, i.e., the finite extensions $K|\mathbb{Q}_p$ of the fields of p -adic numbers $\mathbb{Q}_p$, are called **p -adic number fields**. For them one has an *exponential function* and a *logarithm function*. In contrast to the real and complex case, however, the former is not defined on all of K , whereas the latter is given on the whole multiplicative group K^* . For the definition of the logarithm we make use of the following fact.

(5.3) Proposition. *The multiplicative group of a local field K admits the decomposition*

$$K^* = (\pi) \times \mu_{q-1} \times U^{(1)}.$$

Here π is a prime element, $(\pi) = \{\pi^k \mid k \in \mathbb{Z}\}$, $q = \# \kappa$ is the number of elements in the residue class field $\kappa = \mathcal{O}/\mathfrak{p}$, and $U^{(1)} = 1 + \mathfrak{p}$ is the group of principal units.

Proof: For every $\alpha \in K^*$, one has a unique representation $\alpha = \pi^n u$ with $n \in \mathbb{Z}$, $u \in \mathcal{O}^*$ so that $K^* = (\pi) \times \mathcal{O}^*$. Since the polynomial $X^{q-1} - 1$ splits into linear factors over K by Hensel's lemma, $\mathcal{O}^*$ contains the group μ_{q-1} of $(q-1)$ -th roots of unity. The homomorphism $\mathcal{O}^* \rightarrow \kappa^*$, $u \mapsto u \bmod \mathfrak{p}$, has kernel $U^{(1)}$ and maps μ_{q-1} bijectively onto κ^* . Hence $\mathcal{O}^* = \mu_{q-1} \times U^{(1)}$. $\square$

(5.4) Proposition. *For a p -adic number field K there is a uniquely determined continuous homomorphism*

$$\log : K^* \rightarrow K$$

such that $\log p = 0$ which on principal units $(1+x) \in U^{(1)}$ is given by the series

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots.$$

Proof: By § 4, we can think of the p -adic valuation v_p of $\mathbb{Q}_p$ as extended to K . Observing that $v_p(x) > 0$, so that $c = p^{v_p(x)} > 1$, and $p^{v_p(v)} \leq v$,

giving $v_p(v) \leq \frac{\ln v}{\ln p}$ (with the usual logarithm), we compute the valuation of the terms x^v/v of the series,

$$v_p\left(\frac{x^v}{v}\right) = v v_p(x) - v_p(v) \geq v \frac{\ln c}{\ln p} - \frac{\ln v}{\ln p} = \frac{\ln(c^v/v)}{\ln p}.$$

This shows that x^v/v is a nullsequence, i.e., the logarithm series converges. It defines a homomorphism because

$$\log((1+x)(1+y)) = \log(1+x) + \log(1+y)$$

is an identity of formal power series and all series in it converge provided $1+x, 1+y \in U^{(1)}$.

For every $\alpha \in K^*$, choosing a prime element π , we have a unique representation

$$\alpha = \pi^{v_p(\alpha)} \omega(\alpha) \langle \alpha \rangle,$$

where $v_p = ev_p$ is the normalized valuation of K , $\omega(\alpha) \in \mu_{q-1}$, $\langle \alpha \rangle \in U^{(1)}$. As suggested by the equation $p = \pi^e \omega(p) \langle p \rangle$, we define $\log \pi = -\frac{1}{e} \log \langle p \rangle$ and thus obtain the homomorphism $\log : K^* \rightarrow K$ by

$$\log \alpha = v_p(\alpha) \log \pi + \log \langle \alpha \rangle.$$

It is obviously continuous and has the property that $\log p = 0$. If $\lambda : K^* \rightarrow K$ is any continuation of $\log : U^{(1)} \rightarrow K$ such that $\lambda(p) = 0$, then we find that $\lambda(\xi) = \frac{1}{q-1} \lambda(\xi^{q-1}) = 0$ for each $\xi \in \mu_{q-1}$. It follows that $0 = e\lambda(\pi) + \lambda(\langle p \rangle) = e\lambda(\pi) + \log \langle p \rangle$, so that $\lambda(\pi) = \log \pi$, and thus $\lambda(\alpha) = v_p(\alpha) \lambda(\pi) + \lambda(\langle \alpha \rangle) = v_p(\alpha) \log \pi + \log \langle \alpha \rangle = \log \alpha$, for all $\alpha \in K^*$. $\log$ is therefore uniquely determined and independent of the choice of π . $\square$

(5.5) Proposition. *Let $K|\mathbb{Q}_p$ be a p -adic number field with valuation ring $\mathfrak{o}$ and maximal ideal $\mathfrak{p}$, and let $p\mathfrak{o} = \mathfrak{p}^e$. Then the power series*

$$\exp(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad \text{and} \quad \log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots,$$

yield, for $n > \frac{e}{p-1}$, two mutually inverse isomorphisms (and homeomorphisms)

$$\mathfrak{p}^n \xrightleftharpoons[\log]{\exp} U^{(n)}.$$

We prepare the proof by the following elementary lemma.

(5.6) Lemma. Let $v = \sum_{i=0}^r a_i p^i$, $0 \leq a_i < p$, be the p -adic expansion of the natural number $v \in \mathbb{N}$. Then

$$v_p(v!) = \frac{1}{p-1} \sum_{i=0}^r a_i (p^i - 1).$$

Proof: Let $[c]$ signify the biggest integer $\leq c$. Then we have

$$\begin{aligned} [v/p] &= a_1 + a_2 p + \cdots + a_r p^{r-1}, \\ [v/p^2] &= a_2 + \cdots + a_r p^{r-2}, \\ &\vdots \\ [v/p^r] &= a_r. \end{aligned}$$

Now we count how many numbers $1, 2, \dots, v$ are divisible by p , and then by p^2 , etc. We find

$$v_p(v!) = [v/p] + \cdots + [v/p^r] = a_1 + (p+1)a_2 + \cdots + (p^{r-1} + \cdots + 1)a_r$$

and hence

$$(p-1)v_p(v!) = (p-1)a_1 + (p^2-1)a_2 + \cdots + (p^r-1)a_r = \sum_{i=0}^r a_i (p^i - 1).$$

□

Proof of (5.5): We again think of the p -adic valuation v_p of $\mathbb{Q}_p$ as being extended to K . Then $v_p = e v_p$ is the normalized valuation of K . For every natural number $v > 1$, one has the estimate

$$\frac{v_p(v)}{v-1} \leq \frac{1}{p-1},$$

for if $v = p^a v_0$, with $(v_0, p) = 1$ and $a > 0$, then

$$\frac{v_p(v)}{v-1} = \frac{a}{p^a v_0 - 1} \leq \frac{a}{p^a - 1} = \frac{1}{p-1} \frac{a}{p^{a-1} + \cdots + p + 1} \leq \frac{1}{p-1}.$$

For $v_p(z) > \frac{1}{p-1}$, $z \neq 0$, i.e., $v_p(z) > \frac{e}{p-1}$, this yields

$$v_p\left(\frac{z^v}{v}\right) - v_p(z) = (v-1)v_p(z) - v_p(v) > (v-1)\left(\frac{1}{p-1} - \frac{v_p(v)}{v-1}\right) \geq 0,$$

and thus $v_p(\log(1+z)) = v_p(z)$. For $n > \frac{e}{p-1}$, $\log$ therefore maps $U^{(n)}$ into p^n .

For the exponential series $\sum_{v=0}^{\infty} x^v/v!$, we compute the valuations $v_p(x^v/v!)$ as follows. Writing, for $v > 0$,

$$v = a_0 + a_1 p + \cdots + a_r p^r, \quad 0 \leq a_i < p,$$

we get from (5.6) that

$$v_p(v!) = \frac{1}{p-1} \sum_{i=0}^r a_i(p^i - 1) = \frac{1}{p-1} (v - (a_0 + a_1 + \cdots + a_r)).$$

Putting $s_v = a_0 + \cdots + a_r$ this becomes

$$v_p\left(\frac{x^v}{v!}\right) = v v_p(x) - \frac{v - s_v}{p-1} = v\left(v_p(x) - \frac{1}{p-1}\right) + \frac{s_v}{p-1}.$$

For $v_p(x) > \frac{e}{p-1}$, i.e., $v_p(x) > \frac{1}{p-1}$, this implies the convergence of the exponential series. If furthermore $x \neq 0$ and $v > 1$, then one has

$$v_p\left(\frac{x^v}{v!}\right) - v_p(x) = (v-1)v_p(x) - \frac{v-1}{p-1} + \frac{s_v-1}{p-1} > \frac{s_v-1}{p-1} \geq 0.$$

Therefore $v_p(\exp(x)-1) = v_p(x)$, i.e., for $n > \frac{e}{p-1}$, $\exp$ maps the group $\mathfrak{p}^n$ into $U^{(n)}$. Furthermore, one has for $v_p(x), v_p(z) > \frac{e}{p-1}$ that

$$\exp \log(1+z) = 1+z \quad \text{and} \quad \log \exp x = x,$$

for these are identities of formal power series and all of the series converge. This proves the proposition. $\square$

For an arbitrary local field K , the group of principal units $U^{(1)}$ is a $\mathbb{Z}_p$ -module (where $p = \text{char}(\kappa)$) in a canonical way, i.e., for every $1+x \in U^{(1)}$ and every $z \in \mathbb{Z}_p$, one has the power $(1+x)^z \in U^{(1)}$. This is a consequence of the fact that $U^{(1)}/U^{(n+1)}$ has order q^n for all n (where $q = \#\mathcal{O}/\mathfrak{p}$ — the reason for this is that $U^{(i)}/U^{(i+1)} \cong \mathcal{O}/\mathfrak{p}$, by (3.10), so that $U^{(1)}/U^{(n+1)}$ is a $\mathbb{Z}/q^n\mathbb{Z}$ -module) and of the formulas

$$U^{(1)} = \varprojlim_n U^{(1)}/U^{(n+1)} \quad \text{and} \quad \mathbb{Z}_p = \varprojlim_n \mathbb{Z}/q^n\mathbb{Z}.$$

This obviously extends the $\mathbb{Z}$ -module structure of $U^{(1)}$. The function

$$f(z) = (1+x)^z$$

is continuous because the congruence $z \equiv z' \pmod{q^n\mathbb{Z}_p}$ implies $(1+x)^z \equiv (1+x)^{z'} \pmod{U^{(n+1)}}$, so that the neighbourhood $z + q^n\mathbb{Z}_p$ of z is mapped to the neighbourhood $(1+x)^z U^{(n+1)}$ of $f(z)$. In particular, $(1+x)^z$ may be expressed as the limit

$$(1+x)^z = \lim_{i \rightarrow \infty} (1+x)^{z_i}$$

of ordinary powers $(1+x)^{z_i}$, $z_i \in \mathbb{Z}$, if $z = \lim_{i \rightarrow \infty} z_i$.

After this discussion we can now determine explicitly the structure of the locally compact multiplicative group K^* of a local field K .

(5.7) Proposition. *Let K be a local field and $q = p^f$ the number of elements in the residue class field. Then the following hold.*

(i) *If K has characteristic 0, then one has (both algebraically and topologically)*

$$K^* \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}_p^d,$$

where $a \geq 0$ and $d = [K : \mathbb{Q}_p]$.

(ii) *If K has characteristic p , then one has (both algebraically and topologically)*

$$K^* \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}_p^{\mathbb{N}}.$$

Proof: By (5.3) we have (both algebraically and topologically)

$$K^* = (\pi) \times \mu_{q-1} \times U^{(1)} \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus U^{(1)}.$$

This reduces us to the computation of the $\mathbb{Z}_p$ -module $U^{(1)}$.

(i) Assume $\text{char}(K) = 0$. For n sufficiently big, (5.5) gives us the isomorphism

$$\log : U^{(n)} \longrightarrow \mathfrak{p}^n = \pi^n \mathcal{O} \cong \mathcal{O}.$$

Since $\log$, $\exp$, and $f(z) = (1+x)^z$ are continuous, this is a topological isomorphism of $\mathbb{Z}_p$ -modules. By chap. I, (2.9), $\mathcal{O}$ admits an integral basis $\alpha_1, \dots, \alpha_d$ over $\mathbb{Z}_p$, i.e., $\mathcal{O} = \mathbb{Z}_p\alpha_1 \oplus \dots \oplus \mathbb{Z}_p\alpha_d \cong \mathbb{Z}_p^d$. Therefore $U^{(n)} \cong \mathbb{Z}_p^d$. Since the index $(U^{(1)} : U^{(n)})$ is finite and $U^{(n)}$ is a finitely generated $\mathbb{Z}_p$ -module of rank d , so is $U^{(1)}$. The torsion subgroup of $U^{(1)}$ is the group μ_{p^a} of roots of unity in K of p -power order. By the main theorem on modules over principal ideal domains, there exists in $U^{(1)}$ a free, finitely generated, and therefore closed, $\mathbb{Z}_p$ -submodule V of rank d such that

$$U^{(1)} = \mu_{p^a} \times V \cong \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}_p^d,$$

both algebraically and topologically.

(ii) If $\text{char}(K) = p$, we have $K \cong \mathbb{F}_q((t))$ (see p. 127) and

$$U^{(1)} = 1 + \mathfrak{p} = 1 + t\mathbb{F}_q[[t]].$$

The following argument is taken from the book [79] of *K. IWASAWA*.

Let $\omega_1, \dots, \omega_f$ be a basis of $\mathbb{F}_q|\mathbb{F}_p$. For every natural number n relatively prime to p we consider the continuous homomorphism

$$g_n : \mathbb{Z}_p^f \rightarrow U^{(n)}, \quad g_n(a_1, \dots, a_f) = \prod_{i=1}^f (1 + \omega_i t^n)^{a_i}.$$

This function has the following properties. If $m = np^s$, $s \geq 0$, then

$$(1) \quad U^{(m)} = g_n(p^s \mathbb{Z}_p^f) U^{(m+1)}$$

and, for $\alpha = (a_1, \dots, a_f) \in \mathbb{Z}_p^f$,

$$(2) \quad \alpha \notin p\mathbb{Z}_p^f \iff g_n(p^s \alpha) \notin U^{(m+1)}.$$

Indeed, for $\omega = \sum_{i=1}^f b_i \omega_i \in \mathbb{F}_q$, $b_i \in \mathbb{Z}$, $b_i \equiv a_i \pmod{p}$, we have

$$g_n(\alpha) \equiv \prod_{i=1}^f (1 + \omega_i t^n)^{b_i} \equiv 1 + \omega t^n \pmod{\mathfrak{p}^{n+1}}$$

and hence, since we are in characteristic p ,

$$g_n(p^s \alpha) = g_n(\alpha)^{p^s} \equiv 1 + \omega^{p^s} t^{p^s n} \pmod{\mathfrak{p}^{m+1}}.$$

As α varies over the elements of $\mathbb{Z}_p^f$, ω , and thus also ω^{p^s} , varies over the elements of $\mathbb{F}_q$, and we get (1). Furthermore one has $g_n(p^s \alpha) \equiv 1 \pmod{\mathfrak{p}^{m+1}} \iff \omega = 0 \iff b_i \equiv 0 \pmod{p}$, for $i = 1, \dots, f \iff a_i \equiv 0 \pmod{p}$, for $i = 1, \dots, f \iff \alpha \in p\mathbb{Z}_p^f$, and this amounts to (2).

We now consider the continuous homomorphism of $\mathbb{Z}_p$ -modules

$$g = \prod_{(n,p)=1} g_n : A = \prod_{(n,p)=1} \mathbb{Z}_p^f \longrightarrow U^{(1)},$$

where the product $\prod_{(n,p)=1} \mathbb{Z}_p^f$ is taken over all $n \geq 1$ such that $(n, p) = 1$, each factor being a copy of $\mathbb{Z}_p^f$. Observe that the product $g(\xi) = \prod g_n(\alpha_n)$ converges because $g_n(\alpha_n) \in U^{(n)}$. Let $m = np^s$, with $(n, p) = 1$, be any natural number. As $g_n(\mathbb{Z}_p^f) \subseteq g(A)$, it follows from (1) that each coset of $U^{(m)} / U^{(m+1)}$ is represented by an element of $g(A)$. This means that $g(A)$ is dense in $U^{(1)}$. Since A is compact and g is continuous, g is actually surjective.

On the other hand, let $\xi = (\dots, \alpha_n, \dots) \in A$, $\xi \neq 0$, i.e., $\alpha_n \neq 0$ for some n . Such an α_n is of the form $\alpha_n = p^s \beta_n$ with $s = s(\alpha_n) \geq 0$, and $\beta_n \in \mathbb{Z}_p^f \setminus p\mathbb{Z}_p^f$. It now follows from (2) that

$$g_n(\alpha_n) \in U^{(m)}, \quad g_n(\alpha_n) \notin U^{(m+1)} \quad \text{for } m = m(\alpha_n) = np^s.$$

Since the n are prime to p , all the $m(\alpha_n)$ have to be distinct, for all $\alpha_n \neq 0$. Let n be the natural number, prime to p and such that $\alpha_n \neq 0$, which satisfies $m(\alpha_n) < m(\alpha_{n'})$, for all $n' \neq n$ such that $\alpha_{n'} \neq 0$. Then one has, for all $n' \neq n$, that

$$g_{n'}(\alpha_{n'}) \in U^{(m+1)} \quad \text{where } m = m(\alpha_n) < m(\alpha_{n'}).$$

Consequently

$$g(\xi) \equiv g_n(\alpha_n) \not\equiv 1 \pmod{U^{(m+1)}},$$

and so $g(\xi) \neq 1$. This shows the injectivity of g . Since $A = \mathbb{Z}_p^{\mathbb{N}}$, this proves the claim (ii). $\square$

(5.8) Corollary. *If the natural number n is not divisible by the characteristic of K , then one finds the following indices for the subgroups of n -th powers K^{*n} and U^n in the multiplicative group K^* and in the unit group U :*

$$(K^* : K^{*n}) = n(U : U^n) = \frac{n}{|n|_p} \# \mu_n(K).$$

Proof: The first equality is a consequence of $K^* = (\pi) \times U$. By (5.7), we have

$$U \cong \mu(K) \times \mathbb{Z}_p^d, \quad \text{resp.} \quad U \cong \mu(K) \times \mathbb{Z}_p^{\mathbb{N}},$$

when $\text{char}(K) = 0$, resp. $p > 0$. From the exact sequence

$$1 \longrightarrow \mu_n(K) \longrightarrow \mu(K) \xrightarrow{n} \mu(K) \longrightarrow \mu(K)/\mu(K)^n \longrightarrow 1,$$

one has $\# \mu_n(K) = \# \mu(K)/\mu(K)^n$. When $\text{char}(K) = 0$, this gives:

$$(U : U^n) = \# \mu_n(K) \# (\mathbb{Z}_p/n\mathbb{Z}_p)^d = \# \mu_n(K) p^{dv_p(n)} = \# \mu_n(K)/|n|_p,$$

and when $\text{char}(K) = p$ one gets simply $(U : U^n) = \# \mu_n(K) = \# \mu_n(K)/|n|_p$ because $(n, p) = 1$, i.e., $n\mathbb{Z}_p = \mathbb{Z}_p$. $\square$

Exercise 1. The logarithm function can be continued to a continuous homomorphism $\log : \overline{\mathbb{Q}_p}^* \rightarrow \mathbb{Q}$, and the exponential function to a continuous homomorphism $\exp : \overline{\mathbb{Q}_p}^{\frac{1}{1-p}} \rightarrow \overline{\mathbb{Q}_p}^*$, where $\overline{\mathbb{Q}_p}^{\frac{1}{1-p}} = \{x \in \overline{\mathbb{Q}_p} \mid v_p(x) > \frac{1}{1-p}\}$ and v_p is the unique extension of the normalized valuation on $\mathbb{Q}_p$.

Exercise 2. Let $K|\mathbb{Q}_p$ be a p -adic number field. For $1+x \in U^{(1)}$ and $z \in \mathbb{Z}_p$ one has

$$(1+x)^z = \sum_{v=0}^{\infty} \binom{z}{v} x^v.$$

The series converges even for $x \in K$ such that $v_p(x) > \frac{e}{p-1}$.

Exercise 3. Under the above hypotheses one has

$$(1+x)^z = \exp(z \log(1+x)) \quad \text{and} \quad \log(1+x)^z = z \log(1+x).$$

Exercise 4. For a p -adic number field K , every subgroup of finite index in K^* is both open and closed.

Exercise 5. If K is a p -adic number field, then the groups K^{*n} , for $n \in \mathbb{N}$, form a basis of neighbourhoods of 1 in K^* .

Exercise 6. Let K be a p -adic number field, v_p the normalized exponential valuation of K , and dx the Haar measure on the locally compact additive group K , scaled so that $\int_{\mathcal{O}} dx = 1$. Then one has $v_p(a) = \int_{a\mathcal{O}} dx$. Furthermore,

$$I(f) = \int_{K \setminus \{0\}} f(x) \frac{dx}{|x|_p}$$

is a Haar measure on the locally compact group K^* .

§ 6. Henselian Fields

Most results on complete valued fields can be derived from Hensel's lemma alone, without the full strength of completeness. This lemma is valid in a much bigger class of nonarchimedean valued fields than the complete ones. For example, let (K, v) be a nonarchimedean valued field and $(\widehat{K}, \widehat{v})$ its completion. Let $\mathcal{O}$, resp. $\widehat{\mathcal{O}}$, be the valuation rings of K , resp. $\widehat{K}$. We then consider the separable closure K_v of K in $\widehat{K}$, and the valuation ring $\mathcal{O}_v \subseteq K_v$ with maximal ideal $\mathfrak{p}_v$, which is associated to the restriction of $\widehat{v}$ to K_v ,

$$K \subseteq K_v \subseteq \widehat{K}, \quad \mathcal{O} \subseteq \mathcal{O}_v \subseteq \widehat{\mathcal{O}}.$$

Then Hensel's lemma holds in the ring $\mathcal{O}_v$ as well as in the ring $\widehat{\mathcal{O}}$ even though K_v will not, as a rule, be complete. When K_v is algebraically closed in $\widehat{K}$ — hence in particular $\text{char}(K) = 0$ — this is immediately obvious (otherwise it follows from (6.6) and § 6, exercise 3 below). Indeed, by (4.3) we have

$$\mathcal{O}/\mathfrak{p} = \mathcal{O}_v/\mathfrak{p}_v = \widehat{\mathcal{O}}/\widehat{\mathfrak{p}},$$

and if a primitive polynomial $f(x) \in \mathcal{O}_v[x]$ splits over $\mathcal{O}_v/\mathfrak{p}_v$ into relatively prime factors $\bar{g}(x), \bar{h}(x)$, then we have by Hensel's lemma (4.6) a factorization in $\widehat{\mathcal{O}}$

$$f(x) = g(x)h(x)$$

such that $g \equiv \bar{g} \pmod{\widehat{\mathfrak{p}}}$, $h \equiv \bar{h} \pmod{\widehat{\mathfrak{p}}}$, $\deg(g) = \deg(\bar{g})$. But this factorization already takes place over $\mathcal{O}_v$ once the highest coefficient of g is chosen to be in $\mathcal{O}_v^*$, because the coefficients of f , and therefore also those of g and h are algebraic over K .

The valued field K_v is called the **henselization** of the field K with respect to v . It enjoys all the relevant algebraic properties of the completion $\widehat{K}$, but offers the advantage of being itself an algebraic extension of K which can also be obtained in a purely algebraic manner, without the analytic recourse to the completion (see § 9, exercise 4). The consequence is that taking the henselization of an infinite algebraic extension $L|K$ is possible within the category of algebraic extensions. Let us define in general:

(6.1) Definition. A **henselian field** is a field with a nonarchimedean valuation v whose valuation ring $\mathcal{O}$ satisfies Hensel's lemma in the sense of (4.6). One also calls the valuation v or the valuation ring $\mathcal{O}$ **henselian**.

(6.2) Theorem. *Let K be a henselian field with respect to the valuation $|\cdot|$. Then $|\cdot|$ admits one and only one extension to any given algebraic extension $L|K$. It is given by*

$$|\alpha| = \sqrt[n]{|N_{L|K}(\alpha)|},$$

if $L|K$ has finite degree n . In any case, the valuation ring of the extended valuation is the integral closure of the valuation ring of K in L .

The proof of this theorem is *verbatim* the same as in the case of a complete field (see (4.8)). What is remarkable about our current setting is that, conversely, the unique extendability also characterizes henselian fields. In order to prove this, we appeal to a method which allows us to express the valuations of the roots of a polynomial in terms of the valuations of the coefficients. It relies on the notion of **Newton polygon**, which arises as follows.

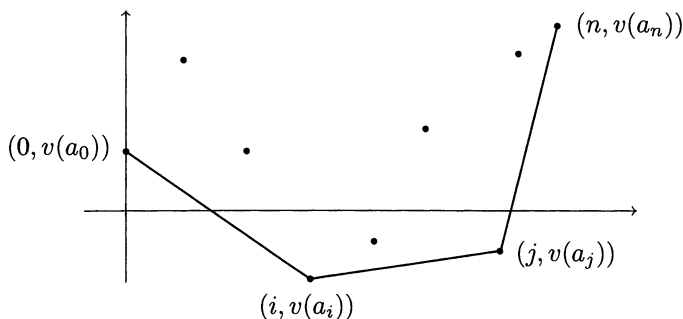
Let v be an arbitrary exponential valuation of the field K and let

$$f(x) = a_0 + a_1x + \cdots + a_nx^n \in K[x]$$

be a polynomial satisfying $a_0a_n \neq 0$. To each term a_ix^i we associate a point $(i, v(a_i)) \in \mathbb{R}^2$, ignoring however the point (i, ∞) if $a_i = 0$. We now take the lower convex envelope of the set of points

$$\{(0, v(a_0)), (1, v(a_1)), \dots, (n, v(a_n))\}.$$

This produces a polygonal chain which is called the **Newton polygon** of $f(x)$.



The polygon consists of a sequence of line segments $S_1, S_2, \dots$ whose slopes are strictly increasing, and which are subject to the following

(6.3) Proposition. Let $f(x) = a_0 + a_1x + \cdots + a_nx^n$, $a_0a_n \neq 0$, be a polynomial over the field K , v an exponential valuation of K , and w an extension to the splitting field L of f .

If $(r, v(a_r)) \leftrightarrow (s, v(a_s))$ is a line segment of slope $-m$ occurring in the Newton polygon of f , then $f(x)$ has precisely $s - r$ roots $\alpha_1, \dots, \alpha_{s-r}$ of value

$$w(\alpha_1) = \cdots = w(\alpha_{s-r}) = m.$$

Proof: Dividing by a_n only shifts the polygon up or down. Thus we may assume that $a_n = 1$. We number the roots $\alpha_1, \dots, \alpha_n \in L$ of f in such a way that

$$\begin{aligned} w(\alpha_1) &= \cdots = w(\alpha_{s_1}) = m_1, \\ w(\alpha_{s_1+1}) &= \cdots = w(\alpha_{s_2}) = m_2, \\ &\dots \quad \dots \quad \dots \quad \dots \\ w(\alpha_{s_t+1}) &= \cdots = w(\alpha_n) = m_{t+1}, \end{aligned}$$

where $m_1 < m_2 < \cdots < m_{t+1}$. Viewing the coefficients a_i as elementary symmetric functions of the roots α_j , we immediately find

$$\begin{aligned} v(a_n) &= v(1) = 0, \\ v(a_{n-1}) &\geq \min_i \{w(\alpha_i)\} = m_1, \\ v(a_{n-2}) &\geq \min_{i,j} \{w(\alpha_i\alpha_j)\} = 2m_1, \\ &\dots \quad \dots \quad \dots \quad \dots \\ v(a_{n-s_1}) &= \min_{i_1, \dots, i_{s_1}} \{w(\alpha_{i_1} \cdots \alpha_{i_{s_1}})\} = s_1 m_1, \end{aligned}$$

the latter because the value of the term $\alpha_1 \cdots \alpha_{s_1}$ is smaller than that of all the others,

$$\begin{aligned} v(a_{n-s_1-1}) &\geq \min_{i_1, \dots, i_{s_1+1}} \{w(\alpha_{i_1} \cdots \alpha_{i_{s_1+1}})\} = s_1 m_1 + m_2, \\ v(a_{n-s_1-2}) &\geq \min_{i_1, \dots, i_{s_1+2}} \{w(\alpha_{i_1} \cdots \alpha_{i_{s_1+2}})\} = s_1 m_1 + 2m_2, \\ &\dots \quad \dots \quad \dots \quad \dots \quad \dots \\ v(a_{n-s_2}) &= \min_{i_1, \dots, i_{s_2}} \{w(\alpha_{i_1} \cdots \alpha_{i_{s_2}})\} = s_1 m_1 + (s_2 - s_1)m_2, \end{aligned}$$

and so on. From this result one concludes that the vertices of the Newton polygon, from right to left, are given by

$$(n, 0), \quad (n - s_1, s_1 m_1), \quad (n - s_2, s_1 m_1 + (s_2 - s_1)m_2), \quad \dots$$

The slope of the extreme right-hand line segment is

$$\frac{0 - s_1 m_1}{n - (n - s_1)} = -m_1,$$

and, proceeding further to the left,

$$\frac{(s_1 m_1 + \cdots + (s_j - s_{j-1}) m_j) - (s_1 m_1 + \cdots + (s_{j+1} - s_j) m_{j+1})}{(n - s_j) - (n - s_{j+1})} = -m_{j+1}.$$

□

We emphasize that, according to the preceding proposition, the Newton polygon consists of precisely one segment if and only if the roots $\alpha_1, \dots, \alpha_n$ of f all have the same value. In general, $f(x)$ factors into a product according to the slopes $-m_r < \cdots < -m_1$,

$$f(x) = a_n \prod_{j=1}^r f_j(x),$$

where

$$f_j(x) = \prod_{w(\alpha_i)=m_j} (x - \alpha_i).$$

Here the factor f_j corresponds to the $(r - j + 1)$ -th segment of the Newton polygon, whose slope equals minus the value of the roots of f_j .

(6.4) Proposition. *If the valuation v admits a unique extension w to the splitting field L of f , then the factorization*

$$f(x) = a_n \prod_{j=1}^r f_j(x)$$

is defined already over K , i.e., $f_j(x) = \prod_{w(\alpha_i)=m_j} (x - \alpha_i) \in K[x]$.

Proof: We may clearly assume that $a_n = 1$. The statement is obvious when $f(x)$ is irreducible because then one has $\alpha_i = \sigma_i \alpha_1$ for some $\sigma_i \in G(L|K)$, and since, for any extension w of v , $w \circ \sigma_i$ is another one, the uniqueness implies that $w(\alpha_i) = w(\sigma_i \alpha_1) = m_1$, hence $f_1(x) = f(x)$.

The general case follows by induction on n . For $n = 1$ there is nothing to show. Let $p(x)$ be the minimal polynomial of α_1 and $g(x) = f(x)/p(x) \in K[x]$. Since all roots of $p(x)$ have the same value m_1 , $p(x)$ is a divisor of $f_1(x)$. Let $g_1(x) = f_1(x)/p(x)$. The factorization of $g(x)$ according to the slopes is

$$g(x) = g_1(x) \prod_{j=2}^r f_j(x).$$

Since $\deg(g) < \deg(f)$, it follows that $f_j(x) \in K[x]$ for all $j = 1, \dots, r$. □

If the polynomial f is irreducible, then, by the above factorization result, there is only *one* slope, i.e., the Newton polygon consists of a single segment. The values of all coefficients lie on or above this line segment and we get the

(6.5) Corollary. *Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in K[x]$ be an irreducible polynomial with $a_n \neq 0$. Then, if $|\cdot|$ is a nonarchimedean valuation of K with a unique extension to the splitting field, one has*

$$|f| = \max\{|a_0|, |a_n|\}.$$

In (4.7) we deduced this result for complete fields from Hensel's lemma and thus obtained the uniqueness of the extended valuation. Here we obtain it, by contrast, as a consequence of the uniqueness of the extended valuation. We now proceed to deduce Hensel's lemma from the unique extendability.

(6.6) Theorem. *A nonarchimedean valued field $(K, |\cdot|)$ is henselian if and only if the valuation $|\cdot|$ can be uniquely extended to any algebraic extension.*

Proof: The fact that a henselian valuation $|\cdot|$ extends uniquely was dealt with in (6.2). Let us assume conversely that $|\cdot|$ admits one and only one extension to any given algebraic extension. We first show:

Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathcal{O}[x]$ be a primitive, irreducible polynomial such that $a_0a_n \neq 0$, and let $\bar{f}(x) = f(x) \bmod \mathfrak{p} \in \kappa[x]$. Then we have $\deg(\bar{f}) = 0$ or $\deg(\bar{f}) = \deg(f)$, and we find

$$\bar{f}(x) = \bar{a} \bar{\varphi}(x)^m,$$

for some irreducible polynomial $\bar{\varphi}(x) \in \kappa[x]$ and a constant $\bar{a}$.

As f is irreducible, the Newton polygon is a single line segment and thus $|f| = \max\{|a_0|, |a_n|\}$. We may assume that a_n is a unit, because otherwise the Newton polygon is a segment which does not lie on the x -axis and this means that $\bar{f}(x) = \bar{a}_0$.

Let $L|K$ be the splitting field of $f(x)$ over K and $\mathcal{O}$ the valuation ring of the unique extension $|\cdot|$ to L , with maximal ideal $\mathfrak{P}$. For an arbitrary K -automorphism $\sigma \in G = G(L|K)$, we have $|\sigma\alpha| = |\alpha|$ for all $\alpha \in L$, because $|\cdot|$ and the composite $|\cdot| \circ \sigma$ extend the same valuation. This shows that $\sigma\mathcal{O} = \mathcal{O}$, $\sigma\mathfrak{P} = \mathfrak{P}$. If α is a zero of $f(x)$ and μ its multiplicity, then $\sigma\alpha \in \mathcal{O}$ for all $\sigma \in G$. Indeed, if $\alpha \notin \mathcal{O}$, then $\prod_{\sigma} |\sigma\alpha|^{\mu} = |\prod_{\sigma} \sigma\alpha|^{\mu} > 1$ would imply that the constant coefficient a_0 could not belong to $\mathcal{O}$. Thus every $\sigma \in G$ induces a κ -automorphism $\bar{\sigma}$ of $\mathcal{O}/\mathfrak{P}$, and the zeroes $\bar{\sigma}\bar{\alpha} = \bar{\sigma}\bar{a}$

of $\bar{f}(x)$ are all conjugate over κ . It follows that $\bar{f}(x) = \bar{a}\bar{\varphi}(x)^m$, if $\bar{\varphi}(x)$ is the minimal polynomial of $\bar{a}$ over κ . Since $a_n \in \mathcal{O}^*$, we have furthermore that $\deg(\bar{f}) = \deg(f)$.

Let now $f(x) \in \mathcal{O}[x]$ be an arbitrary primitive polynomial, and let

$$f(x) = f_1(x) \cdots f_r(x)$$

be its factorization into irreducibles over K . Since $1 = |f| = \prod |f_i|$, multiplying the f_i by suitable constants yields $|f_i| = 1$. The $f_i(x)$ are therefore primitive, irreducible polynomials in $\mathcal{O}[x]$. It follows that

$$\bar{f}(x) = \bar{f}_1(x) \cdots \bar{f}_r(x),$$

where $\deg(\bar{f}_i) = 0$ or $\deg(\bar{f}_i) = \deg(f_i)$, and $\bar{f}_i$ is, up to a constant factor, the power of an irreducible polynomial. If $\bar{f} = \bar{g}\bar{h}$ is a factorization into relatively prime polynomials $\bar{g}, \bar{h} \in \kappa[x]$, then we must have

$$\bar{g} = \bar{a} \prod_{i \in I} \bar{f}_i, \quad \bar{h} = \bar{b} \prod_{j \in J} \bar{f}_j$$

where $\bar{a}, \bar{b} \in \kappa$ and $\{1, \dots, r\} = I \cup J$ and $\deg(\bar{f}_i) = \deg(f_i)$ for $i \in I$. We now put

$$g = a \prod_{i \in I} f_i, \quad h = b \prod_{j \in J} f_j,$$

for $a, b \in \mathcal{O}^*$ such that $a \equiv \bar{a}, b \equiv \bar{b} \pmod{\mathfrak{p}}$ and $f = gh$. □

We have introduced henselian fields by a condition of which the reader will find weaker versions in the literature, restricted to *monic polynomials* only. Both are equivalent as is shown by the following

(6.7) Proposition. *A nonarchimedean field (K, v) is henselian if any monic polynomial $f(x) \in \mathcal{O}[x]$ which splits over the residue class field $\kappa = \mathcal{O}/\mathfrak{p}$ as*

$$f(x) \equiv \bar{g}(x)\bar{h}(x) \pmod{\mathfrak{p}}$$

with relatively prime monic factors $\bar{g}(x), \bar{h}(x) \in \kappa[x]$, admits itself a splitting

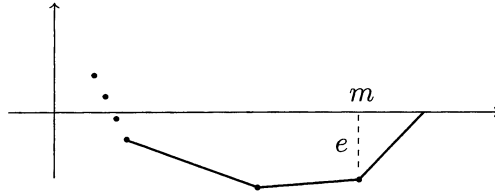
$$f(x) = g(x)h(x)$$

into monic factors $g(x), h(x) \in \mathcal{O}[x]$ such that

$$g(x) \equiv \bar{g}(x) \pmod{\mathfrak{p}} \quad \text{and} \quad h(x) \equiv \bar{h}(x) \pmod{\mathfrak{p}}.$$

Proof (E. NART): We have just seen that the property of K to be henselian follows from the condition that the Newton polygon of every irreducible polynomial $f(x) = a_0 + a_1x + \cdots + a_nx^n \in K[x]$ is a single line segment. It is therefore sufficient to show this. We may assume that $a_n = 1$. Let $L|K$ be the splitting field of f . Then there is always an extension w of v to L . It is obtained for example by taking the completion $\widehat{K}$ of K , extending the valuation of $\widehat{K}$ in a unique way to a valuation $\bar{v}$ of the algebraic closure $\bar{\widehat{K}}$ of $\widehat{K}$, embedding L into $\bar{\widehat{K}}$, and restricting $\bar{v}$ to L . It is also possible to get the extension w directly, without passing through the completion. For this we refer to [93], chap. XII, §4, th. 1.

Assume now that the Newton polygon of f consists of more than one segment:



Let the last segment be given by the points (m, e) and $(n, 0)$. If $e = 0$, we immediately have a contradiction. Because then we have $v(a_i) \geq 0$, so that $f(x) \in \mathcal{O}[x]$, and $a_0 \equiv \cdots \equiv a_{m-1} \equiv 0 \pmod{\mathfrak{p}}$, $a_m \not\equiv 0 \pmod{\mathfrak{p}}$. Therefore $f(x) \equiv (X^{n-m} + \cdots + a_m)X^m \pmod{\mathfrak{p}}$, with $m > 0$ because there is more than one segment. In view of the condition of the proposition this contradicts the irreducibility of f .

We will now reduce to $e = 0$ by a transformation. Let $\alpha \in L$ be a root of $f(x)$ of minimum value $w(\alpha)$ and let $a \in K$ such that $v(a) = e$. We consider the characteristic polynomial $g(x)$ of $a^{-1}\alpha^r \in K(\alpha)$, $r = n - m$. If $f(x) = \prod_{i=1}^n (x - \alpha_i)$, then $g(x) = \prod_{i=1}^n (x - \alpha_i^m a^{-1})$. Proposition (6.3) shows that the Newton polygon of $g(x)$ also has more than one segment, the last one of slope

$$-w(a^{-1}\alpha^r) = v(a) - rw(\alpha) = e - r\frac{e}{r} = 0.$$

Since $g(x)$ is a power of the minimal polynomial of $a^{-1}\alpha^r$, hence of an irreducible polynomial, this produces the same contradiction as before. $\square$

Let K be a field which is henselian with respect to the exponential valuation v . If $L|K$ is a finite extension of degree n , then v extends uniquely to an exponential valuation w of L , namely

$$w(\alpha) = \frac{1}{n} v(N_{L|K}(\alpha)).$$

This follows from (6.2) by taking the logarithm. For the value groups and residue class fields of v and w , one gets the inclusions

$$v(K^*) \subseteq w(L^*) \quad \text{and} \quad \kappa \subseteq \lambda.$$

The index

$$e = e(w | v) = (w(L^*) : v(K^*))$$

is called the **ramification index** of the extension $L|K$ and the degree

$$f = f(w | v) = [\lambda : \kappa]$$

is called the **inertia degree**. If v , and hence $w = \frac{1}{n}v \circ N_{L|K}$, is discrete and if $\mathfrak{o}, \mathfrak{p}, \pi$, resp. $\mathcal{O}, \mathfrak{P}, \Pi$, are the valuation ring, the maximal ideal and a prime element of K , resp. L , then one has

$$e = (w(\Pi)\mathbb{Z} : v(\pi)\mathbb{Z}),$$

so that $v(\pi) = ew(\Pi)$, and we find

$$\pi = \varepsilon \Pi^e,$$

for some unit $\varepsilon \in \mathcal{O}^*$. From this one deduces the familiar (see chap. I) interpretation of the ramification index: $\mathfrak{p}\mathcal{O} = \pi\mathcal{O} = \Pi^e\mathcal{O} = \mathfrak{P}^e$, or

$$\mathfrak{p} = \mathfrak{P}^e.$$

(6.8) Proposition. *One has $[L : K] \geq ef$ and the fundamental identity*

$$[L : K] = ef,$$

if v is discrete and $L|K$ is separable.

Proof: Let $\omega_1, \dots, \omega_f$ be representatives of a basis of $\lambda|\kappa$ and let $\pi_0, \dots, \pi_{e-1} \in L^*$ be elements the values of which represent the various cosets in $w(L^*)/v(K^*)$ (the finiteness of e will be a consequence of what follows). If v is discrete, we may choose for instance $\pi_i = \Pi^i$. We show that the elements

$$\omega_j \pi_i, \quad j = 1, \dots, f, \quad i = 0, \dots, e-1,$$

are linearly independent over K , and in the discrete case form even a basis of $L|K$. Let

$$\sum_{i=0}^{e-1} \sum_{j=1}^f a_{ij} \omega_j \pi_i = 0$$

with $a_{ij} \in K$. Assume that not all $a_{ij} = 0$. Then there exist nonzero sums $s_i = \sum_{j=1}^f a_{ij} \omega_j$, and each time that $s_i \neq 0$ we find $w(s_i) \in v(K^*)$. In

fact, dividing s_i by the coefficient a_{iv} of minimum value, we get a linear combination of the $\omega_1, \dots, \omega_f$ with coefficients in the valuation ring $\mathcal{O} \subseteq K$ one of which equals 1. This linear combination is $\not\equiv 0 \pmod{\mathfrak{P}}$, hence a unit, so that $w(s_i) = w(a_{iv}) \in v(K^*)$.

In the sum $\sum_{i=0}^{e-1} s_i \pi_i$, two nonzero summands must have the same value, say $w(s_i \pi_i) = w(s_j \pi_j)$, $i \neq j$, because otherwise it could not be zero (observe that $w(x) \neq w(y) \Rightarrow w(x+y) = \min\{w(x), w(y)\}$). It follows that

$$w(\pi_i) = w(\pi_j) + w(s_j) - w(s_i) \equiv w(\pi_j) \pmod{v(K^*)},$$

a contradiction. This shows the linear independence of the $\omega_j \pi_i$. In particular, we have $ef \leq [L : K]$.

Assume now that v , and thus also w , is discrete and let Π be a prime element in the valuation ring $\mathcal{O}$ of w . We consider the $\mathcal{O}$ -module

$$M = \sum_{i=0}^{e-1} \sum_{j=1}^f \mathcal{O} \omega_j \pi_i$$

where $\pi_i = \Pi^i$ and show that $M = \mathcal{O}$, i.e., $\{\omega_j \pi_i\}$ is even an integral basis of $\mathcal{O}$ over $\mathcal{O}$. We put

$$N = \sum_{j=1}^f \mathcal{O} \omega_j,$$

so that $M = N + \Pi N + \dots + \Pi^{e-1} N$. We find that

$$\mathcal{O} = N + \Pi \mathcal{O},$$

because, for $\alpha \in \mathcal{O}$, we have $\alpha \equiv a_1 \omega_1 + \dots + a_f \omega_f \pmod{\Pi \mathcal{O}}$, $a_i \in \mathcal{O}$. This implies

$$\mathcal{O} = N + \Pi(N + \Pi \mathcal{O}) = \dots = N + \Pi N + \dots + \Pi^{e-1} N + \Pi^e \mathcal{O},$$

so that $\mathcal{O} = M + \mathfrak{P}^e = M + \mathfrak{p} \mathcal{O}$. Since $L|K$ is separable, $\mathcal{O}$ is a finitely generated $\mathcal{O}$ -module (see chap. I, (2.11)), and we conclude $\mathcal{O} = M$ from Nakayama's lemma (chap. I, § 11, exercise 7). $\square$

Remark: We had already proved the identity $[L : K] = ef$ in a somewhat different way in chap. I, (8.2), also in the case where v was discrete and $L|K$ separable. Both hypotheses are actually needed. But, strangely enough, the separability condition can be dropped once K is complete with respect to the discrete valuation. In this case, one deduces the equality $\mathcal{O} = M$ in the above proof from $\mathcal{O} = M + \mathfrak{p} \mathcal{O}$, not by means of Nakayama's lemma, but rather like this: as $\mathfrak{p}^i M \subseteq M$, we get successively

$$\mathcal{O} = M + \mathfrak{p}(M + \mathfrak{p} \mathcal{O}) = M + \mathfrak{p}^2 \mathcal{O} = \dots = M + \mathfrak{p}^v \mathcal{O}$$

for all $v \geq 1$, and since $\{\mathfrak{p}^v \mathcal{O}\}_{v \in \mathbb{N}}$ is a basis of neighbourhoods of zero in $\mathcal{O}$, M is dense in $\mathcal{O}$. Since $\mathcal{O}$ is closed in K , (4.9) implies that M is closed in $\mathcal{O}$, so that $M = \mathcal{O}$.

Exercise 1. In a henselian field the zeroes of a polynomial are continuous functions of its coefficients. More precisely, one has: let $f(x) \in K[x]$ be a monic polynomial of degree n and

$$f(x) = \prod_{i=1}^r (x - \alpha_i)^{m_i}$$

its decomposition into linear factors, with $m_i \geq 1$, $\alpha_i \neq \alpha_j$ for $i \neq j$. If the monic polynomial $g(x)$ of degree n has all coefficients sufficiently close to those of $f(x)$, then it has r roots $\beta_1, \dots, \beta_r$ which approximate the $\alpha_1, \dots, \alpha_r$ to any previously given precision.

Exercise 2 (Krasner's Lemma). Let $\alpha \in \bar{K}$ be separable over K and let $\alpha = \alpha_1, \dots, \alpha_n$ be its conjugates over K . If $\beta \in \bar{K}$ is such that

$$|\alpha - \beta| < |\alpha - \alpha_i| \quad \text{for } i = 2, \dots, n,$$

then one has $K(\alpha) \subseteq K(\beta)$.

Exercise 3. A field which is henselian with respect to two inequivalent valuations is separably closed (Theorem of F.K. SCHMIDT).

Exercise 4. A separably closed field K is henselian with respect to any nonarchimedean valuation.

More generally, every valuation of K admits a unique extension to any purely inseparable extension $L|K$.

Hint: If $\alpha^p = a \in K$, one is forced to put $w(\alpha) = \frac{1}{p}v(a)$.

Exercise 5. Let K be a nonarchimedean valued field, $\mathfrak{o}$ the valuation ring, and $\mathfrak{p}$ the maximal ideal. K is henselian if and only if every polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in \mathfrak{o}[x]$ such that $a_0 \in \mathfrak{p}$ and $a_1 \notin \mathfrak{p}$ has a zero $a \in \mathfrak{p}$.

Hint: The Newton polygon.

Remark: A local ring $\mathfrak{o}$ with maximal ideal $\mathfrak{p}$ is called *henselian* if Hensel's lemma in the sense of (6.7) holds for it. A characterization of these rings which is important in algebraic geometry is the following:

A local ring $\mathfrak{o}$ is henselian if and only if every finite commutative $\mathfrak{o}$ -algebra A splits into a direct product $A = \prod_{i=1}^r A_i$ of local rings A_i .

The proof is not straightforward, we refer to [103], chap. I, §4, th. 4.2.

§ 7. Unramified and Tamely Ramified Extensions

In this section we fix a base field K which is henselian with respect to a nonarchimedean valuation v or $|\cdot|$. As before, we denote the valuation ring, the maximal ideal and the residue class field by $\mathfrak{o}, \mathfrak{p}, \kappa$, respectively. If $L|K$ is an algebraic extension, then the corresponding invariants are labelled $w, \mathcal{O}, \mathfrak{P}, \lambda$, respectively. An especially important rôle among these extensions is played by the unramified extensions, which are defined as follows.

(7.1) Definition. A finite extension $L|K$ is called **unramified** if the extension $\lambda|\kappa$ of the residue class field is separable and one has

$$[L : K] = [\lambda : \kappa].$$

An arbitrary algebraic extension $L|K$ is called unramified if it is a union of finite unramified subextensions.

Remark: This definition does not require K to be henselian; it applies in all cases where v extends uniquely to L .

(7.2) Proposition. Let $L|K$ and $K'|K$ be two extensions inside an algebraic closure $\bar{K}|K$ and let $L' = LK'$. Then one has

$$L|K \text{ unramified} \implies L'|K' \text{ unramified}.$$

Each subextension of an unramified extension is unramified.

Proof: The notations $\mathcal{O}, \mathfrak{p}, \kappa$; $\mathcal{O}', \mathfrak{p}', \kappa'$; $\mathcal{O}, \mathfrak{P}, \lambda$; $\mathcal{O}', \mathfrak{P}', \lambda'$ are self-explanatory. We may assume that $L|K$ is finite. Then $\lambda|\kappa$ is also finite and, being separable, is therefore generated by a primitive element $\bar{\alpha}$, $\lambda = \kappa(\bar{\alpha})$. Let $\alpha \in \mathcal{O}$ be a lifting, $f(x) \in \mathcal{O}[x]$ the minimal polynomial of α and $\bar{f}(x) = f(x) \bmod \mathfrak{p} \in \kappa[x]$. Since

$$[\lambda : \kappa] \leq \deg(\bar{f}) = \deg(f) = [K(\alpha) : K] \leq [L : K] = [\lambda : \kappa],$$

one has $L = K(\alpha)$ and $\bar{f}(x)$ is the minimal polynomial of $\bar{\alpha}$ over κ .

We thus have $L' = K'(\alpha)$. In order to prove that $L'|K'$ is unramified, let $g(x) \in \mathcal{O}'[x]$ be the minimal polynomial of α over K' and $\bar{g}(x) = g(x) \bmod \mathfrak{p}' \in \kappa'[x]$. Being a factor of $\bar{f}(x)$, $\bar{g}(x)$ is separable and hence irreducible over κ' , because otherwise $g(x)$ is reducible by Hensel's lemma. We obtain

$$[\lambda' : \kappa'] \leq [L' : K'] = \deg(g) = \deg(\bar{g}) = [\kappa'(\bar{\alpha}) : \kappa'] \leq [\lambda' : \kappa'].$$

This implies $[L' : K'] = [\lambda' : \kappa']$, i.e., $L'|K'$ is unramified.

If $L|K$ is a subextension of the unramified extension $L'|K$, then it follows from what we have just proved that $L'|L$ is unramified. Hence so is $L|K$, by the formula for the degree. $\square$

(7.3) Corollary. The composite of two unramified extensions of K is again unramified.

Proof: It suffices to show this for two finite extensions $L|K$ and $L'|K$. $L|K$ is unramified, hence so is $LL'|L'$, by (7.2). This implies that $LL'|K$ is unramified as well because separability is transitive and the degrees of field (and residue field) extensions are multiplicative. $\square$

(7.4) Definition. Let $L|K$ be an algebraic extension. Then the composite $T|K$ of all unramified subextensions is called the **maximal unramified subextension** of $L|K$.

(7.5) Proposition. The residue class field of T is the separable closure λ_s of κ in the residue class field extension $\lambda|\kappa$ of $L|K$, whereas the value group of T equals that of K .

Proof: Let λ_0 be the residue class field of T and assume $\bar{\alpha} \in \lambda$ is separable over κ . We have to show that $\bar{\alpha} \in \lambda_0$. Let $\bar{f}(x) \in \kappa[x]$ be the minimal polynomial of $\bar{\alpha}$ and $f(x) \in \mathcal{O}[x]$ a monic polynomial such that $\bar{f} = f \bmod \mathfrak{p}$. Then $f(x)$ is irreducible and by Hensel's lemma has a root α in L such that $\bar{\alpha} = \alpha \bmod \mathfrak{P}$, i.e., $[K(\alpha) : K] = [\kappa(\bar{\alpha}) : \kappa]$. This implies that $K(\alpha)|K$ is unramified, so that $K(\alpha) \subseteq T$, and thus $\bar{\alpha} \in \lambda_0$.

In order to prove $w(T^*) = v(K^*)$ we may suppose $L|K$ to be finite. The claim then follows from

$$[T : K] \geq (w(T^*) : v(K^*))[\lambda_0 : \kappa] = (w(T^*) : v(K^*)) [T : K]. \quad \square$$

The composite of all unramified extensions inside the algebraic closure $\bar{K}$ of K is simply called the maximal unramified extension $K_{nr}|K$ of K (nr = 'non ramifiée'). Its residue class field is the separable closure $\bar{\kappa}_s|\kappa$. K_{nr} contains all roots of unity of order m not divisible by the characteristic of κ because the separable polynomial $x^m - 1$ splits over $\bar{\kappa}_s$ and hence also over K_{nr} , by Hensel's lemma. If κ is a finite field, then the extension $K_{nr}|K$ is even generated by these roots of unity because they generate $\bar{\kappa}_s|\kappa$.

If the characteristic $p = \text{char}(\kappa)$ of the residue class field is positive, then one has the following weaker notion accompanying that of an unramified extension.

(7.6) Definition. An algebraic extension $L|K$ is called **tamely ramified** if the extension $\lambda|\kappa$ of the residue class fields is separable and one has $([L : T], p) = 1$. In the infinite case this latter condition is taken to mean that the degree of each finite subextension of $L|T$ is prime to p .

As before, in this definition K need not be henselian. We apply it whenever the valuation v of K has a unique extension to L . When the fundamental identity $ef = [L : K]$ holds and $\lambda|\kappa$ is separable, to say that the extension is unramified, resp. tamely ramified, simply amounts to saying that $e = 1$, resp. $(e, p) = 1$.

(7.7) Proposition. *A finite extension $L|K$ is tamely ramified if and only if the extension $L|T$ is generated by radicals*

$$L = T(\sqrt[m_1]{a_1}, \dots, \sqrt[m_r]{a_r})$$

such that $(m_i, p) = 1$. In this case the fundamental identity always holds:

$$[L : K] = ef.$$

Proof: We may assume that $K = T$ because $L|K$ is obviously tamely ramified if and only if $L|T$ is tamely ramified, and if this is the case, then $[T : K] = [\lambda : \kappa] = f$. Let $L|K$ be tamely ramified, so that $\kappa = \lambda$ and $([L : K], p) = 1$. We first show that $e = 1$ implies $L = K$. Let $\alpha \in L \setminus K$. Writing $\alpha = \alpha_1, \dots, \alpha_m$ for the conjugates and $a = \text{Tr}(\alpha) = \sum_{i=1}^m \alpha_i$, the element $\beta = \alpha - \frac{1}{m}a \in L \setminus K$ has trace $\text{Tr}(\beta) = \sum_{i=1}^m \beta_i = 0$. Since $v(K^*) = w(L^*)$, we may choose a $b \in K^*$ such that $v(b) = w(\beta)$ and obtain a unit $\varepsilon = \beta/b \in L \setminus K$ with trace $\sum_{i=1}^m \varepsilon_i = 0$. But the conjugates ε_i have the same residue classes $\bar{\varepsilon}_i$ in λ , because $\lambda = \kappa$. Hence $0 = \sum_{i=1}^m \bar{\varepsilon}_i = m\bar{\varepsilon}$, and thus $m \equiv 0 \pmod{p}$, which contradicts $p \nmid [L : K]$ and $m \mid [L : K]$.

Now let $\omega_1, \dots, \omega_r \in w(L^*)$ be a system of representatives for the quotient $w(L^*)/v(K^*)$ and m_i the order of $\omega_i \pmod{v(K^*)}$. Since $w(L^*) = \frac{1}{n}v(N_{L|K}(L^*)) \subseteq \frac{1}{n}v(K^*)$, where $n = [L : K]$, we have $m_i \mid n$, so that $(m_i, p) = 1$. Let $\gamma_i \in L^*$ be an element such that $w(\gamma_i) = \omega_i$. Then $w(\gamma_i^{m_i}) = v(c_i)$, with $c_i \in K$, so that $\gamma_i^{m_i} = c_i \varepsilon_i$ for some unit ε_i in L . As $\lambda = \kappa$ we may write $\varepsilon_i = b_i u_i$, where $b_i \in K$ and u_i is a unit in L which tends to 1 in λ . By Hensel's lemma the equation $x^{m_i} - u_i = 0$ has a solution $\beta_i \in L$. Putting $\alpha_i = \gamma_i \beta_i^{-1} \in L$, we find $w(\alpha_i) = \omega_i$ and

$$\alpha_i^{m_i} = a_i, \quad i = 1, \dots, r,$$

where $a_i = c_i b_i \in K$, i.e., we have $K(\sqrt[m_1]{a_1}, \dots, \sqrt[m_r]{a_r}) \subseteq L$. By construction, both fields have the same value group and the same residue class field. So, by what we proved first, we have

$$L = K(\sqrt[m_1]{a_1}, \dots, \sqrt[m_r]{a_r}).$$

The inequality $[L : K] \leq e$ and thus, in view of (6.8), the equality $[L : K] = e$, now follows by induction on r . If $L_1 = K(\sqrt[m_1]{a_1})$, then

$\omega_1 \in w(L_1^*)$ yields

$$e(L_1|K) = (w(L_1^*) : v(K^*)) \geq m_1 \geq [L_1 : K].$$

Also $e(L|L_1) \geq [L : L_1]$, because $w(L^*)/w(L_1^*)$ is generated by the residue classes of $\omega_2, \dots, \omega_r$. Thus

$$e = e(L|L_1) e(L_1|K) \geq [L : L_1][L_1 : K] = [L : K].$$

In order to prove that an extension $L = K(\sqrt[m_1]{a_1}, \dots, \sqrt[m_r]{a_r})$ is tamely ramified, it suffices to look at the case $r = 1$, i.e., $L = K(\sqrt[m]{a})$, where $(m, p) = 1$. The general case then follows by induction. We may assume without loss of generality that κ is separably closed. This is seen by passing to the maximal unramified extension $K_1 = K_{nr}$, which has the separable closure $\kappa_1 = \bar{\kappa}_s$ of κ as its residue class field. We obtain the following diagram

$$\begin{array}{ccc} L & \text{---} & L_1 \\ | & & | \\ K & \text{---} & K_1, \end{array}$$

where $L \cap K_1 = T = K$ and $L_1 = K_1(\sqrt[m]{a})$. If now $L_1|K_1$ is tamely ramified, then $\lambda_1|\kappa_1$ is separable; hence $\lambda_1 = \kappa_1$ and $p \nmid [L_1 : K_1] = [L : K] = [L : T]$, i.e., $L|K$ is also tamely ramified.

Let $\alpha = \sqrt[m]{a}$. We may assume that $[L : K] = [K(\sqrt[m]{a}) : K] = m$. In fact, if d is the greatest divisor of m such that $a = a'^d$ for some $a' \in K^*$, and if $m' = m/d$, then $\alpha = \sqrt[m']{a'}$ and $[K(\sqrt[m']{a'}) : K] = m'$. Now let $n = \text{ord}(w(\alpha) \bmod v(K^*))$. Since $mw(\alpha) = v(a) \in v(K^*)$, we have $m = dn$. Consequently $w(\alpha^n) = v(b)$, $b \in K^*$, and $v(b^d) = w(\alpha^m) = v(a)$; thus $\alpha^m = a = \varepsilon b^d$ for some unit ε in K . As $(d, p) = 1$, the equation $x^d - \varepsilon = 0$ splits over the separably closed residue field κ into distinct linear factors, hence also over K by Hensel's lemma. Therefore $\alpha^m = b^d = a$ for some new $b \in K^*$. Since $x^m - a$ is irreducible, we have $d = 1$, and hence $m = n$. Thus

$$e \geq n = [L : K] \geq ef \geq e,$$

in other words $f = 1$, and so $\lambda = \kappa$ and $p \nmid n = e$. This shows that $L|K$ is tamely ramified. $\square$

(7.8) Corollary. *Let $L|K$ and $K'|K$ be two extensions inside the algebraic closure $\bar{K}|K$, and $L' = LK'$. Then we have:*

$$L|K \text{ tamely ramified} \implies L'|K' \text{ tamely ramified.}$$

Every subextension of a tamely ramified extension is tamely ramified.

Proof: We may assume without loss of generality that $L|K$ is finite and consider the diagram

$$\begin{array}{ccc} L & \text{---} & L' \\ | & & | \\ T & \text{---} & T' \\ | & & | \\ K & \text{---} & K'. \end{array}$$

The inclusion $T \subseteq T'$ follows from (7.2). If $L|K$ is tamely ramified, then $L = T(\sqrt[m_1]{a_1}, \dots, \sqrt[m_r]{a_r})$, $(m_i, p) = 1$; hence $L' = LK' = LT' = T'(\sqrt[m_1]{a_1}, \dots, \sqrt[m_r]{a_r})$, so that $L'|K'$ is also tamely ramified, by (7.7).

The claim concerning the subextensions follows exactly as in the unramified case. $\square$

(7.9) Corollary. *The composite of tamely ramified extensions is tamely ramified.*

Proof: This follows from (7.8), exactly as (7.3) followed from (7.2) in the unramified case. $\square$

(7.10) Definition. *Let $L|K$ be an algebraic extension. Then the composite $V|K$ of all tamely ramified subextensions is called the **maximal tamely ramified subextension** of $L|K$.*

Let $w(L^*)^{(p)}$ denote the subgroup of all elements $\omega \in w(L^*)$ such that $m\omega \in v(K^*)$ for some m satisfying $(m, p) = 1$. The quotient group $w(L^*)^{(p)}/v(K^*)$ then consists of all elements of $w(L^*)/v(K^*)$ whose order is prime to p .

(7.11) Proposition. *The maximal tamely ramified subextension $V|K$ of $L|K$ has value group $w(V^*) = w(L^*)^{(p)}$ and residue class field equal to the separable closure λ_s of κ in $\lambda|K$.*

Proof: We may restrict to the case of a finite extension $L|K$. By passing from K to the maximal unramified subextension, we may assume by (7.5) that $\lambda_s = \kappa$. As $p \nmid e(V|K) = \#w(V^*)/v(K^*)$, we certainly have $w(V^*) \subseteq w(L^*)^{(p)}$. Conversely we find, as in the proof of (7.7), for every $\omega \in w(L^*)^{(p)}$ a radical $\alpha = \sqrt[m]{a} \in L$ such that $a \in K$, $(m, p) = 1$ and $w(\alpha) = \omega$, so that one has $\alpha \in V$, and $\omega \in w(V^*)$. $\square$

The results obtained in this section may be summarized in the following picture:

$$\begin{array}{ccccccc} K & \subseteq & T & \subseteq & V & \subseteq & L \\ \kappa & \subseteq & \lambda_s & = & \lambda_s & \subseteq & \lambda \\ v(K^*) & = & w(T^*) & \subseteq & w(L^*)^{(p)} & \subseteq & w(L^*). \end{array}$$

If $L|K$ is finite and $e = e'p^a$ where $(e', p) = 1$, then $[V : T] = e'$. The extension $L|K$ is called **totally** (or **purely**) **ramified** if $T = K$, and **wildly ramified** if it is not tamely ramified, i.e., if $V \neq L$.

Important Example: Consider the extension $\mathbb{Q}_p(\zeta)|\mathbb{Q}_p$ for a primitive n -th root of unity ζ . In the two cases $(n, p) = 1$ and $n = p^s$, this extension behaves completely differently. Let us first look at the case $(n, p) = 1$ and choose as our base field, instead of $\mathbb{Q}_p$, any discretely valued complete field K with finite residue class field $\kappa = \mathbb{F}_q$, with $q = p^r$.

(7.12) Proposition. *Let $L = K(\zeta)$, and let $\mathcal{O}|\mathcal{o}$, resp. $\lambda|\kappa$, be the extension of valuation rings, resp. residue class fields, of $L|K$. Suppose that $(n, p) = 1$. Then one has:*

- (i) *The extension $L|K$ is unramified of degree f , where f is the smallest natural number such that $q^f \equiv 1 \pmod{n}$.*
- (ii) *The Galois group $G(L|K)$ is canonically isomorphic to $G(\lambda|\kappa)$ and is generated by the automorphism $\varphi : \zeta \mapsto \zeta^q$.*
- (iii) $\mathcal{O} = \mathcal{o}[\zeta]$.

Proof: (i) If $\phi(X)$ is the minimal polynomial of ζ over K , then the reduction $\bar{\phi}(X)$ is the minimal polynomial of $\bar{\zeta} = \zeta \pmod{\mathfrak{P}}$ over κ . Indeed, being a divisor of $X^n - 1$, $\bar{\phi}(X)$ is separable and by Hensel's lemma cannot split into factors. ϕ and $\bar{\phi}$ have the same degree, so that $[L : K] = [\kappa(\bar{\zeta}) : \kappa] = [\lambda : \kappa] =: f$. $L|K$ is therefore unramified. The polynomial $X^n - 1$ splits over $\mathcal{O}$ and thus (because $(n, p) = 1$) over λ into distinct linear factors, so that $\lambda = \mathbb{F}_{q^f}$ contains the group μ_n of n -th roots of unity and is generated by it. Consequently f is the smallest number such that $\mu_n \subseteq \mathbb{F}_{q^f}^*$, i.e., such that $n | q^f - 1$. This shows (i). (ii) results trivially from this.

(iii) Since $L|K$ is unramified, we have $\mathfrak{p}\mathcal{O} = \mathfrak{P}$, and since $1, \zeta, \dots, \zeta^{f-1}$ represents a basis of $\lambda|\kappa$, we have $\mathcal{O} = \mathcal{o}[\zeta] + \mathfrak{p}\mathcal{O}$, and $\mathcal{O} = \mathcal{o}[\zeta]$ by Nakayama's lemma. $\square$

(7.13) Proposition. *Let ζ be a primitive p^m -th root of unity. Then one has:*

- (i) $\mathbb{Q}_p(\zeta)|\mathbb{Q}_p$ is totally ramified of degree $\varphi(p^m) = (p-1)p^{m-1}$.
- (ii) $G(\mathbb{Q}_p(\zeta)|\mathbb{Q}_p) \cong (\mathbb{Z}/p^m\mathbb{Z})^*$.
- (iii) $\mathbb{Z}_p[\zeta]$ is the valuation ring of $\mathbb{Q}_p(\zeta)$.
- (iv) $1 - \zeta$ is a prime element of $\mathbb{Z}_p[\zeta]$ with norm p .

Proof: $\xi = \zeta^{p^{m-1}}$ is a primitive p -th root of unity, i.e.,

$$\begin{aligned} \xi^{p-1} + \xi^{p-2} + \cdots + 1 &= 0, \text{ hence} \\ \zeta^{(p-1)p^{m-1}} + \zeta^{(p-2)p^{m-1}} + \cdots + 1 &= 0. \end{aligned}$$

Denoting by ϕ the polynomial on the left, $\zeta - 1$ is a root of the equation $\phi(X+1) = 0$. But this is irreducible because it satisfies Eisenstein's criterion: $\phi(1) = p$ and $\phi(X) \equiv (X^{p^m} - 1)/(X^{p^{m-1}} - 1) = (X - 1)^{p^{m-1}(p-1)} \pmod{p}$. It follows that $[\mathbb{Q}_p(\zeta) : \mathbb{Q}_p] = \varphi(p^m)$. The canonical injection $G(\mathbb{Q}_p(\zeta)|\mathbb{Q}_p) \rightarrow (\mathbb{Z}/p^m\mathbb{Z})^*$, $\sigma \mapsto n(\sigma)$, where $\sigma\zeta = \zeta^{n(\sigma)}$, is therefore bijective, since both groups have order $\varphi(p^m)$. Thus

$$N_{\mathbb{Q}_p(\zeta)|\mathbb{Q}_p}(1 - \zeta) = \prod_{\sigma} (1 - \sigma\zeta) = \phi(1) = p.$$

Writing w for the extension of the normalized valuation v_p of $\mathbb{Q}_p$, we find furthermore that $\varphi(p^m)w(\zeta - 1) = v_p(p) = 1$, i.e., $\mathbb{Q}_p(\zeta)|\mathbb{Q}_p$ is totally ramified and $\zeta - 1$ is a prime element of $\mathbb{Q}_p(\zeta)$. As in the proof of (6.8), it follows that $\mathbb{Z}_p[\zeta - 1] = \mathbb{Z}_p[\zeta]$ is the valuation ring of $\mathbb{Q}_p(\zeta)$. This concludes the proof. $\square$

If ζ_n is a primitive n -th root of unity and $n = n'p^m$, with $(n', p) = 1$, then propositions (7.12) and (7.13) yield the following result for the maximal unramified and the maximal tamely ramified extension:

$$\mathbb{Q}_p \subseteq T = \mathbb{Q}_p(\zeta_{n'}) \subseteq V = T(\zeta_p) \subseteq \mathbb{Q}_p(\zeta_n).$$

Exercise 1. The maximal unramified extension of $\mathbb{Q}_p$ is obtained by adjoining all roots of unity of order prime to p .

Exercise 2. Let K be henselian and $K_{nr}|K$ the maximal unramified extension. Show that the subextensions of $K_{nr}|K$ correspond 1-1 to the subextensions of the separable closure $\bar{\kappa}_s|\kappa$.

Exercise 3. Let $L|K$ be totally and tamely ramified, and let Δ , resp. Γ , be the value group of L , resp. K . Show that the intermediate fields of $L|K$ correspond 1-1 to the subgroups of Δ/Γ .

§ 8. Extensions of Valuations

Having seen that the henselian valuations extend uniquely to algebraic extensions we will now study the question of how a valuation v of a field K extends to an algebraic extension in general. So let v be an arbitrary archimedean or nonarchimedean valuation. There is a little discrepancy in notation here, because archimedean valuations manifest themselves only as absolute values while the letter v has hitherto been used for nonarchimedean exponential valuations. In spite of this, it will prove advantageous, and agrees with current usage, to employ the letter v simultaneously for both types of valuations, to denote the corresponding multiplicative valuation in both cases by $|\cdot|_v$ and the completion by K_v . Where confusion lurks, we will supply clarifying remarks.

For every valuation v of K we consider the completion K_v and an algebraic closure $\bar{K}_v$ of K_v . The canonical extension of v to K_v is again denoted by v and the unique extension of this latter valuation to $\bar{K}_v$ by $\bar{v}$.

Let $L|K$ be an algebraic extension. Choosing a K -embedding

$$\tau : L \longrightarrow \bar{K}_v,$$

we obtain by restriction of $\bar{v}$ to τL an extension

$$w = \bar{v} \circ \tau$$

of the valuation v to L . In other words, if v , resp. $\bar{v}$, are given by the absolute values $|\cdot|_v$, resp. $|\cdot|_{\bar{v}}$, on K , K_v , resp. $\bar{K}_v$, where $|\cdot|_{\bar{v}}$ extends precisely the absolute value $|\cdot|_v$ of K_v , then we obtain on L the multiplicative valuation

$$|x|_w = |\tau x|_{\bar{v}}.$$

The mapping $\tau : L \rightarrow \bar{K}_v$ is obviously continuous with respect to this valuation. It extends in a unique way to a continuous K -embedding

$$\tau : L_w \longrightarrow \bar{K}_v,$$

where, in the case of an infinite extension $L|K$, L_w does not mean the completion of L with respect to w , but the union $L_w = \bigcup_i L_{iw}$ of the completions L_{iw} of all finite subextensions $L_i|K$ of $L|K$. This union will be henceforth called the **localization** of L with respect to w . When $[L : K] < \infty$, τ is given by the rule

$$x = w\text{-}\lim_{n \rightarrow \infty} x_n \quad \longmapsto \quad \tau x := \bar{v}\text{-}\lim_{n \rightarrow \infty} \tau x_n,$$

where $\{x_n\}_{n \in \mathbb{N}}$ is a w -Cauchy sequence in L , and hence $\{\tau x_n\}_{n \in \mathbb{N}}$ a $\bar{v}$ -Cauchy sequence in $\bar{K}_v$. Note here that the sequence τx_n converges in the finite complete extension $\tau L \cdot K_v$ of K_v . We consider the diagram of fields

$$(*) \quad \begin{array}{ccc} L & \xrightarrow{\quad} & L_w \\ | & & | \\ K & \xrightarrow{\quad} & K_v \end{array}$$

The canonical extension of the valuation w from L to L_w is precisely the unique extension of the valuation v from K_v to the extension $L_w|K_v$. We have

$$L_w = LK_v,$$

because if $L|K$ is finite, then the field $LK_v \subseteq L_w$ is complete by (4.8), contains the field L and therefore has to be its completion. If $L_w|K_v$ has degree $n < \infty$, then, by (4.8), the absolute values corresponding to v and w satisfy the relation

$$|x|_w = \sqrt[n]{|N_{L_w|K_v}(x)|_v}.$$

The field diagram $(*)$ is of central importance for algebraic number theory. It shows the passage from the “global extension” $L|K$ to the “local extension” $L_w|K_v$ and thus represents one of the most important methods of algebraic number theory, the so-called **local-to-global principle**. This terminology arises from the case of a function field K , for example $K = \mathbb{C}(t)$, where the elements of the extension L are algebraic functions on a Riemann surface, hence on a *global* object, whereas passing to K_v and L_w signifies looking at power series expansions, i.e., the *local* study of functions. The diagram $(*)$ thus expresses in an abstract manner our original goal, to provide methods of function theory for use in the theory of numbers by means of valuations.

We saw that every K -embedding $\tau : L \rightarrow \bar{K}_v$ gave us an extension $w = \bar{v} \circ \tau$ of v . For every automorphism $\sigma \in G(\bar{K}_v|K_v)$ of $\bar{K}_v$ over K_v , we obtain with the composite

$$L \xrightarrow{\tau} \bar{K}_v \xrightarrow{\sigma} \bar{K}_v$$

a new K -embedding $\tau' = \sigma \circ \tau$ of L . It will be said to be *conjugate to τ over K_v* . The following result gives us a complete description of the possible extensions of v to L .

(8.1) Extension Theorem. *Let $L|K$ be an algebraic field extension and v a valuation of K . Then one has:*

- (i) Every extension w of the valuation v arises as the composite $w = \bar{v} \circ \tau$ for some K -embedding $\tau : L \rightarrow \bar{K}_v$.
- (ii) Two extensions $\bar{v} \circ \tau$ and $\bar{v} \circ \tau'$ are equal if and only if τ and τ' are conjugate over K_v .

Proof: (i) Let w be an extension of v to L and L_w the localization of the canonical valuation, which is again denoted by w . This is the unique extension of the valuation v from K_v to L_w . Choosing any K_v -embedding $\tau : L_w \rightarrow \bar{K}_v$, the valuation $\bar{v} \circ \tau$ has to coincide with w . The restriction of τ to L is therefore a K -embedding $\tau : L \rightarrow \bar{K}_v$ such that $w = \bar{v} \circ \tau$.

(ii) Let τ and $\sigma \circ \tau$, with $\sigma \in G(\bar{K}_v|K_v)$, be two embeddings of L conjugate over K_v . Since $\bar{v}$ is the only extension of the valuation v from K_v to $\bar{K}_v$, one has $\bar{v} = \bar{v} \circ \sigma$, and thus $\bar{v} \circ \tau = \bar{v} \circ (\sigma \circ \tau)$. The extensions induced to L by τ and by $\sigma \circ \tau$ are therefore the same.

Conversely, let $\tau, \tau' : L \rightarrow \bar{K}_v$ be two K -embeddings such that $\bar{v} \circ \tau = \bar{v} \circ \tau'$. Let $\sigma : \tau L \rightarrow \tau' L$ be the K -isomorphism $\sigma = \tau' \circ \tau^{-1}$. We can extend σ to a K_v -isomorphism

$$\sigma : \tau L \cdot K_v \longrightarrow \tau' L \cdot K_v.$$

Indeed, τL is dense in $\tau L \cdot K_v$, so every element $x \in \tau L \cdot K_v$ can be written as a limit

$$x = \lim_{n \rightarrow \infty} \tau x_n$$

for some sequence x_n which belongs to a finite subextension of L . As $\bar{v} \circ \tau = \bar{v} \circ \tau'$, the sequence $\tau' x_n = \sigma \tau x_n$ converges to an element

$$\sigma x := \lim_{n \rightarrow \infty} \sigma \tau x_n$$

in $\tau' L \cdot K_v$. Clearly the correspondence $x \mapsto \sigma x$ does not depend on the choice of a sequence $\{x_n\}$, and yields an isomorphism $\tau L \cdot K_v \xrightarrow{\sigma} \tau' L \cdot K_v$ which leaves K_v fixed. Extending σ to a K_v -automorphism $\tilde{\sigma} \in G(\bar{K}_v|K_v)$ gives $\tau' = \tilde{\sigma} \circ \tau$, so that τ and τ' are indeed conjugate over K_v . $\square$

Those who prefer to be given an extension $L|K$ by an algebraic equation $f(X) = 0$ will appreciate the following concrete variant of the above extension theorem.

Let $L = K(\alpha)$ be generated by the zero α of an irreducible polynomial $f(X) \in K[X]$ and let

$$f(X) = f_1(X)^{m_1} \cdots f_r(X)^{m_r}$$

be the decomposition of $f(X)$ into irreducible factors $f_1(X), \dots, f_r(X)$ over the completion K_v . Of course, the m_i are one if f is separable. The K -embeddings $\tau : L \rightarrow \bar{K}_v$ are then given by the zeroes β of $f(X)$ which lie in $\bar{K}_v$:

$$\tau : L \longrightarrow \bar{K}_v, \quad \tau(\alpha) = \beta.$$

Two embeddings τ and τ' are conjugate over K_v if and only if the zeroes $\tau(\alpha)$ and $\tau'(\alpha)$ are conjugate over K_v , i.e., if they are zeroes of the same irreducible factor f_i . With (8.1), this gives the

(8.2) Proposition. *Suppose the extension $L|K$ is generated by the zero α of the irreducible polynomial $f(X) \in K[X]$.*

Then the valuations $w_1, \dots, w_r$ extending v to L correspond 1-1 to the irreducible factors $f_1, \dots, f_r$ in the decomposition

$$f(X) = f_1(X)^{m_1} \dots f_r(X)^{m_r}$$

of f over the completion K_v .

The extended valuation w_i is explicitly obtained from the factor f_i as follows: let $\alpha_i \in \bar{K}_v$ be a zero of f_i and let

$$\tau_i : L \rightarrow \bar{K}_v, \quad \alpha \mapsto \alpha_i,$$

be the corresponding K -embedding of L into $\bar{K}_v$. Then one has

$$w_i = \bar{v} \circ \tau_i.$$

τ_i extends to an isomorphism

$$\tau_i : L_{w_i} \xrightarrow{\sim} K_v(\alpha_i)$$

on the completion L_{w_i} of L with respect to w_i .

Let $L|K$ be again an arbitrary finite extension. We will write $w|v$ to indicate that w is an extension of the valuation v of K to L . The inclusions $L \hookrightarrow L_w$ induce homomorphisms $L \otimes_K K_v \rightarrow L_w$ via $a \otimes b \mapsto ab$, and hence a canonical homomorphism

$$\varphi : L \otimes_K K_v \longrightarrow \prod_{w|v} L_w.$$

To begin with, the tensor product is taken in the sense of vector spaces, i.e., the K -vector space L is lifted to a K_v -vector space $L \otimes_K K_v$. This latter, however, is in fact a K_v -algebra, with the multiplication $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$, and φ is a homomorphism of K_v -algebras. This homomorphism is the subject of the

(8.3) Proposition. *If $L|K$ is separable, then $L \otimes_K K_v \cong \prod_{w|v} L_w$.*

Proof: Let α be a primitive element for $L|K$, so that $L = K(\alpha)$, and let $f(X) \in K[X]$ be its minimal polynomial. To every $w|v$, there corresponds an irreducible factor $f_w(X) \in K_v[X]$ of $f(X)$, and in view of the separability, we have $f(X) = \prod_{w|v} f_w(X)$. Consider all the L_w as embedded into an algebraic closure $\bar{K}_v$ of K_v and denote by α_w the image of α under $L \rightarrow L_w$. Then we find $L_w = K_v(\alpha_w)$ and $f_w(X)$ is the minimal polynomial of α_w over K_v . We now get a commutative diagram

$$\begin{array}{ccc} K_v[X]/(f) & \longrightarrow & \prod_{w|v} K_v[X]/(f_w) \\ \downarrow & & \downarrow \\ L \otimes_K K_v & \longrightarrow & \prod_{w|v} L_w, \end{array}$$

where the top arrow is an isomorphism by the Chinese remainder theorem. The arrow on the left is induced by $X \mapsto \alpha \otimes 1$ and is an isomorphism because $K[X]/(f) \cong K(\alpha) = L$. The arrow on the right is induced by $X \mapsto \alpha_w$ and is an isomorphism because $K_v[X]/(f_w) \cong K_v(\alpha_w) = L_w$. Hence the bottom arrow is an isomorphism as well. $\square$

(8.4) Corollary. *If $L|K$ is separable, then one has*

$$[L : K] = \sum_{w|v} [L_w : K_v]$$

and

$$N_{L|K}(\alpha) = \prod_{w|v} N_{L_w|K_v}(\alpha), \quad \text{Tr}_{L|K}(\alpha) = \sum_{w|v} \text{Tr}_{L_w|K_v}(\alpha).$$

Proof: The first equation results from (8.3) since $[L : K] = \dim_K(L) = \dim_{K_v}(L \otimes_K K_v)$. On both sides of the isomorphism

$$L \otimes_K K_v \cong \prod_{w|v} L_w$$

let us consider the endomorphism: multiplication by α . The characteristic polynomial of α on the K_v -vector space $L \otimes_K K_v$ is the same as that on the K -vector space L . Therefore

$$\text{char. polynomial}_{L|K}(\alpha) = \prod_{w|v} \text{char. polynomial}_{L_w|K_v}(\alpha).$$

This implies immediately the identities for the norm and the trace. $\square$

If v is a nonarchimedean valuation, then we define, as in the henselian case, the **ramification index** of an extension $w|v$ by

$$e_w = (w(L^*) : v(K^*))$$

and the **inertia degree** by

$$f_w = [\lambda_w : \kappa],$$

where λ_w , resp. κ , is the residue class field of w , resp. v . From (8.4) and (6.8), we obtain the **fundamental identity of valuation theory**:

(8.5) Proposition. *If v is discrete and $L|K$ separable, then*

$$\sum_{w|v} e_w f_w = [L : K].$$

This proposition repeats what we have already seen in chap. I, (8.2), working with the prime decomposition. If K is the field of fractions of a Dedekind domain $\mathcal{O}$, then to every nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}$ is associated the **$\mathfrak{p}$ -adic valuation** $v_{\mathfrak{p}}$ of K , defined by $v_{\mathfrak{p}}(a) = v_{\mathfrak{p}}$, where $(a) = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}}$ (see chap. I, § 11, p. 67). The valuation ring of $v_{\mathfrak{p}}$ is the localization $\mathcal{O}_{\mathfrak{p}}$. If $\mathcal{O}$ is the integral closure of $\mathcal{O}$ in L and if

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$$

is the prime decomposition of $\mathfrak{p}$ in L , then the valuations $w_i = \frac{1}{e_i} v_{\mathfrak{P}_i}$, $i = 1, \dots, r$, are precisely the extensions of $v = v_{\mathfrak{p}}$ to L , e_i are the corresponding ramification indices and $f_i = [\mathcal{O}/\mathfrak{P}_i : \mathcal{O}/\mathfrak{p}]$ the inertia degrees. The fundamental identity

$$\sum_{i=1}^r e_i f_i = [L : K]$$

has thus been established in two different ways. The *raison d'être* of valuation theory, however, is not to reformulate ideal-theoretic knowledge, but rather, as has been stressed earlier, to provide the possibility of passing from the extension $L|K$ to the various completions $L_w|K_v$ where much simpler arithmetic laws apply. Let us also emphasize once more that completions may always be replaced with henselizations.

Exercise 1. Up to equivalence, the valuations of the field $\mathbb{Q}(\sqrt{5})$ are given as follows.

1) $|a + b\sqrt{5}|_1 = |a + b\sqrt{5}|$ and $|a + b\sqrt{5}|_2 = |a - b\sqrt{5}|$ are the archimedean valuations.

2) If $p = 2$ or 5 or a prime number $\neq 2, 5$ such that $\left(\frac{p}{5}\right) = -1$, then there is exactly one extension of $|\cdot|_p$ to $\mathbb{Q}(\sqrt{5})$, namely

$$|a + b\sqrt{5}|_p = |a^2 - 5b^2|_p^{1/2}.$$

3) If p is a prime number $\neq 2, 5$ such that $\left(\frac{p}{5}\right) = 1$, then there are two extensions of $|\cdot|_p$ to $\mathbb{Q}(\sqrt{5})$, namely

$$|a + b\sqrt{5}|_{p_1} = |a + b\gamma|_p, \quad \text{resp.} \quad |a + b\sqrt{5}|_{p_2} = |a - b\gamma|_p,$$

where γ is a solution of $x^2 - 5 = 0$ in $\mathbb{Q}_p$.

Exercise 2. Determine the valuations of the field $\mathbb{Q}(i)$ of the Gaussian numbers.

Exercise 3. How many extensions to $\mathbb{Q}(\sqrt[n]{2})$ does the archimedean absolute value $|\cdot|$ of $\mathbb{Q}$ admit?

Exercise 4. Let $L|K$ be a finite separable extension, $\mathcal{O}$ the valuation ring of a discrete valuation v and $\widehat{\mathcal{O}}$ its integral closure in L . If $w|v$ varies over the extensions of v to L and $\widehat{\mathcal{O}}_w$, resp. $\widehat{\mathcal{O}}_v$, are the valuation rings of the completions K_w , resp. L_v , then one has

$$\mathcal{O} \otimes_{\mathcal{O}} \widehat{\mathcal{O}}_v \cong \prod_{w|v} \widehat{\mathcal{O}}_w.$$

Exercise 5. How does proposition (8.2) relate to Dedekind's proposition, chap. I, (8.3)?

Exercise 6. Let $L|K$ be a finite field extension, v a nonarchimedean exponential valuation, and w an extension to L . If $\mathcal{O}$ is the integral closure of the valuation ring $\mathcal{O}_v$ of v in L , then the localization $\mathcal{O}_{\mathfrak{P}}$ of $\mathcal{O}$ at the prime ideal $\mathfrak{P} = \{\alpha \in \mathcal{O} \mid w(\alpha) > 0\}$ is the valuation ring of w .

§ 9. Galois Theory of Valuations

We now consider Galois extensions $L|K$ and study the effect of the Galois action on the extended valuations $w|v$. This leads to a direct generalization of "Hilbert's ramification theory" — see chap. I, § 9, where we studied, instead of valuations v , the prime ideals $\mathfrak{p}$ and their decomposition $\mathfrak{p} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$ in Galois extensions of algebraic number fields. The arguments stay the same, so we may be rather brief here. However, we formulate and prove all results for extensions that are not necessarily finite, using infinite Galois theory. The reader who happens not to know this theory should feel free to assume all extensions in this section to be finite. On the other hand, we treat infinite Galois theory also in chap. IV, § 1 below. Its main result can be put in a nutshell like this:

In the case of a Galois extension $L|K$ of infinite degree, the main theorem of ordinary Galois theory, concerning the 1–1 correspondence between

the intermediate fields of $L|K$ and the subgroups of the Galois group $G(L|K)$ ceases to hold; there are more subgroups than intermediate fields. The correspondence can be salvaged, however, by considering a canonical topology on the group $G(L|K)$, the **Krull topology**. It is given by defining, for every $\sigma \in G(L|K)$, as a basis of neighbourhoods the cosets $\sigma G(L|M)$, where $M|K$ varies over the *finite* Galois subextensions of $L|K$. $G(L|K)$ is thus turned into a compact, Hausdorff topological group. The main theorem of Galois theory then has to be modified in the infinite case by the condition that the intermediate fields of $L|K$ correspond 1-1 to the *closed* subgroups of $G(L|K)$. Otherwise, everything goes through as in the finite case. So one tacitly restricts attention to *closed* subgroups, and accordingly to *continuous* homomorphisms of $G(L|K)$.

So let $L|K$ be an arbitrary, finite or infinite, Galois extension with Galois group $G = G(L|K)$. If v is an (archimedean or nonarchimedean) valuation of K and w an extension to L , then, for every $\sigma \in G$, $w \circ \sigma$ also extends v , so that the group G acts on the set of extensions $w|v$.

(9.1) Proposition. *The group G acts transitively on the set of extensions $w|v$, i.e., every two extensions are conjugate.*

Proof: Let w and w' be two extensions of v to L . Suppose $L|K$ is finite. If w and w' are not conjugate, then the sets

$$\{w \circ \sigma \mid \sigma \in G\} \quad \text{and} \quad \{w' \circ \sigma \mid \sigma \in G\}$$

would be disjoint. By the approximation theorem (3.4), we would be able to find an $x \in L$ such that

$$|\sigma x|_w < 1 \quad \text{and} \quad |\sigma x|_{w'} > 1$$

for all $\sigma \in G$. Then one would have for the norm $\alpha = N_{L|K}(x) = \prod_{\sigma \in G} \sigma x$ that $|\alpha|_v = \prod_{\sigma} |\sigma x|_w < 1$ and likewise $|\alpha|_v > 1$, a contradiction.

If $L|K$ is infinite, then we let $M|K$ vary over all finite Galois subextensions and consider the sets $X_M = \{\sigma \in G \mid w \circ \sigma|_M = w'|_M\}$. They are nonempty, as we have just seen, and also closed because, for $\sigma \in G \setminus X_M$, the whole open neighbourhood $\sigma G(L|M)$ lies in the complement of X_M . We have $\bigcap_M X_M \neq \emptyset$, because otherwise the compactness of G would yield a relation $\bigcap_{i=1}^r X_{M_i} = \emptyset$ with finitely many M_i , and this is a contradiction because if $M = M_1 \cdots M_r$, then $X_M = \bigcap_{i=1}^r X_{M_i}$. $\square$

(9.2) Definition. *The decomposition group of an extension w of v to L is defined by*

$$G_w = G_w(L|K) = \{\sigma \in G(L|K) \mid w \circ \sigma = w\}.$$

If v is a nonarchimedean valuation, then the decomposition group contains two further canonical subgroups

$$G_w \supseteq I_w \supseteq R_w,$$

which are defined as follows. Let $\mathcal{O}$, resp. $\mathcal{O}$, be the valuation ring, $\mathfrak{p}$, resp. $\mathfrak{P}$, the maximal ideal, and let $\kappa = \mathcal{O}/\mathfrak{p}$, resp. $\lambda = \mathcal{O}/\mathfrak{P}$, be the residue class field of v , resp. w .

(9.3) Definition. The inertia group of $w|v$ is defined by

$$I_w = I_w(L|K) = \{ \sigma \in G_w \mid \sigma x \equiv x \pmod{\mathfrak{P}} \text{ for all } x \in \mathcal{O} \}$$

and the ramification group by

$$R_w = R_w(L|K) = \{ \sigma \in G_w \mid \frac{\sigma x}{x} \equiv 1 \pmod{\mathfrak{P}} \text{ for all } x \in L^* \}.$$

Observe in this definition that, for $\sigma \in G_w$, the identity $w \circ \sigma = w$ implies that one always has $\sigma \mathcal{O} = \mathcal{O}$ and $\sigma x/x \in \mathcal{O}$, for all $x \in L^*$.

The subgroups G_w, I_w, R_w of $G = G(L|K)$, and in fact all canonical subgroups we will encounter in the sequel, are all *closed* in the Krull topology. The proof of this is routine in all cases. Let us just illustrate the model of the argument for the example of the decomposition group.

Let $\sigma \in G = G(L|K)$ be an element which belongs to the closure of G_w . This means that, in every neighbourhood $\sigma G(L|M)$, there is some element σ_M of G_w . Here $M|K$ varies over all finite Galois subextensions of $L|K$. Since $\sigma_M \in \sigma G(L|M)$, we have $\sigma_M|_M = \sigma|_M$, and $w \circ \sigma_M = w$ implies that $w \circ \sigma|_M = w \circ \sigma_M|_M = w|_M$. As L is the union of all the M , we get $w \circ \sigma = w$, so that $\sigma \in G_w$. This shows that the subgroup G_w is closed in G .

The groups G_w, I_w, R_w carry very significant information about the behaviour of the valuation v of K as it is extended to L . But before going into this, we will treat the functorial properties of the groups G_w, I_w, R_w .

Consider two Galois extensions $L|K$ and $L'|K'$ and a commutative diagram

$$\begin{array}{ccc} L & \xrightarrow{\tau} & L' \\ \uparrow & & \uparrow \\ K & \xrightarrow{\tau} & K' \end{array}$$

with homomorphisms τ which will typically be inclusions. They induce a homomorphism

$$\tau^* : G(L'|K') \longrightarrow G(L|K), \quad \tau^*(\sigma') = \tau^{-1}\sigma'\tau.$$

Observe here that, $L|K$ being normal, the same is true of $\tau L|\tau K$, and thus one has $\sigma'\tau L \subseteq \tau L$, so that composing with τ^{-1} makes sense.

Now let w' be a valuation of L' , $v' = w'|_{K'}$ and $w = w' \circ \tau$, $v = w|_K$. Then we have the

(9.4) Proposition. $\tau^* : G(L'|K') \rightarrow G(L|K)$ induces homomorphisms

$$G_{w'}(L'|K') \longrightarrow G_w(L|K),$$

$$I_{w'}(L'|K') \longrightarrow I_w(L|K),$$

$$R_{w'}(L'|K') \longrightarrow R_w(L|K).$$

In the latter two cases, v is assumed to be nonarchimedean.

Proof: Let $\sigma' \in G_{w'}(L'|K')$ and $\sigma = \tau^*(\sigma')$. If $x \in L$, then one has

$$|x|_{w \circ \sigma} = |\sigma x|_w = |\tau^{-1}\sigma'\tau x|_w = |\sigma'\tau x|_{w'} = |\tau x|_{w'} = |x|_w,$$

so that $\sigma \in G_w(L|K)$. If $\sigma' \in I_{w'}(L'|K')$ and $x \in \mathcal{O}$, then

$$w(\sigma x - x) = w(\tau^{-1}(\sigma'\tau x - \tau x)) = w'(\sigma'(\tau x) - (\tau x)) > 0,$$

and $\sigma \in I_w(L|K)$. If $\sigma' \in R_{w'}(L'|K')$ and $x \in L^*$, then

$$w\left(\frac{\sigma x}{x} - 1\right) = w\left(\tau^{-1}\left(\frac{\sigma'\tau x}{\tau x} - 1\right)\right) = w'\left(\frac{\sigma'\tau x}{\tau x} - 1\right) > 0,$$

so that $\sigma \in R_w(L|K)$. □

If the two homomorphisms $\tau : L \rightarrow L'$ and $\tau : K \rightarrow K'$ are isomorphisms, then the homomorphisms (9.4) are of course isomorphisms. In particular, in the case $K = K'$, $L = L'$, we find for each $\tau \in G(L|K)$:

$$G_{w \circ \tau} = \tau^{-1}G_w\tau, \quad I_{w \circ \tau} = \tau^{-1}I_w\tau, \quad R_{w \circ \tau} = \tau^{-1}R_w\tau,$$

i.e., the decomposition, inertia, and ramification groups of conjugate valuations are conjugate.

Another special case arises from an intermediate field M of $L|K$ by the diagram

$$\begin{array}{ccc} L & = & L \\ | & & | \\ K & \hookrightarrow & M. \end{array}$$

τ^* then becomes the inclusion $G(L|M) \hookrightarrow G(L|K)$, and we trivially get the

(9.5) Proposition. *For the extensions $K \subseteq M \subseteq L$, one has*

$$G_w(L|M) = G_w(L|K) \cap G(L|M),$$

$$I_w(L|M) = I_w(L|K) \cap G(L|M),$$

$$R_w(L|M) = R_w(L|K) \cap G(L|M).$$

A particularly important special case of (9.4) occurs with the diagram

$$\begin{array}{ccc} L & \xrightarrow{\quad} & L_w \\ | & & | \\ K & \xrightarrow{\quad} & K_v \end{array}$$

which can be associated to any extension of valuations $w|v$ of $L|K$. If $L|K$ is infinite, then L_w has to be read as the localization in the sense of §8, p. 160. (This distinction is rendered superfluous if we consider, as we may perfectly well do, the henselization of $L|K$.) Since in the local extension $L_w|K_v$ the extension of the valuation is unique, we denote the decomposition, inertia, and ramification groups simply by $G(L_w|K_v)$, $I(L_w|K_v)$, $R(L_w|K_v)$. In this case, the homomorphism τ^* is the restriction map

$$G(L_w|K_v) \longrightarrow G(L|K), \quad \sigma \longmapsto \sigma|_L,$$

and we have the

(9.6) Proposition. $G_w(L|K) \cong G(L_w|K_v)$,

$$I_w(L|K) \cong I(L_w|K_v),$$

$$R_w(L|K) \cong R(L_w|K_v).$$

Proof: The proposition derives from the fact that the decomposition group $G_w(L|K)$ consists precisely of those automorphisms $\sigma \in G(L|K)$ which are continuous with respect to the valuation w . Indeed, the continuity of the $\sigma \in G_w(L|K)$ is clear. For an arbitrary continuous automorphism σ , one has

$$|x|_w < 1 \implies |\sigma x|_w = |x|_{w \circ \sigma} < 1,$$

because $|x|_w < 1$ means that x^n and hence also σx^n is a w -nullsequence, i.e., $|\sigma x|_w < 1$. By §3, p. 117, this implies that w and $w \circ \sigma$ are equivalent, and hence in fact equal because $w|_K = w \circ \sigma|_K$, so that $\sigma \in G_w(L|K)$.

Since L is dense in L_w , every $\sigma \in G_w(L|K)$ extends uniquely to a continuous K_v -automorphism $\hat{\sigma}$ of L_w and it is clear that $\hat{\sigma} \in I(L_w|K_v)$, resp. $\hat{\sigma} \in R(L_w|K_v)$, if $\sigma \in I_w(L|K)$, resp. $\sigma \in R_w(L|K)$. This proves the bijectivity of the mappings in question in all three cases. $\square$

The above proposition reduces the problems concerning a single valuation of K to the local situation. We identify the decomposition group $G_w(L|K)$ with the Galois group of $L_w|K_v$ and write

$$G_w(L|K) = G(L_w|K_v),$$

and similarly $I_w(L|K) = I(L_w|K_v)$ and $R_w(L|K) = R(L_w|K_v)$.

We now explain the concrete meaning of the subgroups G_w, I_w, R_w of $G = G(L|K)$ for the field extension $L|K$.

The **decomposition group** G_w consists — as was shown in the proof of (9.6) — of all automorphisms $\sigma \in G$ that are **continuous** with respect to the valuation w . It controls the extension of v to L in a group-theoretic manner. Denoting by $G_w \backslash G$ the set of all right cosets $G_w \sigma$, by W_v the set of extensions of v to L and choosing a fixed extension w , we obtain a bijection

$$G_w \backslash G \xrightarrow{\sim} W_v, \quad G_w \sigma \mapsto w \sigma.$$

In particular, the number $\#W_v$ of extensions equals the index $(G : G_w)$. As mentioned already in chap. I, §9 — and left for the reader to verify — the decomposition group also describes the way a valuation v extends to an arbitrary separable extension $L|K$. For this, we embed $L|K$ into a Galois extension $N|K$, choose an extension w of v to N , and put $G = G(N|K)$, $H = G(N|L)$, $G_w = G_w(N|K)$, to get a bijection

$$G_w \backslash G/H \xrightarrow{\sim} W_v, \quad G_w \sigma H \mapsto w \circ \sigma|_L.$$

(9.7) Definition. *The fixed field of G_w ,*

$$Z_w = Z_w(L|K) = \{x \in L \mid \sigma x = x \text{ for all } \sigma \in G_w\},$$

is called the decomposition field of w over K .

The rôle of the decomposition field in the extension $L|K$ is described by the following proposition.

(9.8) Proposition.

- (i) *The restriction w_Z of w to Z_w extends uniquely to L .*
- (ii) *If v is nonarchimedean, w_Z has the same residue class field and the same value group as v .*
- (iii) *$Z_w = L \cap K_v$ (the intersection is taken inside L_w).*

Proof: (i) An arbitrary extension w' of w_Z to L is conjugate to w over Z_w ; thus $w' = w \circ \sigma$, for some $\sigma \in G(L|Z_w) = G_w$, i.e., $w' = w$.

(iii) The identity $Z_w = L \cap K_v$ follows immediately from $G_w(L|K) \cong G(L_w|K_v)$.

(ii) Since K_v has the same residue class field and the same value group as K , the same holds true for $Z_w = L \cap K_v$. $\square$

The **inertia group** I_w is defined only if w is a nonarchimedean valuation of L . It is the kernel of a canonical homomorphism of G_w . For if $\mathcal{O}$ is the valuation ring of w and $\mathfrak{P}$ the maximal ideal, then, since $\sigma\mathcal{O} = \mathcal{O}$ and $\sigma\mathfrak{P} = \mathfrak{P}$, every $\sigma \in G_w$ induces a κ -automorphism

$$\bar{\sigma} : \mathcal{O}/\mathfrak{P} \longrightarrow \mathcal{O}/\mathfrak{P}, \quad x \bmod \mathfrak{P} \longmapsto \sigma x \bmod \mathfrak{P},$$

of the residue class field λ , and we obtain a homomorphism

$$G_w \longrightarrow \text{Aut}_\kappa(\lambda)$$

with kernel I_w .

(9.9) Proposition. *The residue class field extension $\lambda|\kappa$ is normal, and we have an exact sequence*

$$1 \longrightarrow I_w \longrightarrow G_w \longrightarrow G(\lambda|\kappa) \longrightarrow 1.$$

Proof: In the case of a finite Galois extension, we have proved this already in chap. I, (9.4). In the infinite case $\lambda|\kappa$ is normal since $L|K$, and hence also $\lambda|\kappa$, is the union of the finite normal subextensions. In order to prove the surjectivity of $f : G_w \rightarrow G(\lambda|\kappa)$ all one has to show is that $f(G_w)$ is dense in $G(\lambda|\kappa)$ because $f(G_w)$, being the continuous image of a compact set, is compact and hence closed. Let $\bar{\sigma} \in G(\lambda|\kappa)$ and $\bar{\sigma}G(\lambda|\mu)$ be a neighbourhood of $\bar{\sigma}$, where $\mu|\kappa$ is a finite Galois subextension of $\lambda|\kappa$. We have to show that this neighbourhood contains an element of the image $f(\tau)$, $\tau \in G_w$. Since Z_w has the residue class field κ , there exists a finite Galois subextension $M|Z_w$ of $L|Z_w$ whose residue class field $\bar{M}$ contains the field μ . As $G(M|Z_w) \rightarrow G(\bar{M}|\kappa)$ is surjective, the composite

$$G_w = G(L|Z_w) \longrightarrow G(M|Z_w) \longrightarrow G(\bar{M}|\kappa) \longrightarrow G(\mu|\kappa)$$

is also surjective, and if $\tau \in G_w$ is mapped to $\bar{\sigma}|_\mu$, then $f(\tau) \in \bar{\sigma}G(\lambda|\mu)$, as required. $\square$

(9.10) Definition. *The fixed field of I_w ,*

$$T_w = T_w(L|K) = \{x \in L \mid \sigma x = x \text{ for all } \sigma \in I_w\},$$

is called the inertia field of w over K .

For the inertia field, (9.9) gives us the isomorphism

$$G(T_w|Z_w) \cong G(\lambda|\kappa).$$

It has the following significance for the extension $L|K$.

(9.11) Proposition. *$T_w|Z_w$ is the maximal unramified subextension of $L|Z_w$.*

Proof: By (9.6), we may assume that $K = Z_w$ is henselian. Let $T|K$ be the maximal unramified subextension of $L|K$. It is Galois, since the conjugate extensions are also unramified. By (7.5), T has the residue class field λ_s , and we have an isomorphism

$$G(T|K) \xrightarrow{\sim} G(\lambda_s|\kappa).$$

Surjectivity follows from (9.9) and the injectivity from the fact that $T|K$ is unramified: every finite Galois subextension has the same degree as its residue class field extension. An element $\sigma \in G(L|K)$ therefore induces the identity on λ_s , i.e., on λ , if and only if it belongs to $G(L|T)$. Consequently, $G(L|T) = I_w$, hence $T = T_w$. $\square$

If in particular K is a henselian field and $\bar{K}_s|K$ its separable closure, then the inertia field of this extension is *the* maximal unramified extension $T|K$ and has the separable closure $\bar{\kappa}_s|\kappa$ as its residue class field. The isomorphism

$$G(T|K) \cong G(\bar{\kappa}_s|\kappa)$$

shows that the unramified extensions of K correspond 1–1 to the separable extensions of κ .

Like the inertia group, the **ramification group** R_w is the kernel of a canonical homomorphism

$$I_w \rightarrow \chi(L|K),$$

where

$$\chi(L|K) = \text{Hom}(\Delta/\Gamma, \lambda^*),$$

where $\Delta = w(L^*)$, and $\Gamma = v(K^*)$. If $\sigma \in I_w$, then the associated homomorphism

$$\chi_\sigma : \Delta/\Gamma \rightarrow \lambda^*$$

is given as follows: for $\bar{\delta} = \delta \bmod \Gamma \in \Delta/\Gamma$, choose an $x \in L^*$ such that $w(x) = \delta$ and put

$$\chi_\sigma(\bar{\delta}) = \frac{\sigma x}{x} \bmod \mathfrak{P}.$$

This definition is independent of the choice of the representative $\delta \in \bar{\delta}$ and of $x \in L^*$. For if $x' \in L^*$ is an element such that $w(x') \equiv w(x) \bmod \Gamma$, then $w(x') = w(xa)$, $a \in K^*$. Then $x' = xau$, $u \in \mathcal{O}^*$, and since $\sigma u/u \equiv 1 \bmod \mathfrak{P}$ (because $\sigma \in I_{\mathfrak{P}}$), one gets $\sigma x'/x' \equiv \sigma x/x \bmod \mathfrak{P}$.

One sees immediately that mapping $\sigma \mapsto \chi_\sigma$ is a homomorphism $I_w \rightarrow \chi(L|K)$ with kernel R_w .

(9.12) Proposition. *R_w is the unique p -Sylow subgroup of I_w .*

Remark: If $L|K$ is a finite extension, then it is clear what this means. In the infinite case it has to be understood in the sense of **profinite groups**, i.e., all finite quotient groups of R_w , resp. I_w/R_w , by closed normal subgroups have p -power order, resp. an order prime to p . In order to understand this better, we refer the reader to chap. IV, §2, exercise 3–5.

Proof of (9.12): By (9.6), we may assume that K is henselian. We restrict to the case where $L|K$ is a finite extension. The infinite case of the proposition follows formally from this.

If R_w were not a p -group, then we would find an element $\sigma \in R_w$ of prime order $\ell \neq p$. Let K' be the fixed field of σ and κ' its residue class field. We show that $\kappa' = \lambda$. Since $R_w \subseteq I_w$, we have that $T \subseteq K'$. Thus $\lambda_s \subseteq \kappa'$, so that $\lambda|\kappa'$ is purely inseparable and of p -power degree. On the other hand, the degree has to be a power of ℓ , for if $\bar{\alpha} \in \lambda$ and if $\alpha \in L$ is a lifting of $\bar{\alpha}$, and $f(x) \in K'[x]$ is the minimal polynomial of α over K' , then $\bar{f}(x) = \bar{g}(x)^m$, where $\bar{g}(x) \in \kappa'[x]$ is the minimal polynomial of $\bar{\alpha}$ over κ' , which has degree either 1 or ℓ , as this is so for $f(x)$. Thus we have indeed $\kappa' = \lambda$, so that $L|K'$ is tamely ramified, and by (7.7) is of the form $L = K'(\alpha)$ with $\alpha = \sqrt[\ell]{a}$, $a \in K'$. It follows that $\sigma\alpha = \zeta\alpha$, with a primitive ℓ -th root of unity $\zeta \in K'$. Since $\sigma \in R_w$, we have on the other hand $\sigma\alpha/\alpha = \zeta \equiv 1 \bmod \mathfrak{P}$, a contradiction. This proves that R_w is a p -group.

Since $p = \text{char}(\lambda)$, the elements in λ^* have order prime to p , provided they are of finite order. The group $\chi(L|K) = \text{Hom}(\Delta/\Gamma, \lambda^*)$ therefore has order prime to p . This also applies to the group $I_w/R_w \subseteq \chi(L|K)$, so that R_w is indeed the unique p -Sylow subgroup. $\square$

(9.13) Definition. *The fixed field of R_w ,*

$$V_w = V_w(L|K) = \{x \in L \mid \sigma x = x \text{ for all } \sigma \in R_w\},$$

is called the ramification field of w over K .

(9.14) Proposition. *$V_w|Z_w$ is the maximal tamely ramified subextension of $L|Z_w$.*

Proof: By (9.6) and the fact that the value group and residue class field do not change, we may assume that $K = Z_w$ is henselian. Let V_w be the fixed field of R_w . Since R_w is the p -Sylow subgroup of I_w , V_w is the union of all finite Galois subextensions of $L|T$ of degree prime to p . Therefore V_w contains the maximal tamely ramified extension V of T (and thus of Z_w). Since the degree of each finite subextension $M|V$ of $V_w|V$ is not divisible by p , the residue field extension of $M|V$ is separable (see the argument in the proof of (9.12)). Therefore $V_w|V$ is tamely ramified, and $V_w = V$. $\square$

(9.15) Corollary. *We have the exact sequence*

$$1 \longrightarrow R_w \longrightarrow I_w \longrightarrow \chi(L|K) \longrightarrow 1.$$

Proof: By (9.6) we may assume, as we have already done several times before, that K is henselian. We restrict to considering the case of a finite extension $L|K$. In the infinite case the proof follows as in (9.9). We have already seen that R_w is the kernel of the arrow on the right. It therefore suffices to show that

$$(I_w : R_w) = [V_w : T_w] = \#\chi(L|K).$$

As $T_w|K$ is the maximal unramified subextension of $V_w|K$, $V_w|T_w$ has inertia degree 1. Thus, by (7.7),

$$[V_w : T_w] = \#(w(V_w^*)/w(T_w^*)).$$

Furthermore, by (7.5), we have $w(T_w^*) = v(K^*) =: \Gamma$, and putting $\Delta = w(L^*)$, we see that $w(V_w^*)/v(K^*)$ is the subgroup $\Delta^{(p)}/\Gamma$ of Δ/Γ consisting of all elements of order prime to p , where $p = \text{char}(\kappa)$. Thus

$$[V_w : T_w] = \#(\Delta^{(p)}/\Gamma).$$

Since λ^* has no elements of order divisible by p , we have on the other hand that

$$\chi(L|K) = \text{Hom}(\Delta/\Gamma, \lambda^*) = \text{Hom}(\Delta^{(p)}/\Gamma, \lambda^*).$$

But (7.7) implies that λ^* contains the m -th roots of unity whenever $\Delta^{(p)}/\Gamma$ contains an element of order m , because then there is a Galois extension generated by radicals $T_w(\sqrt[m]{a})|T_w$ of degree m . This shows that $\chi(L|K)$ is the Pontryagin dual of the group $\Delta^{(p)}/\Gamma$ so that indeed

$$[V_w : T_w] = \#(\Delta^{(p)}/\Gamma) = \#\chi(L|K). \quad \square$$

Exercise 1. Let K be a henselian field, $L|K$ a tamely ramified Galois extension, $G = G(L|K)$, $I = I(L|K)$ and $\Gamma = G/I = G(\lambda|K)$. Then I is abelian and becomes a Γ -module by letting $\bar{\sigma} = \sigma I \in \Gamma$ operate on I via $\tau \mapsto \sigma\tau\sigma^{-1}$.

Show that there is a canonical isomorphism $I \cong \chi(L|K)$ of Γ -modules. Show furthermore that every tamely ramified extension can be embedded into a tamely ramified extension $L|K$, such that G is the semi-direct product of $\chi(L|K)$ with $G(\lambda|K)$: $G \cong \chi(L|K) \rtimes G(\lambda|K)$.

Hint: Use (7.7).

Exercise 2. The maximal tamely ramified abelian extension V of $\mathbb{Q}_p$ is finite over the maximal unramified abelian extension T of $\mathbb{Q}_p$.

Exercise 3. Show that the maximal unramified extension of the power series field $K = \mathbb{F}_p((t))$ is given by $T = \overline{\mathbb{F}_p}((t))$, where $\overline{\mathbb{F}_p}$ is the algebraic closure of $\mathbb{F}_p$, and the maximal tamely ramified extension by $T(\{\sqrt[m]{t} \mid m \in \mathbb{N}, (m, p) = 1\})$.

Exercise 4. Let v be a nonarchimedean valuation of the field K and let $\bar{v}$ be an extension to the separable closure $\bar{K}$ of K . Then the decomposition field $Z_{\bar{v}}$ of $\bar{v}$ over K is isomorphic to the henselization of K with respect to v , in the sense of §6, p. 143.

§ 10. Higher Ramification Groups

The inertia group and the ramification group inside the Galois group of valued fields are only the first terms in a whole series of subgroups that we are now going to study. We assume that $L|K$ is a finite Galois extension and that v_K is a discrete normalized valuation of K , with positive residue field characteristic p , which admits a *unique* extension w to L . We denote by $v_L = ew$ the associated normalized valuation of L .

(10.1) Definition. For every real number $s \geq -1$ we define the **s -th ramification group** of $L|K$ by

$$G_s = G_s(L|K) = \{ \sigma \in G \mid v_L(\sigma a - a) \geq s + 1 \text{ for all } a \in \mathcal{O} \}.$$

Clearly, $G_{-1} = G$, G_0 is the inertia group $I = I(L|K)$, and G_1 the ramification group $R = R(L|K)$ which have already been defined in (9.3). As

$$v_L(\tau^{-1}\sigma\tau a - a) = v_L(\tau^{-1}(\sigma\tau a - \tau a)) = v_L(\sigma(\tau a) - \tau a)$$

and $\tau\mathcal{O} = \mathcal{O}$, the ramification groups form a chain

$$G = G_{-1} \supseteq G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots$$

of normal subgroups of G . The quotients of this chain satisfy the

(10.2) Proposition. *Let $\pi_L \in \mathcal{O}$ be a prime element of L . For every integer $s \geq 0$, the mapping*

$$G_s/G_{s+1} \longrightarrow U_L^{(s)}/U_L^{(s+1)}, \quad \sigma \longmapsto \frac{\sigma\pi_L}{\pi_L},$$

is an injective homomorphism which is independent of the prime element π_L . Here $U_L^{(s)}$ denotes the s -th group of principal units of L , i.e., $U_L^{(0)} = \mathcal{O}^$ and $U_L^{(s)} = 1 + \pi_L^s \mathcal{O}$, for $s \geq 1$.*

We leave the elementary proof to the reader. Observe that one has $U_L^{(0)}/U_L^{(1)} \cong \lambda^*$ and $U_L^{(s)}/U_L^{(s+1)} \cong \lambda$, for $s \geq 1$. The factors G_s/G_{s+1} are therefore abelian groups of exponent p , for $s \geq 1$, and of order prime to p , for $s = 0$. In particular, we find again that the ramification group $R = G_1$ is the unique p -Sylow subgroup in the inertia group $I = G_0$.

We now study the behaviour of the higher ramification groups under change of fields. If only the base field K is changed, then we get directly from the definition of the ramification groups the following generalization of (9.5).

(10.3) Proposition. *If K' is an intermediate field of $L|K$, then one has, for all $s \geq -1$, that*

$$G_s(L|K') = G_s(L|K) \cap G(L|K').$$

Matters become much more complicated when we pass from $L|K$ to a Galois subextension $L'|K$. It is true that the ramification groups of $L|K$ are mapped under $G(L|K) \rightarrow G(L'|K)$ into the ramification groups of $L'|K$, but the indexing changes. For the precise description of the situation we need some preparation. We will assume for the sequel that the residue field extension $\lambda|\kappa$ of $L|K$ is **separable**.

(10.4) Lemma. *The ring extension $\mathcal{O}$ of $\mathfrak{o}$ is monogenous, i.e., there exists an $x \in \mathcal{O}$ such that $\mathcal{O} = \mathfrak{o}[x]$.*

Proof: As the residue field extension $\lambda|\kappa$ is separable by assumption, it admits a primitive element $\bar{x}$. Let $f(X) \in \mathfrak{o}[X]$ be a lifting of the minimal polynomial $\bar{f}(X)$ of $\bar{x}$. Then there is a representative $x \in \mathcal{O}$ of $\bar{x}$ such that $\pi = f(x)$ is a prime element of $\mathcal{O}$. Indeed, if x is an arbitrary representative, then we certainly have $v_L(f(x)) \geq 1$ because $\bar{f}(\bar{x}) = 0$. If x itself is not as required, i.e., if $v_L(f(x)) \geq 2$, the representative $x + \pi_L$ will do. In fact, from Taylor's formula

$$f(x + \pi_L) = f(x) + f'(x)\pi_L + b\pi_L^2, \quad b \in \mathcal{O},$$

we obtain $v_L(f(x + \pi_L)) = 1$ since $f'(x) \in \mathcal{O}^*$, because $\bar{f}'(\bar{x}) \neq 0$. In the proof of (6.8), we saw that the

$$x^j \pi^i = x^j f(x)^i, \quad j = 0, \dots, f-1, \quad i = 0, \dots, e-1,$$

form an integral basis of $\mathcal{O}$ over $\mathfrak{o}$. Hence indeed $\mathcal{O} = \mathfrak{o}[x]$. $\square$

For every $\sigma \in G$ we now put

$$i_{L|K}(\sigma) = v_L(\sigma x - x),$$

where $\mathcal{O} = \mathfrak{o}[x]$. This definition does not depend on the choice of the generator x and we may write

$$G_s(L|K) = \{\sigma \in G \mid i_{L|K}(\sigma) \geq s+1\}.$$

Passing to a Galois subextension $L'|K$ of $L|K$, the numbers $i_{L|K}(\sigma)$ obey the following rule.

(10.5) Proposition. *If $e' = e_{L|L'}$ is the ramification index of $L|L'$, then*

$$i_{L'|K}(\sigma') = \frac{1}{e'} \sum_{\sigma|_{L'} = \sigma'} i_{L|K}(\sigma).$$

Proof: For $\sigma' = 1$ both sides are infinite. Let $\sigma' \neq 1$, and let $\mathcal{O} = \mathfrak{o}[x]$ and $\mathcal{O}' = \mathfrak{o}[y]$, with $\mathcal{O}'$ the valuation ring of L' . By definition, we have

$$e' i_{L'|K}(\sigma') = v_L(\sigma' y - y), \quad i_{L|K}(\sigma) = v_L(\sigma x - x).$$

We choose a fixed $\sigma \in G = G(L|K)$ such that $\sigma|_{L'} = \sigma'$. The other elements of G with image σ' in $G' = G(L'|K)$ are then given by $\sigma\tau$, $\tau \in H = G(L|L')$. It therefore suffices to show that the elements

$$a = \sigma y - y \quad \text{and} \quad b = \prod_{\tau \in H} (\sigma\tau x - x)$$

generate the same ideal in $\mathcal{O}$.

Let $f(X) \in \mathcal{O}'[X]$ be the minimal polynomial of x over L' . Then $f(X) = \prod_{\tau \in H} (X - \tau x)$. Letting σ act on the coefficients of f , we get the polynomial $(\sigma f)(X) = \prod_{\tau \in H} (X - \sigma \tau x)$. The coefficients of $\sigma f - f$ are all divisible by $a = \sigma y - y$. Hence a divides $(\sigma f)(x) - f(x) = \pm b$.

To show that conversely b is a divisor of a , we write y as a polynomial in x with coefficients in $\mathcal{O}$, $y = g(x)$. As x is a zero of the polynomial $g(X) - y \in \mathcal{O}'[X]$, we have

$$g(X) - y = f(X)h(X), \quad h(X) \in \mathcal{O}'[X].$$

Letting σ operate on the coefficients of both sides and then substituting $X = x$ yields $y - \sigma y = (\sigma f)(x)(\sigma h)(x) = \pm b(\sigma h)(x)$, i.e., b divides a . $\square$

We now want to show that the ramification group $G_s(L|K)$ is mapped onto the ramification group $G_t(L'|K)$ by the projection

$$G(L|K) \longrightarrow G(L'|K),$$

where t is given by the function $\eta_{L|K} : [-1, \infty) \rightarrow [-1, \infty)$,

$$t = \eta_{L|K}(s) = \int_0^s \frac{dx}{(G_0 : G_x)}.$$

Here $(G_0 : G_x)$ is meant to denote the inverse $(G_x : G_0)^{-1}$ when $-1 \leq x \leq 0$, i.e., simply 1, if $-1 < x \leq 0$. For $0 < m \leq s \leq m+1$, $m \in \mathbb{N}$, we have explicitly

$$\eta_{L|K}(s) = \frac{1}{g_0} (g_1 + g_2 + \cdots + g_m + (s - m)g_{m+1}), \quad g_i = \#G_i.$$

The function $\eta_{L|K}$ can be expressed in terms of the numbers $i_{L|K}(\sigma)$ as follows:

$$(10.6) \text{ Proposition. } \eta_{L|K}(s) = \frac{1}{g_0} \sum_{\sigma \in G} \min\{i_{L|K}(\sigma), s+1\} - 1.$$

Proof: Let $\theta(s)$ be the function on the right-hand side. It is continuous and piecewise linear. One has $\theta(0) = \eta_{L|K}(0) = 0$, and if $m \geq -1$ is an integer and $m < s < m+1$, then

$$\theta'(s) = \frac{1}{g_0} \#\{\sigma \in G \mid i_{L|K}(\sigma) \geq m+2\} = \frac{1}{(G_0 : G_{m+1})} = \eta'_{L|K}(s).$$

Hence $\theta = \eta_{L|K}$. $\square$

(10.7) Theorem (HERBRAND). *Let $L'|K$ be a Galois subextension of $L|K$ and $H = G(L|L')$. Then one has*

$$G_s(L|K)H/H = G_t(L'|K) \quad \text{where} \quad t = \eta_{L|L'}(s).$$

Proof: Let $G = G(L|K)$, $G' = G(L'|K)$. For every $\sigma' \in G'$, we choose an preimage $\sigma \in G$ of maximal value $i_{L|K}(\sigma)$ and show that

$$(*) \quad i_{L'|K}(\sigma') - 1 = \eta_{L|L'}(i_{L|K}(\sigma) - 1).$$

Let $m = i_{L|K}(\sigma)$. If $\tau \in H$ belongs to $H_{m-1} = G_{m-1}(L|L')$, then $i_{L|K}(\tau) \geq m$, and $i_{L|K}(\sigma\tau) \geq m$, so that $i_{L|K}(\sigma\tau) = m$. If $\tau \notin H_{m-1}$, then $i_{L|K}(\tau) < m$ and $i_{L|K}(\sigma\tau) = i_{L|K}(\tau)$. In both cases we therefore find that $i_{L|K}(\sigma\tau) = \min\{i_{L|K}(\tau), m\}$. Applying (10.5), this gives

$$i_{L'|K}(\sigma') = \frac{1}{e'} \sum_{\tau \in H} \min\{i_{L|K}(\tau), m\}.$$

Since $i_{L|K}(\tau) = i_{L|L'}(\tau)$ and $e' = e_{L|L'} = \#H_0$, (10.6) gives the formula (*), which in turn yields

$$\begin{aligned} \sigma' \in G_s H/H &\iff i_{L|K}(\sigma) - 1 \geq s \iff \eta_{L|L'}(i_{L|K}(\sigma) - 1) \geq \eta_{L|L'}(s) \\ &\iff i_{L'|K}(\sigma') - 1 \geq \eta_{L|L'}(s) \\ &\iff \sigma' \in G_t(L'|K), \quad t = \eta_{L|L'}(s). \end{aligned} \quad \square$$

The function $\eta_{L|K}$ is by definition strictly increasing. Let the inverse function be $\psi_{L|K} : [-1, \infty) \rightarrow [-1, \infty)$. One defines the **upper numbering** of the ramification groups by

$$G^t(L|K) := G_s(L|K) \quad \text{where} \quad s = \psi_{L|K}(t).$$

The functions $\eta_{L|K}$ and $\psi_{L|K}$ satisfy the following transitivity condition:

(10.8) Proposition. *If $L'|K$ is a Galois subextension of $L|K$, then*

$$\eta_{L|K} = \eta_{L'|K} \circ \eta_{L|L'} \quad \text{and} \quad \psi_{L|K} = \psi_{L|L'} \circ \psi_{L'|K}.$$

Proof: For the ramification indices of the extensions $L|K$, $L'|K$, $L|L'$ we have $e_{L|K} = e_{L'|K} e_{L|L'}$. From (10.7), we obtain $G_s/H_s = (G/H)_t$, $t = \eta_{L|L'}(s)$; thus

$$\frac{1}{e_{L|K}} \#G_s = \frac{1}{e_{L'|K}} \#(G/H)_t \frac{1}{e_{L|L'}} \#H_s.$$

This equation is equivalent to

$$\eta'_{L|K}(s) = \eta'_{L'|K}(t)\eta'_{L|L'}(s) = (\eta_{L'|K} \circ \eta_{L|L'})'(s).$$

As $\eta_{L|K}(0) = (\eta_{L'|K} \circ \eta_{L|L'})(0)$, it follows that $\eta_{L|K} = \eta_{L'|K} \circ \eta_{L|L'}$ and the formula for ψ follows. $\square$

The advantage of the upper numbering of the ramification groups is that it is invariant when passing from $L|K$ to a Galois subextension.

(10.9) Proposition. *Let $L'|K$ be a Galois subextension of $L|K$ and $H = G(L|L')$. Then one has*

$$G^t(L|K)H/H = G^t(L'|K).$$

Proof: We put $s = \psi_{L'|K}(t)$, $G' = G(L'|K)$, apply (10.7) and (10.8), and get

$$\begin{aligned} G^t H/H &= G_{\psi_{L|K}(t)} H/H = G'_{\eta_{L|L'}(\psi_{L|K}(t))} = G'_{\eta_{L|L'}(\psi_{L|L'}(s))} \\ &= G'_s = G'^t. \end{aligned} \quad \square$$

Exercise 1. Let $K = \mathbb{Q}_p$ and $K_n = K(\zeta)$, where ζ is a primitive p^n -th root of unity. Show that the ramification groups of $K_n|K$ are given as follows:

$$\begin{aligned} G_s &= G(K_n|K) && \text{for } s = 0, \\ G_s &= G(K_n|K_1) && \text{for } 1 \leq s \leq p-1, \\ G_s &= G(K_n|K_2) && \text{for } p \leq s \leq p^2-1, \\ &\dots \\ G_s &= 1 && \text{for } p^{n-1} \leq s. \end{aligned}$$

Exercise 2. Let K' be an intermediate field of $L|K$. Describe the relation between the ramification groups of $L|K$ and of $L|K'$ in the upper numbering.

